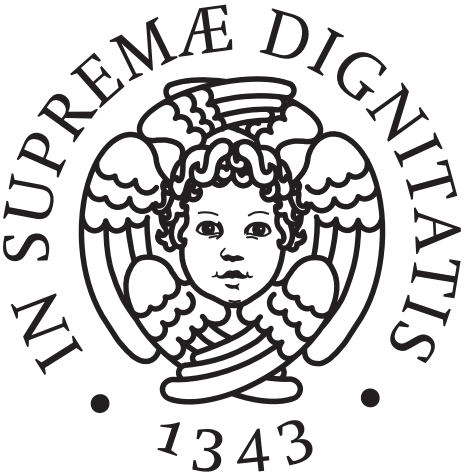
Geometrical frustration and its effect in signed social networks

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October 20, 2021

Acknowledgements

I thank Professor Frank Schweitzer for the opportunity to carry out my master thesis at the Chair of Systems Design at ETH Zürich. It was a unique experience for me that made me grow both personally and professionally.

I also want to thank Dr Giacomo Vaccario for his careful supervision and helpful advice, Georges Andres for his comments and suggestions and my Italian supervisor, Professor Riccardo Mannella, for his comments on the thesis.

I want to thank my old friends from Siena, the Bujo e Bollore, my old high school class, and my university friends from Siena and Pisa. But I also want to thank the new friends of Culmannstrasse, particularly the floor number 1, with whom I spent a pleasant time in Zürich.

Most importantly, I would like to thank my family: my mother Emma, my father Wally, my sister Eleonora and my grandparents Nino and Marisa, for believing in me and supporting me always.

Abstract

The thesis studies frustrated random networks and proposes a model for generating interactions from them. In particular, the proposed model generates interaction patterns within social systems.

The thesis is divided into four parts. 1) We first introduce the necessary definitions and concepts from network theory, geometrical frustration in physics and balance theory in social science. With a network perspective, we abstract from the many different details of physical and social systems and focus instead on the shared elements: i) the nodes representing individuals (or particles) and ii) the signed links representing the relations between individuals (or interactions between particles). Moreover, thanks to balance theory, we bridge the concepts of geometrical frustration from physics and cognitive dissonance in social science.

2) By taking a statistical physics approach, we introduce a pseudo-Hamiltonian function that quantifies the "cost" of a given set of signed dyadic relations in a system. This function is composed of three different terms: the preference of having positive relations, the preference of being included in balanced triadic relations, and the preference to interact with nodes with similar characteristics. We study these terms analytically and numerically to characterise the energy landscape and provide exact values of the energy levels.

3) Using the proposed pseudo-Hamiltonian, we develop a procedure to generate interactions within a social system. By carefully analysing the pseudo-Hamiltonian, we find that it cannot be directly used to model the formation of social interactions. Therefore, we modify the procedure by proposing a new set of rules. We then analyse the topology of the networks that emerge by aggregating the generated interactions.

4) The last part of this thesis is devoted to the extensions and applications of the presented model. We first discuss how to introduce a dynamic for signed dyadic relations by using a Lagrangian formulation. Then we create a connection between our model and the Exponential family of Random Graph Models (ERGMs). With this connection, we provide a way to infer the parameters of the presented model starting from real-world data.

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Chapter 1

Introduction

Since the 19th century, social sciences have always been related to quantitative models inspired by mathematics, probability theory and physics. Even before, Scottish philosopher David Hume (1711–76), in "A Treatise of Human Nature" [13], proposed the study of a new "science of man" with the spirit of mathematics and physics. Later, French philosopher Auguste Comte (1798–1857) started thinking that general laws could be applied to society much as the physical world does, giving birth to the field of *sociology*. In the following century, further developments in physics led to the formulation of models that shed new light on both physics and sociology. This is the case of the Ising model [14], put forward in 1924 by Ernst Ising, which gives an abstract representation of spin systems to study magnetism. It assigns a spin value $+1$ or -1 to each element on a lattice, and the strength of the pairwise coupling between neighbouring spins controls the state of the system. It is possible to observe a ferromagnetic phase with all the spins oriented in the same direction or, in other conditions of the couplings, an antiferromagnetic phase where the spins are antiparallel. This kind of lattice models was later generalised as *cellular automata* (CA) [23] and was one first example of *agent-based models*. While some cellular automata models have been employed to study social segregation, the Ising model has also been applied to early studies of opinion dynamics. Positive or negative spins represented opinions on a given question and the coupling the pairwise social influence. Different versions of the model have been proposed to better study phenomena such as consensus formation, polarisation or a stable coexistence of clusters of different opinions. The voter model [8] is an example of this and has also been extended to topologies different from lattices, including networks.

In the latest years, important developments in agent-based modelling and network science provided fundamental tools for quantitative studies of social systems, giving birth to new fields such as computational social science. By taking a network perspective, it is possible to abstract from the many different facets of physical and social systems and to focus instead on the shared elements: i) the nodes representing individuals (or particles) and ii) the signed links representing the relations between individuals (or interactions between particles). Obviously, physical particles and people are far from being alike. Single particles can be represented by simple variables, like their spins, and their pair interactions can follow simple rules too. The complexity arises when considering large numbers of interacting particles, as new emergent behaviours of the global system arise. For humans, this representation is not possible as they are not

simply representable in the first place. The social behaviour of a person is the result of complex mental processes that simple rules cannot represent.

In the spirit of the agent-based modelling methods, we can try to characterise the agents with few properties and simple interaction rules and reproduce a specific global behaviour of the system. Comparing the simulation results to the expected behaviour considered by the model allows to study the driving mechanisms in the system.

In the case of the model proposed in this thesis, we will consider constant attributes $s_i \in \{+1, -1\}$ assigned to every agent, represented as nodes in a network. Every link in the network between two nodes can assume two values $A_{ij} \in \{+1, -1\}$ representing the valence of the relation between agent i and agent j: $+1$ for friendship, -1 for enmity. We will consider every agent in the system to have a relation with every other agent, and therefore the network to have a fully connected topology.

This thesis will study the properties of frustrated random networks and propose a model for generating interactions from them. In particular, the proposed model will be deployed to generate interaction patterns within social systems. In this Chapter 1 we present the outline of the thesis.

In Chapter 2 we will introduce the necessary formal definitions from network theory and the basic concepts that will allow us to study the topology of systems quantitatively. We will present the characteristic features of different types of networks that will be used when discussing the model and the relative observables with which the outputs of the proposed process will be studied.

We will then introduce the concept of geometrical frustration as first formulated by J. Vannimenus and G. Toulouse in [22] for the study of magnetic systems. These systems are characterised by competing interactions, where both ferromagnetic and antiferromagnetic couplings can connect any pair of elements of the system. The representation of these systems is provided by models similar to the Ising and is fundamental for the studies on spin glasses.

The formulation developed in the field of frustrated magnetic systems fits naturally into the study of signed social networks, where links represent positive or negative relations between agents. The social-psychological theory of structural balance, first introduced by F. Heider in [12], suggests that certain triadic relations are more stable than others, which can also induce balancing dynamics on the system. This theory has been generalised and mathematically formalised by D. Cartwright and F. Harary in [5] to take into account multiple agents, in the form of signed networks, and to provide valuable theorems and definitions to characterise the overall cognitive dissonance in the system.

A triadic relation consists of three individuals with mutual opinions towards one another, that can be either positive or negative, i.e. they can be friends or enemies. It is considered *balanced* when the three agents are all friends with each other or when two friends have a common enemy, which also corresponds to the situation described by the proverb "the enemy of my enemy is my friend". In those cases, the number of negative relations is 0 or 2. In the other cases, the triadic relation is considered *unbalanced*. The number of negative relations can be 1, for situations as "the friend of my friend is my enemy", or 3 for "the enemy of my enemy is *not* my friend". This kind of situations can lead to tensions within the three agents. We refer to this tension as

cognitive dissonance that can be produced by unbalanced triadic relations.

Chapter 3 quantitatively describes signed social networks by means of a utility function that measures the "cost" of a given configuration of relations. The cost represents the degree of cognitive dissonance present in the system. The use of this function is similar to the Hamiltonian formalism in statistical physics. We will assume the cost function to be analogous to the energy in magnetic systems and, therefore, proceed to call it pseudo-Hamiltonian, or Hamiltonian for simplicity. We will present existing analyses of similar systems by using this approach to measure the degree of structural balance of the system, as in the work by Seth A. Marvel and collaborators [15] or G. Facchetti and collaborators [9].

Subsequently, we will define a different pseudo-Hamiltonian composed of the contribution of three separate terms. They represent the preference of having positive relations, the preference of being included in balanced triadic relations, and the preference to interact with agents with similar characteristics. We will study the properties of each term analytically and numerically. We will show their properties of symmetry and the calculation of their energy levels. In particular, for the structural balance term, we will develop three different numerical processes to study its dependence on the topology, which is non-trivial. Therefore, we will provide the analytical form of specific energy levels and characterise the local minimum states. At the end of the Chapter, we will present the results of the energies for the global Hamiltonian combining all the terms.

The aim of Chapter 4 is to define a procedure to simulate the formation of interactions within a social system, starting from a given network of the relations. Such interactions could represent exchanging messages, emails, a chat, a phone call, or other social interactions. A careful analysis of the proposed pseudo-Hamiltonian shows that it cannot be directly used to model this process, and we will need to define a different yet related formulation. Then we will define a probability of interaction based on the pseudo-Hamiltonian. Two different algorithms will be used to generate the interactions aggregated as multi-edge networks, where every edge between two nodes represents a social interaction. We will identify relevant observables on these networks and compute them for many different sets of parameters. With this, we will study how the systemic properties of the networks change with reference to the weights given to the "social effects". Moreover, we will present some connections between the interaction networks and the values of the energy of the signed networks that generate them.

The last Chapter 5 will focus on the possible applications and extensions of the model. We will start by introducing an analytical formulation for the dynamics of signed networks by defining a Lagrangian function based on the pseudo-Hamiltonian introduced in Chapter 2. We will deduce the equations of motion for the signed networks and a solution approach. The last part will be devoted to making a connection between our model and the Exponential Random Graph Models (ERGMs), which are common models to study real-world networks. The aim will be to provide a procedure to infer the model parameters starting from an empirical interaction network by using a Maximum Likelihood Estimation algorithm (MLE) combined with a Markov Chain Monte-Carlo sampling (MCMC).

Chapter 2

Signed networks and structural balance

In this Chapter, we will introduce the necessary definitions and concepts from network theory, geometrical frustration in physics and balance theory in social science.

In the first section, the standard network definitions will be presented. By taking a network perspective, we abstract from the many different details of physical and social systems and focus instead on the shared elements: i) the nodes representing individuals (or particles) and ii) the signed links representing the relations between individuals (or interactions between particles). We will discuss the characteristic features of different types of networks and the relative observables.

In the second section, we will introduce the concept of geometrical frustration as first formulated by J. Vannimenus and G. Toulouse in [22] for the study of magnetic systems characterised by competing interactions, where both ferromagnetic and anti-ferromagnetic couplings can connect any pair of elements of the system.

In the third and last section, we will present the social-psychological theory of structural balance, first introduced by F. Heider in [12] and the generalisation provided by D. Cartwright and F. Harary in [5] together with some of their results and new important concepts.

This Chapter is essentially all based on the existing literature, and it serves as the basis for building our novel model.

2.1 Network theory

The concepts and definitions that we will provide now follow the notes of a series of lectures on Complex Networks [18] held at ETH Zürich by Professor F. Schweitzer during the spring semester 2021, in particular lectures 2 and 3.

In order to model and analyse a system in general and a social system in particular, it is often convenient to adopt a network perspective. In network theory a network, or graph, can be defined as an ordered pair $G := (V, E)$ where $V := \{v_1, v_2, ..., v_N\}$ is the set of nodes and $E := \{(v_i, v_j), ..., (v_k, v_l) : v_i, v_j, v_k, v_l \in V\}$ is a set of edges connecting the nodes in V and it is a subset of all the possible edges: $E \subseteq V \times V$ where $\times$ is the Cartesian product. The nodes are sometimes called vertices and represent

the elements in the system, while the edges, often called links or bonds, represent the connection between them. By considering a network $G = (V, E)$ to represent a system, we abstract from its details and take into account only information about its elements represented as nodes and the topology of their interactions. Despite the simplicity of the representation, the network analysis can highlight important properties of the system under consideration. For this reason, it has been successfully applied to different problems ranging from information retrieval, statistical physics, ecology, and sociology.

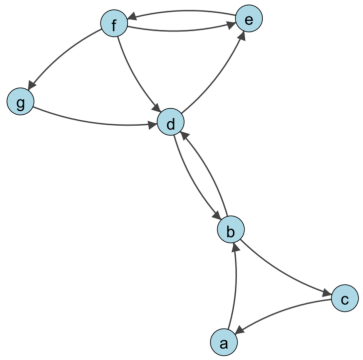
An important property of the connections between elements is the direction. In fact, sometimes it is necessary to specify that certain connection happens only in one direction, and not in the opposite one. This is the case, for example, of citation data where a document i citing a document j is represented as a node i with a direct link towards node j. The convention is to represent this link as (i, j), where i points to j and $(i, j) \in E \implies (j, i) \in E$ in general. Directed links are represented graphically by means of an arrow, as seen in figure 2.1.

Some other situations instead do not need to have a direction in the representation of the connection. For example, in an online social network, like Facebook, when two users declare to be friends, their relation of friendship does not need to be indicated as directed; both (i, j) and (j, i) are present in the network and represent the same link. In this case, when all the links (i, j) in a network G are such that $\forall\,(i, j) \in E \implies (j, i) \in E$ the network is said "undirected", otherwise it is called "directed".

In any case, the complete representation of a network is given by the adjacency matrix $\mathbf{A}$, whose entries are defined as:

$$A_{ij} = \begin{cases} 1 & \text{if } (i, j) \in E \\ 0 & \text{otherwise} \end{cases} \tag{2.1}$$

We can consider as an example the following directed network on the left and its relative adjacency matrix on the right of figure 2.1:



(a) Directed network of $N = 7$ nodes

$$\mathbf{A} = \begin{array}{c} \\ a \\ b \\ c \\ d \\ e \\ f \\ g \end{array} \begin{array}{c} \begin{array}{ccccccc} a & b & c & d & e & f & g \end{array} \\ \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix} \end{array}$$

(b) Adjacency matrix 7×7 of the network on the left

Figure 2.1: Example of a directed unweighted network and its relative adjacency matrix

The presence of a link (i, j) from i to j is indicated by entry A_{ij}: in row i and column j of matrix $\mathbf{A}$. The outgoing links of a node i can be found in the entries of row i. Conversely, the incoming links of a node j are reported in the entries of the column

j. Adjacency matrices are necessarily square matrices, and in most practical cases, they are sparse, i.e. the number of edges is of the order of the number of nodes: $|E| \approx |V| = N$. Networks with this property are called *sparse networks*.

It is clear that the adjacency matrix referred to an undirected network is symmetric, i.e. $\mathbf{A} = \mathbf{A}^T$, and the graphical representation of the links can be done just with lines, without arrows.

Sometimes the analysis of the system requires taking into account an additional characteristic of the links, namely their weight w. We, therefore, define the function $w : E \to \mathbb{R}$ to assign weights to links. *Weighted networks* have real-valued adjacency matrices defined by $A_{ij} = w(i, j)$. In this case, in order to include weights for each link, the set of edges E changes from being a set of ordered pairs of the form (i, j), to a set of triplets (i, j, A_{ij}), where A_{ij} represents the edge weight. The link weights can capture different aspects of "link strength" between two nodes: they can describe, for example, the frequency or duration of contacts between two agents in a social network, the number of co-authored papers in a co-authorship network, or the trade volume in a network of financial transactions between economic actors, and many others.

A particular subset of weighted networks is represented by networks whose links are weighted by only two possible values of $w \in \{+1, -1\}$. The networks with this property are called *signed networks*, and they will be fundamental in the description of relations in a social system as they could represent their valence, i.e. positive/friendship or negative/enmity relations, as we will see.

Some systems can be characterised by elements with self-interactions, called *self-loops*, which can be represented by non-zero entries in the diagonal of the adjacency matrix, even if we will not consider such systems in this work.

Although in most practical cases real networks are sparse, for future analysis, we will consider systems characterised by all-to-all connections: each element of the system has a connection with all the others. Such networks are called complete graphs or *fully connected networks* and are considered in order to simplify the analysis. The number of edges $|E|$ depends on the type of network we are considering. Let us consider a fully connected directed network with N nodes and self-loops, in this case, the number of edges $|E| = N^2$, i.e. the total number of entries in the adjacency matrix. If the directed network does not have self-loops, the number of edges is decreased by N and $|E| = N(N - 1)$. In the case of a fully connected undirected network, we can avoid counting every link twice and the number of edges becomes $|E| = \frac{N(N+1)}{2}$ for the case with self-loops and $|E| = \frac{N(N-1)}{2}$ without, which are basically the numbers of independent elements of the respective adjacency matrices. The number of edges $|E|$ is an important characteristic of the network and it will be indicated with M. For the future study, we will extensively use the case of fully connected undirected networks without self-loops, and therefore an $M = \frac{N(N-1)}{2} = \binom{N}{2}$.

The last type of network we are going to consider is the *multi-edge network* or multigraph. As the name suggests, it represents systems where there can be multiple connections between the same two elements. In order to take into account the information regarding the number of edges between two nodes, we have to define a function $r : V \times V \to \mathbb{N}$. The adjacency matrix of multi-edge networks is therefore defined by $A_{ij} = r(i, j)$. The set of edges is modified accordingly as the set of triplets of the form (i, j, A_{ij}), where $A_{ij} \in \mathbb{N}$ represents the number of edges between nodes i and j.

Despite the similarity between the representation of the multi-edge and the weighted networks, they are conceptually different as well as being represented by functions that have different codomain. Depending on the system under consideration, the edges of a multi-edge network can be either directed or undirected and can even represent self-loops. It is important to say that in this case the number of edges $M = |E|$ can be even larger than N^2, regardless of the natures directed or undirected of the edges.

In this work, we will make intensive use of these notions, in particular the concepts of signed, weighted and multi-edge networks. In the following sections, signed networks will play an important role as they will be used to represent the interpersonal relations in social systems, while weighted and multi-edge networks will be used in Chapter 4 when we will start to generate and study the patterns of social interactions.

Degree sequence

In order to characterise the topology of different networks, we will introduce some statistics of the networks and some observables to measure their topological properties. The first statistics that we are going to introduce is the degree sequence S, defined as a vector of length N, where every entry represents the number of links that are connected to the node i, i.e. its degree d_i:

$$S := \Big(d_i\Big)_{i \in V} \tag{2.2}$$

The degree can be calculated starting from the adjacency matrix A_{ij}. When we consider an undirected network, thanks to the symmetry of A_{ij}, we can define the degree as:

$$d_i = \sum_j A_{ij} = \sum_j A_{ji} \tag{2.3}$$

While, instead, if we consider a directed network we can differentiate between the *indegree* and the *outdegree* of a node, i.e. the number of incoming and outgoing links connected to the node i and define:

$$\begin{cases} d_{in}(i) = \sum_j A_{ji} \\ d_{out}(i) = \sum_j A_{ij} \end{cases} \tag{2.4}$$

Sometimes, for directed networks a total degree $d_{total}(i)$ is defined as $d_{total}(i) = d_{in}(i) + d_{out}(i)$, i.e. the total degree counts all links (both incoming and outgoing). As, in the following, we will always consider undirected networks, and therefore a symmetric adjacency matrix, from now on, we will deal only with the properties for the undirected case, without mentioning the directed corresponding, which is anyway often straightforward.
For weighted networks, the same definitions apply as above. However, we have to remember that in this case $A_{ij} \in \mathbb{R}$, and similar reasoning can be used for multi-edges networks, where instead we can consider $A_{ij} \in \mathbb{N}$, as multiple (unweighted) edges can exist between node i and node j.

Degree distribution

Another important statistics, related to the degree sequence S, is the degree distribution $P(k)$, that can be defined as:

$$\begin{aligned} P(k) &:= \frac{N_k}{\sum_k N_k} = \frac{N_k}{N} \\ \text{where} \quad N_k &:= |\{i \in V : d_i = k\}| \end{aligned} \tag{2.5}$$

For each degree k we can calculate the number of nodes N_k with degree k, then divide N_k by the sum over N_k for all k, i.e. the number of nodes N in the network, as $\sum_k N_k = N$. For each k we obtain the fraction of all nodes that have degree k and, interpreted in a probabilistic way, $P(k)$ is the probability that a node chosen at random has a given degree k.

We can use the degree distribution to compute the mean node degree $\langle k \rangle$ in the network, i.e. the expected degree of a node chosen at random:

$$\langle k \rangle = \sum_k k \cdot P(k) = \sum_{i \in V} \frac{d_i}{N} = \frac{2 \cdot |E|}{|V|} \tag{2.6}$$

The mean degree $\langle k \rangle$ is a good estimation of the degree distribution only if the distribution is unimodal, and this, as we will see, could not always be the case.

The degree distribution is a useful aggregate statistic that can encode many properties of the network. It is actually a good compromise between the complexity of the adjacency matrix and the excessive simplicity of the mean degree. It is, therefore, an important quantity to study, but it comes with the complication that it is not a single number but rather a density function, and we will need to deal with it.

Path, distance and average path length

One of the most important concepts in a network $G = (V, E)$ is that of a path. Two nodes i and j can be in fact connected by a direct link (i, j) or through a succession of links. We can therefore define a path as a sequence of nodes $(p_0, ..., p_l)$ where any pair of consecutive nodes is connected by a link and:

$$\begin{cases} p_i \in V \\ p_0 = i \ , \ p_l = j \\ (p_i, p_{i+1}) \in E \quad \forall i \in [0, l-1] \end{cases} \tag{2.7}$$

We also define the path length as the number of links traversed by a path, in this case $\text{len}(p_0, ..., p_l) = l$.

There can be many different paths (actually infinitely many) that connect any two nodes. Note that a path between two nodes does not necessarily need to follow the shortest. The distance $\text{dist}(i, j)$ between nodes i and j is defined as the minimum length of any path between i and j. If there is no path between i and j we define the distance to be infinite: $\text{dist}(i, j) := \infty$. The definition of distance between two nodes induces to define a path $(p_0, ..., p_l)$ to be a shortest path if $\text{len}(p_0, ..., p_l) = \text{dist}(p_0, p_l)$. Again,

it is important to stress that the shortest path is not necessarily unique, i.e. different shortest paths of the same length can exist for a given pair of nodes. The distribution of shortest path lengths between all pairs of nodes is an important topological feature of complex networks.

Two estimators for this distribution can be introduced. The network diameter $\mathrm{diam}(G)$ is the longest shortest path, defined as:

$$\mathrm{diam}(G) := \max_{(i,j)\in V\times V} \Big(\mathrm{dist}(i,j)\Big) \tag{2.8}$$

The diameter is essentially the worst-case shortest path length, but it does not say anything about typical shortest path lengths.

The average shortest path length $\langle l \rangle$ (sometimes called average path length or simply APL) is therefore introduced as the average distance between any pair of nodes:

$$\langle l \rangle := \frac{1}{N^2} \sum_{(i,j)\in V\times V} \mathrm{dist}(i,j) \tag{2.9}$$

In fact, we could find that for some networks the diameter can be large, even if most shortest path lengths are actually very small; the most important example of networks with this property are the so-called *small-world networks*. In these cases, the average shortest path length is a better characterisation for the shortest path structures in a network.

Connectedness and largest connected component

One of the most important characteristics of a network is whether all nodes in the network are actually connected to each other via a path. If this is the case (i.e. when all distances are finite), we say that the network is connected and $\mathrm{dist}(i,j) < \infty \; \forall i,j \in V$. If a network is not connected we can always find a set of maximally connected subgraphs $G^{(i)} = (V^{(i)}, E^{(i)})$ with $V^{(i)} \subset V$ and $E^{(i)} \subset E$ called connected components, such that the union of all the connected components is equal to the starting network $G = \bigcup_i G^{(i)}$. For networks that are not connected, we are often interested in the largest connected component (LCC), i.e. the largest subset of nodes where all pairs of nodes are connected via paths. We often say that the largest connected component is a giant connected component if it contains almost all of the nodes (which implies that it is much larger than the second-largest component). How exactly we define this depends on the context: for theoretical studies of graph models with a variable number of nodes N we typically call the largest connected component G' a giant connected component if $\frac{|V'|}{|V|} \to 1$ for $N \to \infty$. For finite-size empirical networks, we typically use a reasonable threshold (e.g. 0.95 or 0.99), i.e. we accept that there is a tiny fraction of disconnected nodes. In these cases, we sometimes disregard those nodes and study the largest connected component as a network representation of the system in question.

Clustering coefficient

Another important feature of networks, in particular of social networks, is the property of clustering. The local clustering coefficient C_i is defined for a focal node $i \in V$, and it

measures how connected are the neighbours of i. In fact, it counts how many links are between the neighbours of i, divided by the number of edges between the neighbours:

$$C_i = \frac{k(i)}{\binom{d_i}{2}} = \frac{2 \cdot k(i)}{d_i(d_i - 1)} \tag{2.10}$$

where d_i is the degree of node i and $k(i)$ is the number of couples, neighbours of i, that have a link between them, and it is defined as:

$$k(i) := |\{(j,k) \in E : (i,j) \in E \wedge (i,k) \in E\}| \tag{2.11}$$

The local clustering coefficient basically measures the number of closed triads in which node i is included, normalised with the maximum number of triangles in which it can be included. It is clear that in a fully connected network, all the nodes in the system have a local clustering coefficient of exactly 1: $C_i = 1 \ \forall i \in V$. In the interaction networks, where the topology is less trivial, the distribution of the values of the local clustering coefficient is different. An estimator of this distribution is given by the global clustering coefficient, which is basically the mean local clustering coefficient over all the nodes, and it is defined as:

$$C = \frac{1}{N} \sum_{i \in V} C_i \tag{2.12}$$

The global clustering coefficient C can be seen as the expected local clustering coefficient of a node chosen at random. From a statistical point of view, it can be seen as a conditional probability which captures the probability of a link (j,k), given the condition that the two links (i,j) and (i,k) exist:

$$C = P[(j,k) \in E \mid (i,j) \in E, \ (i,k) \in E] \tag{2.13}$$

Modularity

The last important concept that we will use in order to study networks is the *modularity*. We could define modularity as the quantity that measures the strength of division of a network into *communities*. Communities, also called modules or clusters, are partitions $C_i \subseteq V$ such that $\bigcup_i C_i = V$. Communities are said *overlapping* if $C_i \cap C_j \neq 0$. In order to measure how well an undirected network G with N nodes, M links and adjacency matrix $\boldsymbol{A}$ can be divided into non overlapping communities, we introduce the *partition quality* $Q(G,C)$ [4]:

$$Q(G,C) = \frac{1}{2M} \sum_{i,j} \left(A_{ij} - \frac{d_i d_j}{2M} \right) \delta_{c_i,c_j} \tag{2.14}$$

where $C = \{C_1, ..., C_p\}$ is a non overlapping community partition of V and for a node i we define $c_i := k \iff i \in C_k$. $\delta_{x,y}$ is the Kronecker delta function defined such that $\delta_{x,y} = 1 \iff x = y$. For example c_i and c_j are equal if node i and j belong to the same community, in this case δ_{c_i,c_j} in equation 4.37 would be 1. We can understand better equation 4.37 if we consider the contribution of a single edge (i,j) to Q: if $c_i = c_j$, the

term $A_{ij} - \frac{d_i d_j}{2M}$ is positive if link (i, j) exists ($A_{ij} = 1$) and it is negative if the link does not exist ($A_{ij} = 0$). In other words: existing links between pairs of nodes in the same community increase Q, while missing links between nodes in the same C_i decrease Q. The resulting measure assumes values between a minimum of $Q = -0.5$ (partitions are contrary to "natural" partitions in the network) and a maximum of $Q = 1$ (partitions perfectly capture "natural" partitions in the network). For the special case of a fully connected network with N nodes without self-loops, all nodes are in a single community C_1 and the partition quality is 0.

The problem to find the partition $\tilde{C}$ that correspond to the optimal partition quality can be seen as an optimisation problem. We can define the optimal partition quality as:

$$Q_{opt}(G) := \max_{C} Q(G, C) \tag{2.15}$$

While Q is a property of both the network and the partition Q_{opt} is a property of the network only and is called *modularity* of a network. Several different algorithms address the community detection problem (though not necessarily based on the partition quality function introduced above).

Sometimes the modularity is normalised the theoretical maximum of partition quality Q_{max} of network G, which is calculated pretending that all links are between nodes that are in the same community, i.e. $A_{ij} \implies \delta(c_i, c_j) = 1$:

$$Q_{max} = \frac{1}{2M}\Big(2M - \sum_{i,j} \delta(c_i, c_j)\frac{d_i d_j}{2M}\Big) \tag{2.16}$$

This leads to the definition of *community assortativity coefficient* that, given a community partition $\tilde{C}$ with optimal partition quality Q_{opt}, is defined as:

$$\frac{Q_{opt}(G)}{Q_{max}(G, \tilde{C})} \tag{2.17}$$

The community assortativity coefficient captures the nodes' preference to connect to nodes in the same community. It normalises the partition quality to a range between 0 and 1. Also, this quantity is independent of the number of links in the network; this allows us to judge more easily whether a network exhibits community structures or not.

The definitions and the concepts introduced in these sections will be used to model the systems that we are going to study and to analyse the topological properties of the outputs of our model.

2.2 Geometrical frustration

Geometrical frustration is a concept that comes from condensed matter physics, and it plays an important role in the study of spin glasses. It is fundamental to the description of disordered systems where a long-range ordered state is prohibited. It can produce two different types of effects, depending on the system under consideration. Geometrical frustration can lead, in fact, to non-trivial spatial dispositions of the elements in

the system [10] or, when considering the case on a regular crystal lattice, conflicting inter-element forces can lead to complex spin, or pseudo-spin, configurations. In the latter case, constraints imposed by the topology of the inter-element forces can prevent adjacent spins from reaching a configuration that minimises all bonds simultaneously. As a consequence of the frustration, self-organising mechanisms in the interactions of elements can produce local patterns, called motifs, which are in discord with global order [2]. Such dynamics can result in a plenitude of structurally distinct yet nearly degenerate ground states and rough energy landscapes.

The first time the term "frustration" was used in the context of magnetic systems was by J. Vannimenus and G. Toulouse in [22]. Competing interactions characterise this type of systems; in fact, both ferromagnetic and antiferromagnetic couplings, which will be formally defined shortly, can be present between spin pairs or magnetic moments. Although the definition of frustration appeared for the first time in [22], frustrated magnetic systems had been studied even earlier, for example, in 2D Ising systems on a triangular lattice with antiferromagnetic coupling between spins [24].

In order to model magnetic systems and subsequently define a function that takes into account the geometrical frustration in that context, we have first to define the "total energy" of such systems. In statistical physics, this quantity is widely used and called "Hamiltonian"; a discussion about this terminology can be found at the beginning of Chapter 3. It is defined as follows:

$$H = \sum_{G} J_{ij} \; s_i \cdot s_j \tag{2.18}$$

where G is the topology of the system that we are considering, represented by a graph or network, discussed in section 2.1. The quantities J_{ij} represent the so-called "exchange energies" between two elements of the system i and j while s_i and s_j are their spin, or pseudo-spin, i.e. variables that represent the internal state of the elements. It can be shown that this formulation is invariant under a "gauge transformation":

$$\begin{cases} J_{ij} \to \varepsilon_i J_{ij} \varepsilon_j \\ s_i \to \varepsilon_i s_i \\ s_j \to \varepsilon_j s_j \end{cases} \tag{2.19}$$

where ε_i and ε_j are arbitrary signs $+1$ or -1. In fact, under this set of transformations the "total energy" H do not change, nor the other measurable quantities that we will introduce do.

While J_{ij} is usually represented by real scalar values, s_i can have a scalar or vector representation depending on the model considered; that is the reason for the presence of the inner product $s_i \cdot s_j$. For this discussion, we will only consider system elements with an internal degree of freedom represented by a binary scalar value, therefore considering pseudo-spins and not actual quantum spins.

In the light of the principle of minimum energy, we can say that two elements of the system are in a stable configuration when they assume values of the internal state (s_i, s_j) that minimise H in equation 2.18. At this point, the sign of J_{ij} plays a fundamental role. In fact, if $J_{ij} > 0$, the minimum energy configuration of the spins corresponds to the state with parallel spins, with the same signs, and the coupling is

said "ferromagnetic". In reverse, when $J_{ij} < 0$, the stable configuration corresponds to the state with antiparallel spins and opposite signs, and the energy increases.

Given these conditions, we can introduce the mathematical definition of frustration analogous to the Wilson loop in gauge theory, used, for example, in quantum chromodynamics. It is defined on a "plaquette", i.e. an elementary closed path in the considered graph G. The so-called *frustration function* is defined as the product of the couplings signs over the plaquette:

$$\phi_P = \prod_P \text{sign}(J_{ij}) \tag{2.20}$$

The possible outputs of this function are either $+1$ or -1. If $\phi_P = -1$ the plaquette P is said to be geometrically frustrated or "curved", otherwise P is said "flat". It is clear that the necessary and sufficient condition for P to be frustrated is that the number of negative J_{ij} along the plaquette is odd.

In the work presented by Vannimenus and Toulouse in [22] the frustration function is computed considering a simple system formed by Ising spins on a 2D square lattice with nearest-neighbour interactions of random sign but fixed strength: $J_{ij} \in \{+J, -J\}$, with J constant. We can therefore assume J a normalized constant and consider $J_{ij} \in \{+1, -1\}$, then obtaining again equation 2.18 with an overall multiplication for J.

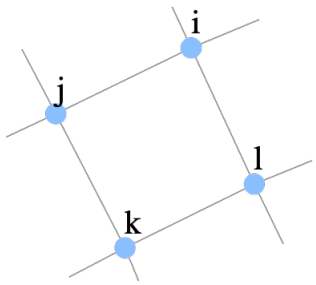


Figure 2.2: Graphic representation of a square plaquette, i.e. the elementary closed path in a 2D square lattice with nearest-neighbour interactions

In this case $\phi_P = J_{ij}J_{jk}J_{kl}J_{li}$ and, in order for the plaquette to be frustrated, the number of antiferromagnetic couplings have to be 1 or 3; it is flat otherwise. It can be simply verified that the quantity ϕ_P is invariant under the gauge transformation 2.19. In their work, Vannimenus and Toulouse studied the thermodynamic effects of the concentration x of randomly distributed antiferromagnetic connections throughout the system. They considered the average concentration of curved plaquettes c_p to be given by:

$$c_p = 4[x^3(1-x) + x(1-x)^3] \tag{2.21}$$

In the case of a 2D Ising system on a triangular lattice, in figure 2.3, we have instead $\phi_P = J_{ij}J_{jk}J_{ki}$, which is again gauge invariant.

There is a clear connection between the frustration of the plaquettes and the configuration of the spins around them: if $\phi_P = +1$ all spins around the flat plaquette can

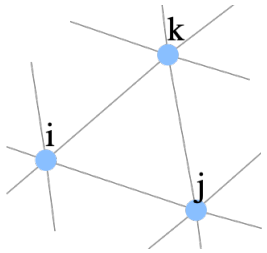


Figure 2.3: Graphic representation of a triangular plaquette, i.e. the elementary closed path in a 2D triangular lattice with nearest-neighbour interactions

be arranged so as to minimise the contribution to H of all the bonds in P; if $\phi_P = -1$ this is not possible and at least one bond is not satisfied in the ground state. As we can see from the example in figure 2.4, the plaquette under consideration has three negative couplings, depicted as red edges. We consider the couplings to be fixed, as we will do throughout the thesis. If we focus on the couple (i, k), thanks to the antiferromagnetic connection, we expect the spins to be antiparallel and we can set $s_i = -1$ and $s_k = +1$. At this point, we can continue the same reasoning with s_j, but we observe the impossibility for the node j to assume a state that is simultaneously opposite to s_i and s_j. With this configuration of the couplings, for every possible value of s_j, H would always be larger than the case of a flat plaquette. That is the underlying meaning of the geometrical frustration on regular lattices.

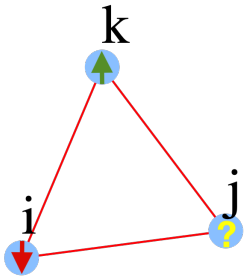


Figure 2.4: Graphic representation of an example of frustrated triangular plaquette. In this case, all the couplings are antiferromagnetic and depicted in red. Similar behaviour can be observed in a plaquette with one negative and two positive connections.

The concepts developed in the field of frustrated solid state systems fit naturally into the study of signed social networks, where links represent positive or negative relations between agents. The connection is supported by some social-psychological considerations that we are going to explore in the following section.

2.3 Structural balance

The structural balance theory is a social-psychological theory that suggests that certain triadic relations are more stable than others, which may induce balancing dynamics on

the system. The theory was first proposed by Fritz Heider in "Attitudes and Cognitive Organization" [11] and then formalised in his work "The Psychology of Interpersonal Relations" [12].

"His studies focus on the so-called P-O-X units of a cognitive configuration, where P represents a person, O another person and X an impersonal object. Every relation among the unit is considered interdependent, and if we consider P to have a relation of affection towards O and O to be responsible for the X, we will therefore observe a tendency for P to like or approve X. Conversely, if X is such that it is normally considered bad, the entire P-O-X unit is placed in an unbalanced state, and pressures will arise to bring it into a more balanced state. These pressures may lead to a change in the relation of affection between P and O, the relation of responsibility between O and X, or the valuation relationship between P and X. From a psychological perspective, Heider introduced two different forms of relations, one for liking or evaluating: $\mathbf{L}$, and one for similarity, possession, causality, proximity, or belonging: $\mathbf{U}$. The negation of such relations are represented respectively by $\sim\mathbf{L}$ and $\sim\mathbf{U}$. Starting from them, in his work "Attitudes and cognitive organisation" [11], Heider provided the definition of "balanced state", stated separately for two and three entities:

> In the case of two entities, a balanced state exists if the relation between them is positive (or negative) in all respects, i.e., in regard to all meanings of **L** and **U**... . In the case of three entities, a balanced state exists if all three relations are positive in all respects, or if two are negative and one positive (9, p. 110).[1]

Multiple balanced situation can be listed as an example:

- P likes something X he made ($P\mathbf{U}X$, $P\mathbf{L}X$)
- P likes what his friend O likes ($P\mathbf{L}O$, $O\mathbf{L}X$, $P\mathbf{L}X$)
- P dislikes what his friend O dislikes ($P\mathbf{L}O$, $O \sim\mathbf{L}X$, $P \sim\mathbf{L}X$)
- P likes what his enemy O dislikes ($P \sim\mathbf{L}O$, $O \sim\mathbf{L}X$, $P\mathbf{L}X$)
- and, for example, P's son likes what P likes ($P\mathbf{U}O$, $P\mathbf{L}X$, $O\mathbf{L}X$)

Heider's primary hypothesis asserts that there is a tendency for cognitive units to achieve a balanced state. Pressures toward balance may produce various effects, as he explained in the same work:

> If no balanced state exists, then forces towards this state will arise. Either the dynamic characters will change, or the unit relations will be changed through action or through cognitive reorganisation. If a change is not possible, the state of imbalance will produce tension (9, pp. 107-109).

[1] F. Heider. Attitudes and cognitive organisation. The Journal of Psychology, 21(1):107–112, 1946 [11]

The theory of structural balance, as formulated by Heider, presents certain ambiguities and has been defined so as to apply to a relatively limited range of situations, one above all, the fact of considering a system of up to three elements. In order to provide a well defined and generalised theory of structural balance, Dorwin Cartwright and Frank Harary, in their work "Structural balance: a generalisation of Heider's theory" [5], identified a list of specific problems of the former formulation, and proposed a suitable solution by using the concepts of linear graphs.

In fact, in their work, they defined a relation (of any kind) between two people to be represented as an unweighted link between two nodes. They extended this concept to an arbitrary number of nodes that form a network. They further represented directed relations through directed graphs and, most importantly, positive and negative relations through signed networks. They also generalised them to signed directed networks in case of signed and directed relations. Lastly, they introduced multi-type networks in order to account for different kinds of relations such as **L** and **U**. Multi-type networks are basically multiple different networks defined on the same set of nodes. They introduced the concept of *path* consisting of a sequence of connected links, and, most importantly, the concept of *cycle* as a closed path of any length (and semi-cycle for the directed case).

Through these definitions, they were able to develop a rigorous generalisation of Heider's concept of balance by considering an "entity", such as P, O or X, as a node of the signed network, a "relation" as a link, and the "sign of a relation" as the sign of the link. Using these definitions, and the concept of cycle, they defined the sign of a cycle as the product of the sign of the links in the cycle. Therefore they defined a cycle as *balanced* if the sign of the cycle is positive, and *unbalanced* if the sign is negative." [2] Note that the formulation for the sign of a cycle is equivalent to the definition 2.20 of the frustration function over a plaquette, and therefore balanced cycles correspond to flat plaquettes and unbalanced cycles to frustrated plaquettes. Thanks to these definitions, several theorems have been presented in the paper [5] and the most relevant one is the following:

> "A signed network is balanced if and only if its nodes can be separated into two mutually exclusive subsets such that each positive link joins two points of the same subset and each negative link joins points from different subsets."

It can be viewed as two factions of friends that are "in war" with each other. Sometimes, in fact, this kind of state will be called a "war state".

When considering a signed network G with multiple cycles, G is defined as *balanced* if all of its cycles are balanced. If there is at least one unbalanced cycle, the network is not balanced, but clearly, there should be a difference between networks with a large number of unbalanced cycles and networks with only a few ones. For this reason, they introduced an important quantity, i.e. the degree of balance $b(G)$ of a signed network G. It is defined as the ratio between the number of balanced cycles and the total number of cycles in G. As it is defined, $b(G)$ can range from 0 to 1, and it is conceptually similar to the quantity c_P introduced by Vannimenus and Toulouse to consider the concentration of frustrated plaquettes.

[2] This part was taken and adapted from the original paper by D. Cartwright and F. Harary, "Structural balance: a generalisation of Heider's theory", 1956 [5]

In the definition of $b(G)$, all the cycles of every length are counted, while it is reasonable to assume that the balance of large cycles does not influence the stability of a social system. For this reason, it is lastly introduced the definition of *N-balanced* signed network if all its cycles of length not exceeding N are balanced. Clearly, the lowest N such that an *N-balanced* signed network is well defined is 3, as cycles of length 2 do not exist in undirected networks. 3-cycles actually correspond to the structure that most influences the stability of the system. Let us consider the three nodes as people having some relation with each other. Four kinds of configuration are possible, depending on the number of negative edges, and they are shown in figure 2.5.



(a) Balanced (b) Unbalanced (c) Balanced (d) Unbalanced

Figure 2.5: All the possible sign configurations of a triangular relation.

In 2.5a the configuration is balanced as all the people involved are friends, while in 2.5b it represents an uncomfortable situation as n_0 and n_2, which are friends, have an opposite "opinion" on n_1, that could create some tensions. State 2.5c is again considered balanced as it represents the situation where "the enemy of my enemy is my friend", that can be also seen as n_0 and n_2, which are friends, have the same (negative) "opinion" on n_1. The forth and last situation in figure 2.5c is considered unbalanced since happen that n_0 and n_2, which are enemies, have the same (negative) "opinion" on n_1 and also this situation can be perceived as uncomfortable.

Starting from the formalisation provided by the work of Cartwright and Harary, in the following chapter, we will try to study this kind of social signed networks from a statistical physics perspective, also considering additional contributions due to other socio-psychological effects, and then study their consequences.

Chapter 3

Global Hamiltonian

The following Chapter is devoted to the representation and characterisation of social systems and the pattern of relations within them. The analysis is carried on by representing the system as a signed network and defining a "cost function" based on it.

In section 3.1 we will define the systems that we are going to study as well as the cost function to represent them. We will also explain the role of such function and why we can call it *Hamiltonian*. In the following section 3.2 we will present Hamiltonian formulations that have been previously introduced in order to consider structural balance contributions and some significant results that have been deduced starting from them. This section will be, therefore, essentially based on the existing literature. In section 3.3 we will describe our contribution to the topic by introducing a novel Hamiltonian. It is formulated in order to consider structural balance contributions as well as other social effects. Each term of this formulation will be explained, and the analytical expression of the energy levels will be calculated when available. We will see that this is not always possible for the structural balance term, for this reason, in section 3.4 three different numerical processes will be presented to study it, while in section 3.5 the analytical results for some of the energy levels of the same term will be presented. In section 3.6 we will characterise and study the configurations corresponding to local minima of the energy landscape, and in the last section, the contributions of all the three terms will be presented together.

Therefore, all the work in this Chapter, apart from section 3.2 and where explicitly declared, is our original contribution.

3.1 Definition of the system and the Hamiltonian function

In the following, we will model a social system as a set of N agents connected to each other by friendship or enmity relationships. We will consider every agent to have a relationship with every other agent, apart from himself. This particular way to represent a social system can naturally be formalised by a network $G = (V, E)$ where V is the set of the N agents, represented as nodes, and $E \subseteq V \times V$ is the set of all the relationships between agents, represented as edges between nodes. Since we consider the case where every agent has a relationship with every other agent but himself, the network generated is a signed undirected fully connected network without self-loops.

The nature of the relationship between two agents i and j is characterised by a scalar binary value $A_{ij} \in \{-1, +1\}$, where -1 represent the relationship of enmity while $+1$ represent friendship. We will additionally consider every agent to have a binary scalar internal variable $s_i \in \{-1, +1\}$. This variable is meant to be constant in time, and it is introduced to represent some characteristic of individuals in the social system, such as belonging to a certain social class, age class, city, to a certain social context or having some personal characteristics.

Although the topology of the network is fixed and fully connected, different systems can differ from each other because of the different dispositions of positive and negative links in the network. One of these configurations can be seen in the following figure 3.1. It is an example of the type of network that we will study in this section.

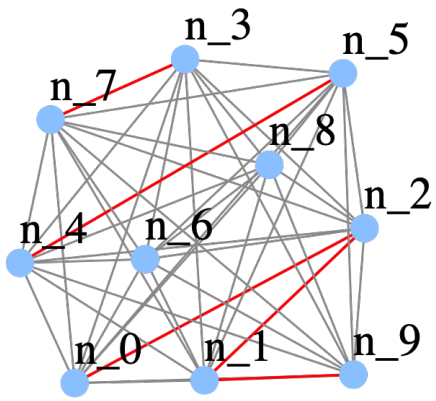


Figure 3.1: Representation of a fully connected signed network of 10 nodes and 5 negative edges, without self-loops. The negative edges are depicted in red while the positive ones in grey.

Structural balance theory suggests that some configurations of relations are more probable to appear in real social networks than others. In order to classify and quantify the differences between configurations of relations, it is convenient to introduce a utility function that measures the "cost" of a given configuration that quantify the degree of cognitive dissonance of the social system. This formalism can be seen in analogy with some models in statistical physics, such as the Ising model, and therefore identify this utility function as the energy corresponding to the configuration of the system or, alternatively, as its Hamiltonian function H.

Continuing the analogy with the Ising model, we can say that, at the basic level, both models are not defined over a "phase space", but rather over a "configuration space", that is, the space of all spin configurations for the Ising, and the space of all relation configurations for the social system model. There is no explicit time, and there are no canonically conjugate variables. In the following, we will refer to this quantity as the "energy" or the "Hamiltonian" of the system, always keeping in mind the true meaning of this definition. It is clear that configurations that are more probable to exist correspond to lower energies, while more unlikely configurations have a higher energy.

In the following section, we will present Hamiltonian formulations that have been previously introduced in order to consider structural balance contributions and some important results that have been deduced starting from them.

3.2 Previous Hamiltonian formulations

Starting from the seminal work of Heider [12] and the following formalisation and generalisation introduced by Cartwright and Harary [5], many studies have been conducted in order to consider the structural balance theory in quantitative studies of social systems. Some of them have introduced a Hamiltonian formulation, or equivalently an energy function, H, that takes into account the structural balance to measure the cost of the configuration of the system under consideration.

Among the literature read during the preparation of the thesis, we have selected two papers that make use of two different versions of such a Hamiltonian and produced interesting results that have to be taken under consideration for the subsequent studies. The first paper is "Energy Landscape of Social Balance" from Seth A. Marvel and collaborators [15], where they introduced an energy function U that basically counts how many balanced triangular relations are there more than unbalanced ones, in a fully connected network with positive and negative signs on its edges:

$$U = -\frac{1}{\binom{n}{3}} \sum_{i,j,k} s_{ij} s_{jk} s_{ik} \tag{3.1}$$

where n is the number of nodes in the network and s_{ij} is the sign of the edge between nodes i and j. We can notice the presence of the normalisation respect all the possible triangles in the network, which, in the case of a fully connected network without self-loops, is equal to the binomial coefficient $\binom{n}{3}$. The quantity U measures, therefore, the fraction of the difference between the number of balanced and unbalanced triangles in a system and its value can range from -1 (minimum energy configuration) to +1 (maximum energy configuration).

"The configuration in which all node pairs are friends has the lowest possible energy $U = -1$. Hence, no additional structure is necessary to define the global minima; they are just the sign configurations for which $U = -1$. Cartwright and Harary [5] identified all such ground states, finding that they consist of two warring factions: internally friendly cliques with only antagonistic edges between them. (The all-friends configuration represents the extreme in which one clique is the empty set.)."

The aim of this paper was to study the so-called "jammed states", local minima in the energy landscape, with particular attention to higher energy local minima.

"To define the concept of a local minimum, however, we need to specify what it means for two states to be adjacent. The most natural choice is to define two sign configurations to be adjacent if each can be reached from the other by a single sign flip. Then a jammed state, as defined by Antal et al., is a sign configuration for which all adjacent sign configurations have higher energy [1]. Here, however, we will slightly vary their terminology by calling this a "strict jammed state", reserving the term "jammed state" for the weaker concept of a sign configuration with no adjacent sign configurations of lower energy."[1]

The first result shown in the paper is the absence of jammed states for $U > 0$, i.e. when the number of unbalanced triangles is greater than the number of balanced ones.

[1]These parts were taken and adapted from the original paper: Seth A. Marvel, Steven H. Strogatz, and Jon M. Kleinberg. Energy landscape of social balance. Phys. Rev. Lett., 103:198701, Nov 2009. [15]

They also provided the description of a family of arbitrarily large jammed states with $U = 0$. They can be derived from the undirected Paley graphs and, using the theory of two-graphs, an infinite families of strict jammed states that approach $U = 0$ from below as n grows large. Such constructions can be carried out using bilinear forms modulo 2 [19], and projective planes in finite vector spaces [20]. They found these families to be the graphs that can be decomposed in k internally balanced cliques.

The second interesting paper is "Computing global structural balance in large-scale signed social networks" from G. Facchetti and collaborators [9]. The quantity that they introduced to measure the global balance of the system is:

$$h(\mathbf{s}) = \sum_{(i,j)} (1 - J_{ij} s_i s_j)/2 \tag{3.2}$$

where $J_{ij} \in \{+1, -1\}$ is the relation between agent i and agent j, while in this case $s_i \in \{+1, -1\}$ is a binary variable assigned to every agent i. The function $h(\mathbf{s})$ is built in such a way that if the system is completely balanced $h(\mathbf{s}) = 0$, that is the reason why they included the constant $+1$ in the formula. It is possible to observe a similarity with the formulation of the Ising model. The main difference, apart from the different topology of the system, is the possibility for J_{ij} to assume values different from 0 and 1, in particular, it can be equal to -1. For this reason, the model presents more similarities with the models introduced for the study of spin-glass systems, where $J_{ij} = +1$ correspond to ferromagnetic bonds and $J_{ij} = -1$ to anti-ferromagnetic ones, however actual quantum spin-glass models obviously present differences regarding the nature of the variable s_i. In the case of the structural balance, in fact, s_i is represented as a binary scalar value, while in spin-glass models is often represented as Pauli matrices of the quantum particles that form the system.

We can notice that, unlike the Marvel energy function, in this case, the triangular relation considered for the structural balance is formed by the relationship between two agents and their opinion s_i on a certain question and not between three different agents. Another difference is the presence of the opinions s_i themselves. As we will see, we will introduce a similar but different term to consider the homophily effect in the population.

Even though the structure is not the same, the paper provides and proves a connection between their formulation and the formulation without opinions (considering every agent having the same opinion, e.g. $s_i = 1 \ \forall i \in V$). Thanks to some interesting original methods, they verified that real social networks, with many individuals, actually present a high degree of global social balance. This result allows us to conduct and continue our studies on structural balance in social systems based on robust experimental findings rather than just a theoretical hypothesis.

3.3 Hamiltonian of the model

For our studies, we have introduced a novel Hamiltonian to include social effects other than the structural balance. We, therefore, define the following Hamiltonian:

$$H = -\frac{\alpha}{2}\sum_{i,j} A_{ij} - \frac{\beta}{6}\sum_{i,j,k} A_{ij}A_{jk}A_{ki} - \frac{\gamma}{2}\sum_{i,j} s_i|A_{ij}|s_j \tag{3.3}$$

It is possible to recognise similarities between some terms of this Hamiltonian and the previous formulations presented. The second term is, in fact, similar to the energy function presented by Marvel and collaborators, but the normalisation, and the third term is similar to the quantity presented by Facchetti and collaborators, but as we will see, it plays a different role in this case.

In the following paragraphs, we will explain in detail each of the terms, respectively: the first term, the third term and the second term, since the second term, representing the structural balance, is the one that deserves more in-depth studies.

3.3.1 Positivity term

The first term of the Hamiltonian will be called in the following "positivity term" H_p since it refers to the preference for the agents in a social system of having positive relationships. The term is computed by summing all the entries of the adjacency matrix of the signed network. Since we always consider symmetric relationships, we divide by 2 to not sum the same link twice. The parameter α measures the importance of the social system under consideration of having positive relations.

From now on, we will try to express the relevant quantities as a function of the number of negative edges; as we will see, this will simplify some calculations. We can define the number of negative edges M_- as:

$$M_- := \frac{1}{2}\sum_{i,j} \delta_{A_{ij},-1} \tag{3.4}$$

Where the $\delta_{A_{ij},-1}$ function is the Kronecker delta defined as:

$$\delta_{ij} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases} \tag{3.5}$$

Or alternatively:

$$M_- := \frac{1}{2}\sum_{i,j}(1 - A_{ij})/2 \tag{3.6}$$

And therefore:

$$M_+ := M - M_- = \binom{N}{2} - M_- \tag{3.7}$$

where M is the number of total edges, that for our case of a fully connected network without self-loops is equal to the binomial coefficient $\binom{N}{2}$, with N being the number of agents in the system.

By summing all the elements in the adjacency matrix, we obtain the difference between the number of positive and negative links in the network multiplied by two because we count every link twice in the adjacency matrix.
As correspondence with the "gauge transformation" shown in the case of the geometrical frustration in the equations 2.19, if we apply the transformation where the value of every edge is reversed, $A_{ij} \to -A_{ij}\ \forall (i,j) \in E$ we expect this time the parity of H_p to be odd:

$$H_p(-\mathbf{A}) = -\frac{\alpha}{2}\sum_{i,j} -A_{ij} = -H_p(\mathbf{A}) \tag{3.8}$$

Where $\mathbf{A} := \{A_{ij} :\ i,j \in V\}$. By defining $\Delta M := M_+ - M_-$ we can therefore express:

$$\sum_{i,j} A_{ij} = 2\Delta M \implies H_p = -\frac{\alpha}{2} 2\Delta M = -\alpha \Delta M \tag{3.9}$$

As we can see the energy is a linear function of ΔM and the symmetry of the function is therefore confirmed, i.e. $H_p(\Delta M) = -H_p(-\Delta M)$. The term can be rewritten as a function of the number of negative links as follows:

$$\begin{aligned} H_p &= -\alpha(M_+ - M_- + M_- - M_-) = -\alpha(M - 2M_-) = \\ &= -\alpha\left[\binom{N}{2} - 2M_-\right] = -\alpha\left[\frac{N(N-1)}{2} - 2M_-\right] \end{aligned} \tag{3.10}$$

Since M_- can range from 0 to M, the value of energy due to the positivity term H_p can range from $-\alpha M$ to $+\alpha M$, that in the case of the following Figure 3.2 with $N = 10$ and $\alpha = 1$ corresponds to -45 to 45. In general the maximum and minimum energy of the system scale with the size of the system N as $M = \binom{N}{2} \sim N^2$. We expect the energy minimum to correspond to the state where all the signs of the network links are $+1$, i.e. every agent shares a friendship relationship with every other agent in the system. This particular state will be called "paradise state", corresponding to $M_- = 0$. Due to odd symmetry, we also expect the maximum to correspond to $M_- = M$.

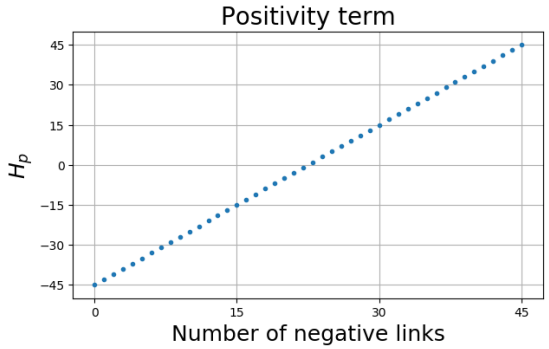


Figure 3.2: H_p for a system of $N = 10$ agents and $\alpha = 1$ as a function of the number of negative links

It is important to underline the fact that the value of H_p is uniquely determined by the number of negative links M_- and not by their position on the network. In this case, the topology of the signed network does not play any role, unlike the structural balance term, but likewise the next term that we are going to present.

3.3.2 Homophily term

The third term H_h will be called "homophily term" since it accounts for agents' preference in the social system to interact with other agents with similar attributes. We suppose every agent to have a binary scalar internal variable $s_i \in \{-1, +1\}$. This variable is meant to be constant in time, and it is introduced to represent some characteristic of individuals in the social system, such as belonging to a certain social class, age class, city, to a certain social context or having some personal characteristics. The term H_h is built in order to count how many pairs sharing the same value of attribute are there more than pairs with different attributes, regardless of their relationship. Therefore, we expect systems where homophily is considered important to have smaller values of H_h. In our case, where we consider fully connected networks without self-loops, the adjacency matrix element A_{ij} is just put in order to avoid counting self-terms $s_i s_i$ which otherwise would contribute to the energy value. By inserting the absolute value of the adjacency matrix, we consider only the existence of a relationship between agent i and agent j, not its nature. In this way, the homophily term is also ready to study even non fully connected networks. These considerations allow us to express H_h as a function of the difference ΔN_s between the number of positive and negative attributes of the agents in the system:

$$\begin{aligned} H_h = -\frac{\gamma}{2}\sum_{i,j} s_i |A_{ij}| s_j = -\frac{\gamma}{2}\Big[\sum_{i,j} s_i s_j - \sum_i s_i s_i\Big] = \\ = -\frac{\gamma}{2}\Big[\sum_i s_i \sum_j s_j - N\Big] = -\frac{\gamma}{2}\Big[\Delta N_s^2 - N\Big] \end{aligned} \tag{3.11}$$

The way we decided to build the homophily term clearly suggests two properties: 1) since the contributions depend only on the relative values of the attribute and not on the actual values, H_h should be symmetric for the transformation $s_i \to -s_i \; \forall i$, and an even function of ΔN_s such that $H_h(\Delta N_s) = H_h(-\Delta N_s)$, as the parabolic equation above confirm, 2) the configurations that correspond to an energy minimum are the states where all the agents share the same attribute value, regardless of its actual value. In those cases, the homophily is maximal and

$$\Delta N_s = \pm N \implies \min(H_h) = -\frac{\gamma}{2}\Big[N^2 - N\Big] = -\gamma\binom{N}{2} = -\gamma M \tag{3.12}$$

The opposite situation, i.e. when half of the agents have the opposite value of the other half of the agents, leads to the largest possible energy as ΔN_s is minimal. Here we can have two different situations depending on the number N of agents in the system:

$$\begin{cases} N \text{ even} \implies \min(\Delta N_s) = 0 \implies \max(H_h) = \gamma\frac{N}{2} \\ N \text{ odd} \implies \min(\Delta N_s) = \pm 1 \implies \max(H_h) = \gamma\frac{N-1}{2} \end{cases} \tag{3.13}$$

In order to be coherent with the previous section, where we studied H_p as a function of the number of negative links, we can rewrite H_h as a function of negative attributes and find:

$$\begin{aligned} H_h &= -\frac{\gamma}{2}\Big[\Delta N_s^2 - N\Big] = -\frac{\gamma}{2}\Big[(N_+ - N_- + N_- - N_-)^2 - N\Big] = \\ &= -\frac{\gamma}{2}\Big[(N - 2N_-)^2 - N\Big] = -\frac{\gamma}{2}\Big[N^2 - 4NN_- + 4N_-^2 - N\Big] \end{aligned} \tag{3.14}$$

As expected, we can see from the figure below the presence of two global minima of H_h corresponding to:

$$\begin{cases} \Delta N_s = N \implies N_- = 0 \\ \Delta N_s = -N \implies N_- = N \end{cases} \tag{3.15}$$

We can also find one or two maxima depending on the number of nodes at $\Delta N_s = 0 \implies N_- = N/2$ if N is even and $\Delta N_s = \pm 1 \implies N- = (N \pm 1)/2$ if N is odd:

$$\begin{cases} \Delta N_s = 0 \implies N_- = N/2 \quad \text{if } N \text{ is even} \\ \Delta N_s = \pm 1 \implies N- = (N \pm 1)/2 \quad \text{if } N \text{ is odd} \end{cases} \tag{3.16}$$

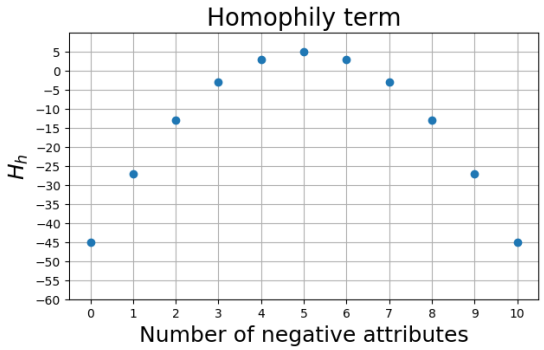


Figure 3.3: H_h for a system of $N = 10$ (even) agents and $\gamma = 1$ as a function of the number of negative attributes. The maximum energy level is unique.

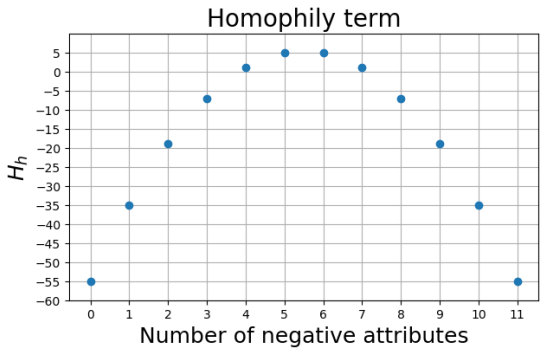


Figure 3.4: H_h for a system of $N = 11$ (odd) agents and $\gamma = 1$ as a function of the number of negative attributes. Two different states correspond to the maximum energy level.

3.3.3 Structural balance term

The second term H_{SB} accounts for the structural balancedness of the system. The form of the term is similar to the energy function U introduced by Marvel and collaborators in [15]. The term is built to count the actual difference of the number of balanced and unbalanced triangular relationships present in the system. The numerical coefficient $\frac{1}{6}$ is added in order to avoid counting the same triangle multiple times: 2 times because of the undirected network and 3 times because the same triangle will be counted starting from all the three vertices that form it. The parameter β measures the importance for the members of the social system under consideration to belong to balanced triangular relationships. We can say that for a social system where the structural balance is important, we expect smaller values of H_{SB}. The main difference between the term presented in [15] and this term is that in this case it is not normalised with the total number of possible triangles in the network $\binom{N}{3}$. Therefore, as we will see, the maximum and the minimum energy provided by H_{SB} scale with N as the number of possible triangles $\binom{N}{3} \sim N^3$.

We can use the properties of the adjacency matrix in order to provide a more compact form for the structural balance term that will also be very convenient in the implementation, as it will be much faster to compute. In fact, we can write:

$$\begin{aligned} H_{SB} &= -\frac{\beta}{6}\sum_{i,j,k} A_{ij}A_{jk}A_{ki} = -\frac{\beta}{6}\sum_{i,j} A_{ij}\sum_{k} A_{jk}A_{ki} = \\ &= -\frac{\beta}{6}\sum_{i,j} A_{ij}(A^2)_{ji} = -\frac{\beta}{6}\sum_{i}(A^3)_{ii} = -\frac{\beta}{6}Tr(A^3) \end{aligned} \tag{3.17}$$

Applying the same transformation as before, where the value of every edge is reversed, i.e. $A_{ij} \to -A_{ij}\ \forall (i,j) \in E$, every balanced triangle become unbalanced and the opposite, implying an odd parity $H_{SB}(-\mathbf{A}) = -H_{SB}(\mathbf{A})$, as confirmed by the direct computation:

$$H_{SB}(-\mathbf{A}) = -\frac{\beta}{6}\sum_{i,j,k}(-A_{ij})(-A_{jk})(-A_{ki}) = +\frac{\beta}{6}\sum_{i,j,k} A_{ij}A_{jk}A_{ki} = -H_{SB}(\mathbf{A}) \tag{3.18}$$

As previously anticipated, in this case, the topology of the signed network is crucial, as the same number of negative edges can form either balanced or unbalanced triangles. For example, we can have situations where all the negative links are included only in balanced triangles so that the whole system is overall balanced, but the number of negative links is different from zero. Alternatively, we can have a different situation where the same number of negative links forms some unbalanced triangles, leading H_{SB} to assume higher values. The possible energy outcomes of this term are thus more difficult to study as the energy is not uniquely determined by the number of negative links and also because of the large number of the possible configurations that the system can assume. For these reasons, in the following section, three different approaches will be introduced in order to study the possible energy levels of the structural balance term.

3.4 Computational analysis of H_{SB}

Since we cannot simply study the energy of a signed network with a specific number of negative edges, we should explore all the possible configurations of the signed network. As we said, every link (i, j) can assume two possible values $A_{ij} \in \{+1, -1\}$. For a fully connected undirected network of N nodes without self-loops, the number of possible configurations is, therefore, $2^M = 2^{\binom{N}{2}} = 2^{\frac{N(N-1)}{2}}$ which quickly diverge as we increase the size N of the system. For a quite small network of $N = 10$ agents, the number of possible configurations is already $2^{45} \simeq 3.5 \cdot 10^{13}$. The number of possible configurations for a system of N agents, M edges and M_- negative edges is computed by counting how many different combinations we can obtain by choosing M_- edges from the total M without replacement and without considering the order. The number of configurations is therefore given by the binomial coefficient $\binom{M}{M_-}$ and $\sum_{M_-=0}^{M} \binom{M}{M_-} = 2^M$.

Since these numbers can become quickly unmanageable, a stochastic approach will be used instead of computing the energy for all of them. In this way, only a certain number of configurations will be explored, with the hope to capture the essential features. As we will see, this will not always be the case because some of the configurations are very unlikely to happen at random, but we will try to address them differently.

The stochastic study of the possible outcomes will be carried out in several ways: we will first present the distributions of energies for random signed networks with a fixed number of negative edges, then we will introduce a random process to explore the possible energy landscapes as a function of the number of negative edges, and lastly, we will show a particular realisation of this process in order to study the lowest energy levels and particular configurations that are very unlikely to happen at random, but that will give us important insights on those states.

3.4.1 Energy distribution

As said above, the number of different configurations for a system of N agents, M edges and M_- negative edges is given by $\binom{M}{M_-}$, but the number of different energy outputs coming from these possible configurations are way less. For a network of 10 agents and 3 negative edges, the number of possible configurations is $\binom{M}{M_-} = \binom{\binom{10}{2}}{3} = 14190$ but the number of topologically different states is 5, represented in the following figures:

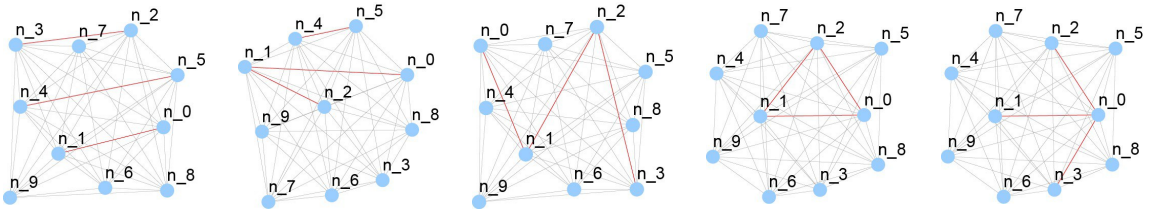


Figure 3.5: 5 topologically different configurations of 3 negative edges (red) in a fully connected network of 10 nodes

By exploring all the possible configurations and computing the corresponding energy H_{SB}, it is possible to build a histogram of the energy levels and the corresponding numbers of occurrences. As we can see from the figure 3.6a below in this case, in a network of $N = 10$, $M_- = 3$, where we have 14190 different configurations, of

which only 5 topologically different, the corresponding energies are only 4; in fact, the configurations 2 and 4 lead to the same energy.

Due to the large number of possible configurations, even for small systems, instead of computing the value of H_{SB} for each possible state, we decided to address this problem with a stochastic approach. The exploration of the configurations is carried out through a random signed network generator,

Similarly to the Erdős–Rényi random graph models[2], which is formulated in two different versions $G(N, p)$ and $G(N, M)$, also the random signed network generator can be formulated considering the probability p_- of an edge to be negative or, alternatively, the total number of negative edges M_-, in a system with a fixed number of N nodes. In both cases, the parameters define a macrostate, the ensemble of networks with the specified characteristics mentioned above. In the first case we can define RSNG(N, p_-) and in the second RSNG(N, M_-). Let us describe both the generative models in the following algorithms:

Algorithm 1: Random Signed Network Generator 1: RSNG(N, p_-)

```
- Define the size of the system N and the probability of negative edges p_-
- Define an empty network of N nodes
for i in set of nodes do
    for j in set of nodes greater than i do
        - Generate a random number from a uniform distribution r ∈ [0, 1]
        if r < p_- then
            - Add an edge between nodes i and j with weight −1
        else
            - Add an edge between nodes i and j with weight +1
        end
    end
end
```

In order to compute the probability $P(G)$ of obtaining a particular microstate from the generative model just presented, we can assume the maximum entropy principle, i.e., in the absence of any further information, the probability distribution that maximises entropy (and thus minimises our information on the system) is our best guess. In our particular case, this corresponds to the assumption that all possible realisations are equally likely. From a sociological perspective, this assumption is obviously not true; in fact, we know from the studies of Facchetti and collaborators [9] that real-world social systems tend to maximise the amount of structural balance, leading some configurations to happen with a higher probability than others. Therefore if we consider a network with a fixed number of negative edges, the probability $P(G)$ of the configuration of links that provide a smaller energy is larger than the one that provides a higher energy.

[2]The $G(N, M)$ model generates a random network of N nodes and exactly M unweighted and undirected edges. The edges are displaced randomly among the set $V \times V$ of possible edges. The probability of obtaining a specific realisation of the model is constant among all the realisations and equal to $P(G) = 1/Z(N, M)$ where $Z(N, M) = \binom{\binom{N}{2}}{M}$ for the case undirected without self-loops.

The $G(N, p)$ model relaxes the constraint of having exactly M links and creates an edge between every pair of nodes with a fixed probability $p \in [0, 1]$. The expected number of links is $\langle M \rangle = p \cdot \binom{N}{2}$. The probability of a given network with M_G links is $P(G) = p^{M_G} \cdot (1-p)^{\binom{N}{2} - M_G}$.

Suppose we consider the same problem from a stochastic point of view instead. In that case, the two configurations have the same probability as we do not have any reason to prefer one configuration over the other. We can start from this perspective in order to have a "null model" and have an idea of what to expect at random. The number of negative edges is not fixed, but its expectation $\langle M_- \rangle = p_- \cdot M = p_- \cdot \binom{N}{2}$. The probability of the network $P(G)$ depends, in this case, on the number of negative edges actually created. We can see it as a repeated M_- times a success of a trial that succeeds with probability p_- and all other trials ($\binom{N}{2} - M_-$) are not successful (which happens with probability $(1 - p_-)$). The probability of obtaining *at random* a certain signed network coming from the RSNG(N, p_-) is, therefore:

$$P(G) = p_-^{M_-} \cdot (1 - p_-)^{\binom{N}{2} - M_-} \tag{3.19}$$

And the probabilities of all microstates that have the same number of negative links are the same (i.e. the model does not differentiate between microstates with the same number of negative links).

This is different from the second model that we have introduced, the RSNG(N, M_-), where, similarly to the $G(n, m)$ model, the number of negative links is fixed and, thanks to the principle of maximum entropy, the probability of obtaining *at random* a certain signed network is uniform among all possible outputs:

$$P(G) = \frac{1}{Z(N, M_-)} = \frac{1}{\binom{\binom{N}{2}}{2}} \tag{3.20}$$

where $Z(N, M_-)$ is the size of the configuration space $\Omega(N, M_-)$ and it is called Z as an analogy with the partition function in statistical mechanics ("Zustandssumme" means "summing over all possible states" in German). The algorithm that generates such signed networks is the following:

Algorithm 2: Random Signed Network Generator 2: RSNG(N, M_-)

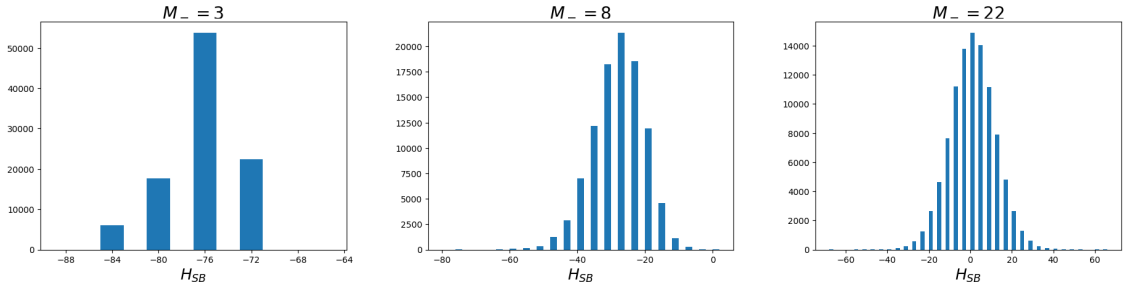
- Define the size of the system N and the total number of negative edges M_-
- Define an empty network of N nodes
- Compute the number of edges $M = \binom{N}{2}$
- Define the vector for the weights of the edges, of length M. A vector with M_- entries equal to -1 displaced randomly and the remaining ones equal to $+1$

for *i in set of nodes* **do**
  for *j in set of nodes greater than i* **do**
    - Add an edge between nodes i and j with a weight equals to the corresponding entry of the vector of weights
  end
end

Both algorithms can be simply modified in order to produce directed signed networks, even with self-loops. Since we will only consider undirected signed networks without self-loops in this thesis, we decided to omit their algorithmic representations.

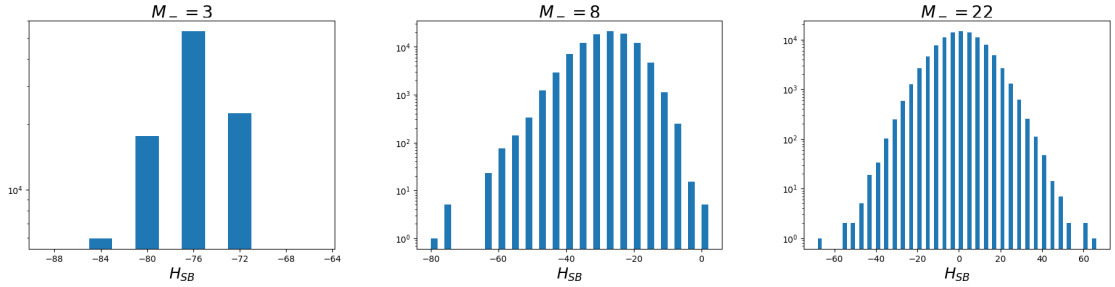
Additionally, since we consider attributes defined on the nodes in our model, it is possible to implement a method to add those values to the nodes in a specified manner or in a random way.

For a given macrostate, a certain number of realisations N_r of networks is generated, and H_{SB} is computed for each of them. The results of these calculations can be viewed by means of histograms showing the occurrences of a given value of the energy H_{SB}. From the histograms, we can therefore read the possible energy values for a fixed number of negative edges and the frequencies expected *at random*. Of course, it is possible that some of the energy levels are not explored due to the enormous number of possibilities, but nevertheless, interesting characteristics can be deduced from them.



(a) $M_- = 3$ (b) $M_- = 8$ (c) $M_- = 22$

Figure 3.6: Energy distribution for a system with $N = 10$ and $N_r = 10^5$



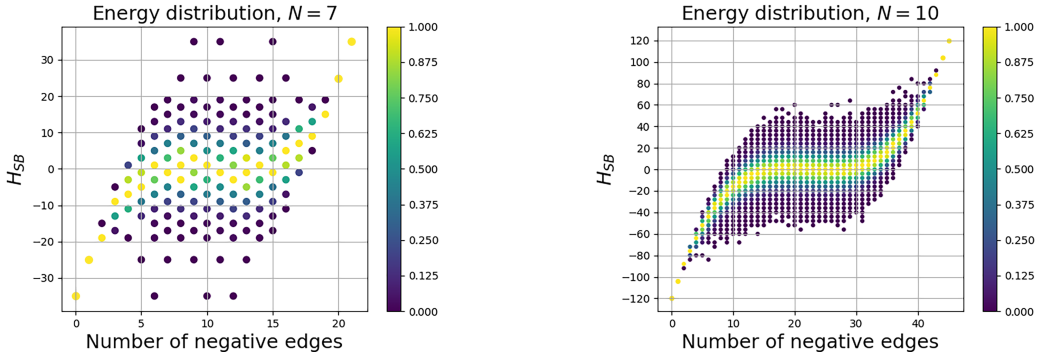
(a) $M_- = 3$ (b) $M_- = 8$ (c) $M_- = 22$

Figure 3.7: Energy distribution for a system with $N = 10$ and $N_r = 10^4$ in a logarithmic scale

It is possible to notice the asymmetrical shape of the distributions for a relatively small number of negative links, figures 3.6b and 3.6c. In fact, for larger values of M_-, as in figure 3.6c, we can observe a much more symmetrical shape, very similar to a normal distribution, also confirmed by the shape of the same distribution but on a logarithmic scale in figure 3.7c.

Apart from the few lowest energy levels, the higher energy levels of a network with the same number of nodes and negative edges always differ by a value of 4β between each other. This is due intuitively to the fact that two adjacent energy levels, corresponding to a network with the same number of negative edges, can be obtained by "moving" a negative edge from a triangular relation to another triangular relation. Let us consider the first triangular relation to be unbalanced, containing an odd number of negative edges and flip one negative link to positive. For sure, the number of negative links in the triangle becomes even, and the triangular relation becomes balanced. The contribution to the energy given by the triangle under consideration goes from -1 to $+1$, producing

a difference in the total energy of $\Delta E = +2\beta$. In order to keep the number of negative edges constant, we have to introduce a negative link to another triangular relation that we can assume to be unbalanced, i.e. with an odd number of negative edges. With the same reasoning as before, we can say that the difference in the total energy is again $\Delta E = +2\beta$ and therefore, the total energy shift is $\Delta E_{tot} = +4\beta$



(a) $N = 7$ (b) $N = 10$

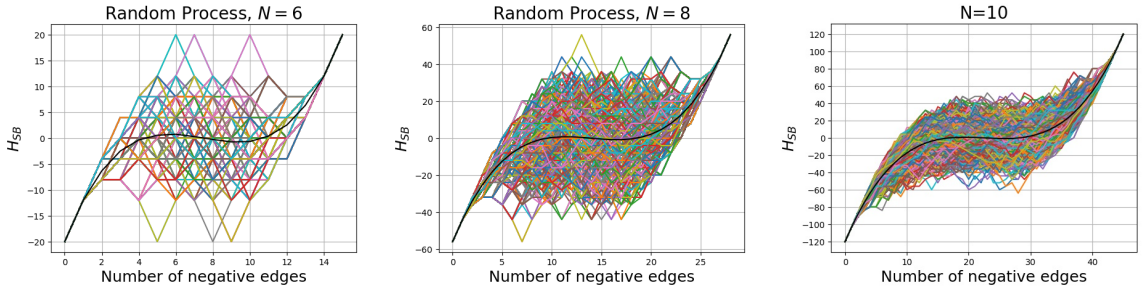
Figure 3.8: Energy distributions as a function of the number of negative edges for a system with $N = 7$ and $N = 10$. The number of realisations performed for every number of negative edges is $N_r = 2 \cdot 10^4$

In the figures 3.8 above, the energy distributions are represented as a function of the number of negative edges for two different sizes of the system. The colours of the dots represent the occurrences of that specific value of energy, normalised with the number of realisations created for that specific value of M_-. As we can see, the majority of the realisations assume a value of H_{SB} that is pretty narrow throughout all the graph, while the actual distribution is very broad. As we can see from figure 3.8a, sometimes it can happen that some configurations have the same energy of the initial or final state, meaning that the total distribution basically covers the whole interval between $\min(H_{SB})$ and $\max(H_{SB})$. We will study these configurations later in this chapter.

3.4.2 Random process

In order to study the energy landscape of the configurations as a function of the number of negative links in a way that highlights the adjacent configurations, and therefore some local minima, we introduced a random process to explore different configurations in a sequential manner. The process always starts with a fully connected network with all the values of edges $A_{ij} = +1$, the so-called "paradise state", then one edge is chosen randomly, and its value is set to -1. In the next step another edge, different from the previous one, is chosen and set to -1. The process is then carried out in this way until every edge is set to -1. At each step, the energy H_{SB} is computed and saved in a vector of length $M + 1$ that refers to that specific realisation of the process. The process always ends with an all-enmity network, and the results of a large number of these realisations can be seen in the following figures:

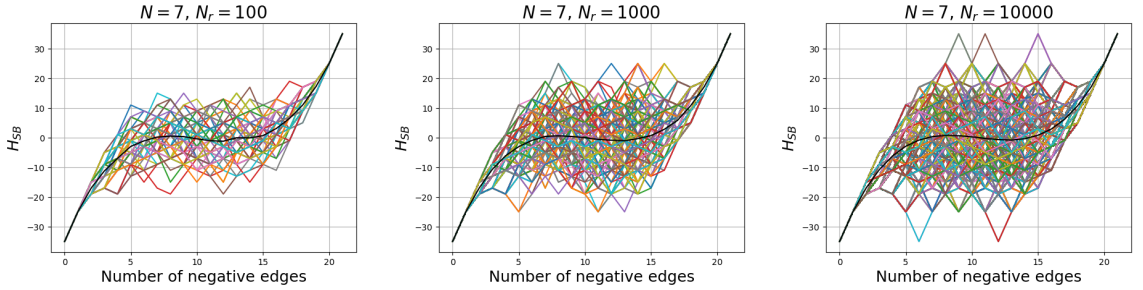
In the figures above, we can notice that the different realisations of the process with the same N can assume a wide range of values, compatible with the findings of the previous section. We can see that some of the realisations reach the same energy value



(a) $N_r = 10^3$ (b) $N_r = 10^5$ (c) $N_r = 10^4$

Figure 3.9: Representation of the process explain above for the computation of H_{SB}, considering networks with an increasing number of nodes and a value of $\beta = 1$. The black line represent the mean of H_{SB} over all the realisations. The different numbers of realisations come from a compromise between the maximal exploration of the configuration space and the computational time.

as the initial paradise state, hence they are global minima themselves. These states have a very low degeneracy and are very unlikely to appear at random (but they are also very common in real social networks, as discussed before). The probability of obtaining such realisations is inversely proportional to the number of possible configurations. Therefore, it becomes almost negligible for large networks, in our case for a size $N > 8$. That is the reason why in order to obtain a second global minimum in figure 3.9b we had to perform $N_r = 10^5$ realisations of the random process. For higher N, it is ineffective to increase N_r as the number of realisations needed to obtain global minima configuration other than the paradise state would require a computational time that is too long for our possibilities. By increasing N_r in figure 3.10 we can see that the energy distribution become larger and larger, as it explores zones of the configuration space which are much more unlikely to obtain at random.



(a) $N_r = 10^2$ (b) $N_r = 10^3$ (c) $N_r = 10^4$

Figure 3.10: Representation of the random process for the computation of H_{SB}, considering networks with the same number of 7 nodes and a value of $\beta = 1$. The black line represent the mean of H_{SB} over all the realisations. The different numbers of realisations are related to the increasing in the configuration space exploration, leading to wider distributions.

The average over all the realisations, shown in black, presents an explicit cubic behaviour as a function of the number of negative edges. It is interesting to study the behaviour of such a mean for different sizes of the system. In fact, as shown in the following figure, we can observe a non-monotonic behaviour for small values of N.

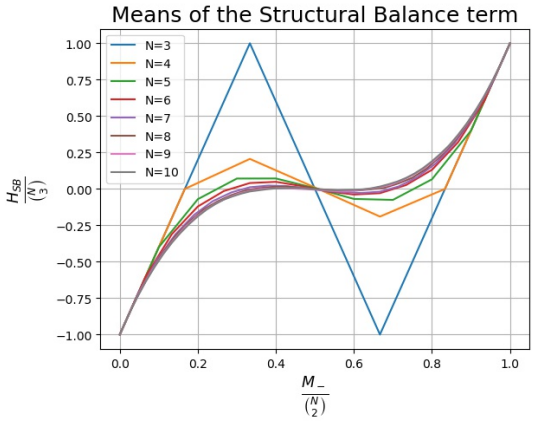


Figure 3.11: Normalised mean of 1000 realisations of the random process for different sizes N of the system

In order to compare the different means, both the axis M_- and H_{SB} have been normalised with the number of edges and the number of triangles in a complete network, respectively.

The non-monotonic behaviour is due to the fact that for intermediate numbers of negative edges between 0 and half of the possible edges, it is probable the formation of triangles with 1 negative and 2 positive edges that are unbalanced. For intermediate numbers of negative edges between half and the total possible edges, it is probable the formation of triangles, with 2 negative and 1 positive edges, considered balanced, as it is clear in the limit of $N = 3$ in the figure above. By increasing the size of the system, this effect becomes less and less probable and, as we can see from the figure 3.11 above, already for $N = 10$ the behaviour is basically monotonic.

As we know that in the limit of $N \to \infty$ the behaviour of $\langle H_{SB} \rangle$ is a cubic function of the number of negative edges, we can deduce its shape by plotting $\Delta M^3 = (M - 2M_-)^3$. In the limit of large N if we consider the average of the possible configurations for a fixed number of negative edges, we can therefore make use of the approximation $\langle \sum_{i,j,k} A_{ij} A_{jk} A_{ki} \rangle \propto (\sum_{i,j} A_{ij})^3$ which translates in the following calculation:

$$\lim_{N\to\infty} \langle H_{SB} \rangle \propto \lim_{N\to\infty} -\left\langle \sum_{i,j,k} A_{ij} A_{jk} A_{ki} \right\rangle \propto -\left(\sum_{i,j} A_{ij} \right)^3 = -\Delta M^3 \tag{3.21}$$

In order to capture the non-monotonic behaviour, we have to include a linear term in the equation. The calculation is carried out considering the average of H_{SB} normalised with β and the number of possible triangles so that the function always starts at -1 and ends at $+1$. This translates in the normalisation of ΔM with the total number of edges on the left-hand side of the equation, and it allows us to set a simple relation between the coefficient of the cubic term and the coefficient of the linear term: they have to sum to 1.

$$\frac{\langle H_{SB} \rangle}{\beta \binom{N}{3}} = -k(N) \left(\frac{\Delta M}{M} \right)^3 + [k(N) - 1] \left(\frac{\Delta M}{M} \right) \tag{3.22}$$

As we can see from the following figures, the expected behaviour computed using equation 3.22 practically coincides perfectly with the average over the realisations. This confirms the accuracy of the equation 3.22 and the correct coefficient $k(7) = 1.16$. The error μ is defined for every M_- as the difference between the average computed over the realisations and the value expected by the equation 3.22. Figure 3.13 shows that the error is always lower than 0.1%, and it is probably due to the stochastic nature of the average.

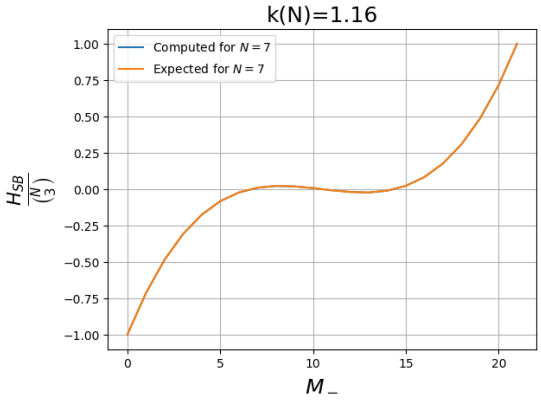


Figure 3.12: Comparison between the computed values of the average over $N_r = 10^4$ random processes and the expectations coming from equation 3.22 for a network of size $N = 7$. It is actually two lines, but the second overlaps the first almost perfectly.

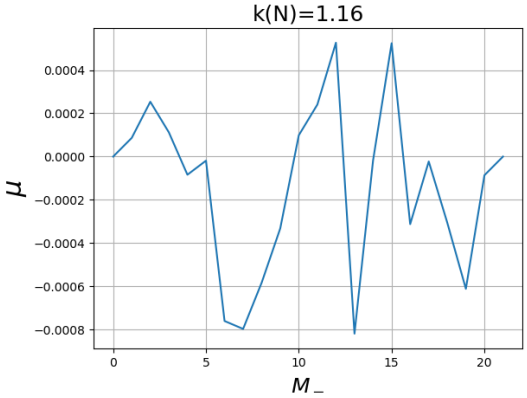


Figure 3.13: Representation of the error μ, defined for every M_- as the difference between the average computed over the realisations and the value expected by the equation 3.22

The parameter $k(N)$ depends on the number of agents in the system, and it can be computed numerically by fitting the averages with the expected values. As we can see from the figure below, $k(N)$ assumes values closer to 1 as the size of the system increase, making the linear term vanishing for large values of N as expected.

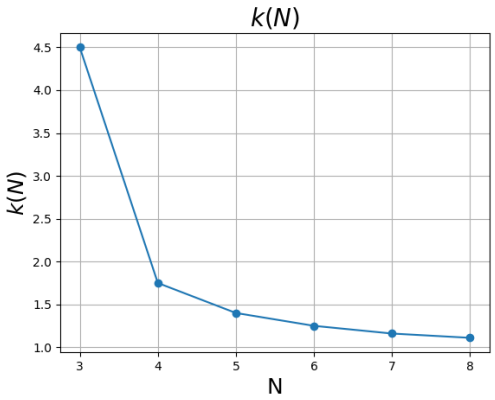


Figure 3.14: Trend of $k(N)$ as a function of the size of the system. The values have been computed numerically by fitting the results of figure 3.11 with the equation 3.22

3.4.3 Ordered process

In order to explore some of the rare configurations that provide the lowest energy levels, a specific order of the process above is chosen. In this ordered process, one agent is chosen fixed and all of its $N-1$ edges attached to it are sequentially set to -1. It is not important which agent is chosen, but it is important to change only the sign of the edges that are attached to it. Also, the order of the edges is not relevant since, apart from the chosen agent, all the other agents have the same properties and are therefore equivalent.

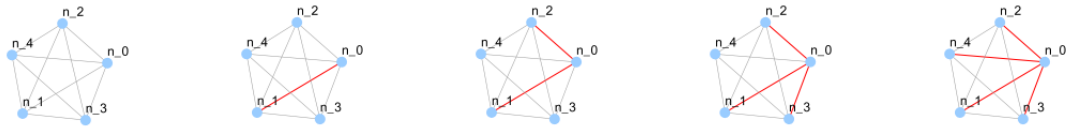


Figure 3.15: Visualisation of the first 5 steps of the ordered process for a network of $N = 5$ nodes. Negative links are highlighted in red, while positive links are grey.

Once all the edges referred to the first node are set to -1, a second node is picked, and all its remaining $N-2$ positive links are set to -1. This process continues until every edge attached to every node is set to -1.

From figure 3.16 we can observe that following this specific ordering, the behaviour of H_{SB} is no more symmetric, and the values of the energy are almost always lower than the ones in the previous section.

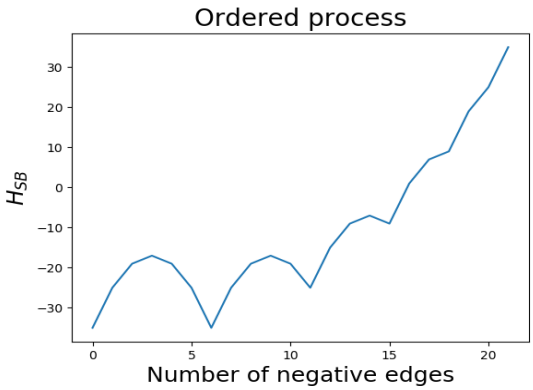


Figure 3.16: Energy output coming from the ordered process for a network of $N = 7$ nodes

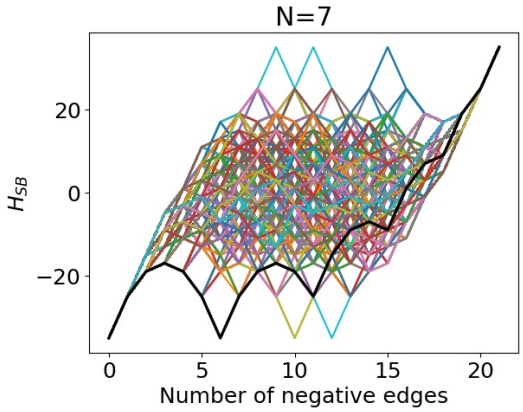


Figure 3.17: Comparison between random realisations and ordered realisation of the process for a network of $N = 7$ nodes

We can see from figure 3.16 that for $M_- = 9 = N - 1$, we found a second global minimum which corresponds to the configuration where there is an enemy with all the other nodes. We are in the presence of a common enemy.

From the comparison between the ordered process and the random process in figure 3.19 we can clearly deduce that, for a number of negative edges smaller than at least the number of nodes, the ordered process gives as output the lowest possible energy that a signed network with a certain number of nodes and negative edges can assume.

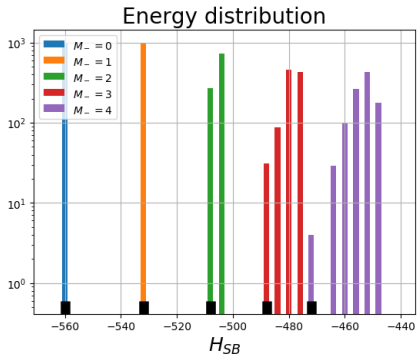


Figure 3.18: Visualisation of the energy distributions for a network of $N = 16$ nodes and a number of negative edges ranging from 0 to 4. The lowest possible energy for every M_-, given by equation 3.24, is highlighted in black.

3.5 Analytical results for energy levels

While the focus of Marvel's paper [15] was to compute the configurations and energy levels corresponding to high energy local minima, the goal of this section is to derive some of the possible energy levels analytically, with particular attention to those with low energy or with a low number of negative edges.

In this first part, we will derive the formula to compute the lowest possible energy given a fixed number of negative edges $M_- < N$.

As we said before, the value of H_{SB} depends on the whole topology of the signed network and not only on the number of negative edges M_-, but we can define a quantity:

$$H_{SB}^{(\min)}(M_-) := \frac{1}{\beta} \min_{G_{M_-}} (H_{SB}(G_{M_-})) \tag{3.23}$$

where G_{M_-} is every possible signed network with N nodes and M_- negative edges. The factor $\frac{1}{\beta}$ is added in order to simplify the formula.

As we observed in the previous section, the ordered process introduced before gives as output the minimum possible energy for a number of negative edges M_- smaller than the number of nodes N. We, therefore, need to compute the number of balanced triangles in the case where all the edges referred to one node are set to -1 sequentially.

When the first edge $(0, 1)$ is set to -1, $N-2$ balanced triangles become unbalanced, hence H_{SB}, i.e. the difference between the number of unbalanced and balanced triangles, increases by $2(N-2)$. By following the same reasoning, we can set the second edge $(0, 2)$ to -1, and this causes $N-3$ triangle to become unbalanced and 1 triangle, formed by $(0, 1, 2)$, to become from unbalanced to balanced. The difference in H_{SB} is therefore $2(N-4)$. Continuing this procedure, we can infer the following rule, which is valid only for $M_- < N$:

$$H_{SB}^{(\min)}(M_-) = -\binom{N}{3} + \sum_{i=1}^{M_-} 2(N-2i) = -\binom{N}{3} - 2\Big[M_-^2 - (N+1)M_- + N\Big] \tag{3.24}$$

The energy outputs of equation 3.24 correspond to the first N steps of the ordered process in figure 3.16, as we can see from the comparison in the figure 3.19.

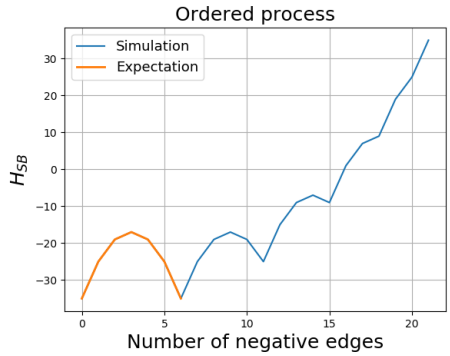


Figure 3.19: Comparison between the simulation and the expected value of the lowest energy possible for $M_- < N$ in a network of $N = 7$. The expected values perfectly overlap the simulation.

3.6 Studies on local minima

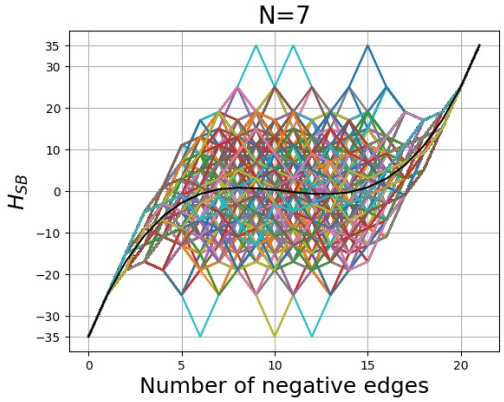


Figure 3.20: Figure showing 10^4 realisations of the process explain above, considering a network with $N = 7$ nodes. The figure shows four global minima corresponding to $M_- = 0, 6, 10, 12$

Due to the low degeneracy of the low energy configurations to the large number of possible configurations, it is very unlikely to explore at random low energy configurations with a large number of negative edges, but, as said before and showed by Cartwright and Harary, we know their existence. In fact, they correspond to the states where a clique of nodes that share only positive links (they are all friends) shares only negative links to another clique of friends. These states are sometimes called "war states". We can see an example of such a state in the figure 3.21 below:

The number of global minima of H_{SB} that can be found in a fully connected signed network is equal to the number of different ways that we can divide in two the set of nodes that is:

$$N_{minima} = \begin{cases} \frac{N+2}{2} & \text{if N is even} \\ \frac{N+1}{2} & \text{if N is odd} \end{cases} \tag{3.25}$$

As previously found by Cartwright and Harary these minima correspond to the case where two cliques of agents that are friends with each other have only negative edges

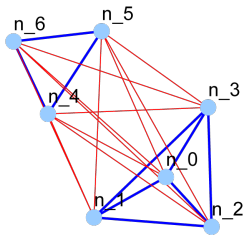


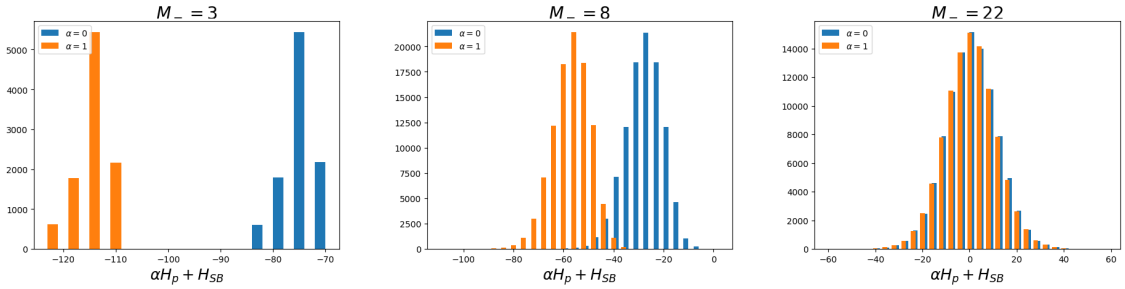
Figure 3.21: Network with $N = 7$ representing a global minimum configuration with $N_1 = 4$, $N_2 = 3$ and therefore $M_{12} = 12$

between them. If we consider N as the number of agents in the system, N_1 the number of agents in the first clique and $N_2 = N - N_1$ the number of agents in the second clique, the global minima correspond to the number of edges M_{12} that connects the set of N_1 nodes and the set of N_2 nodes:

$$M_{12}(N, N_1) = N_1 \cdot N_2 = N_1(N - N_1) \tag{3.26}$$

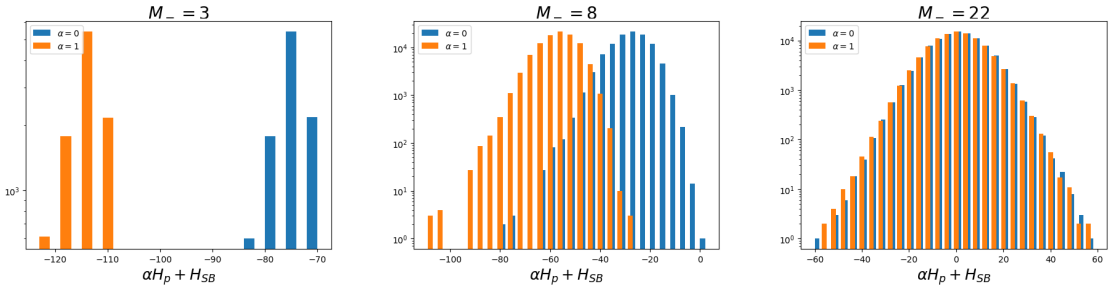
3.7 Total Hamiltonian

Now that we worked out the properties of every term, the first and the third analytically, the second numerically, we can sum them and study the behaviour for different values of the coefficients α, β and γ. As the three terms sum linearly, we do not expect the energy to present any phase transition or strange behaviour as the parameters change. Nevertheless, it is worth studying the total Hamiltonian H as it will tell us the overall properties of the systems we will study. Let us first start by summing the first and second terms since they all strongly depend on the adjacency matrix and can be studied through the number of negative edges. As we saw from the previous study of H_p, the contribution depends linearly on the number of negative edges, and it does not depend on the topology of the network. We, therefore, expect the sum to follow the same behaviour as H_{SB} but shifted on every M_- by a quantity proportional to $\Delta M = M_+ - M_-$. As we can see from the figures below, our expectations are confirmed.



(a) $\Delta M = 39$ (b) $\Delta M = 29$ (c) $\Delta M = 1$

Figure 3.22: Energy distribution for a system with $N = 10$ and $N_r = 10^5$, with (orange) and without (blue) the additional contribution of H_p with $\alpha = 1$. $\Delta M = M_+ - M_- = M - 2M_- = \binom{N}{2} - 2M_-$



(a) $\Delta M = 39$ (b) $\Delta M = 29$ (c) $\Delta M = 1$

Figure 3.23: Energy distribution for a system with $N = 10$ and $N_r = 10^4$ with (orange) and without (blue) the additional contribution of H_p with $\alpha = 1$, in a logarithmic scale.

The shift in the energy distributions linearly depends on the difference between the number of positive and negative edges. In the case considered for the figures above, the shift is exactly ΔM as we fixed the value $\alpha = 1$. As the number of negative edges increases, the energy shift becomes smaller and smaller. When M_- reaches half of the number of possible edges the shift is minimal and after that it starts increasing in the same manner, on the opposite direction, thanks to the odd parity of both H_p and H_{SB}.

If we perform the random process explained in section 3.4.2, considering the sum between H_p and H_{SB}, the results can be seen in the following figure 3.24.

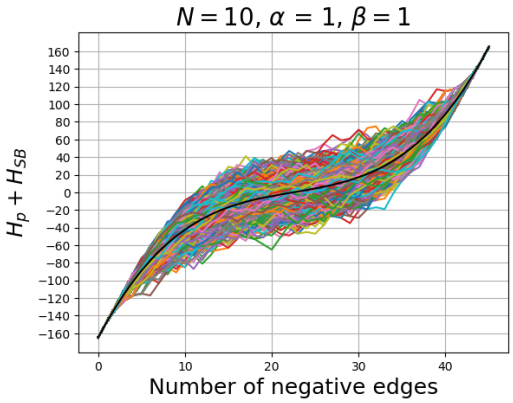
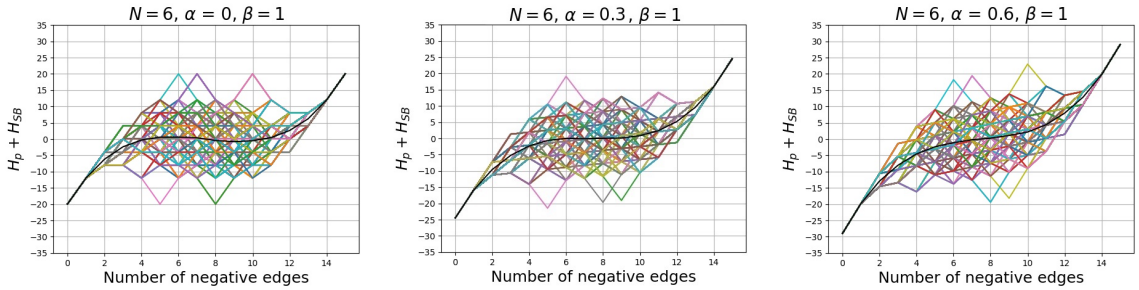


Figure 3.24: Representation of the random process for the computation of $H_p + H_{SB}$, considering a network with $N = 10$, $\alpha = 1$, $\beta = 1$ and $N_r = 10^4$. The black line represents the mean over all the realisations.

Let us look at the average of the realisations. We notice that at the center of the figure 3.24, corresponding to a $M_- = M/2$, the slope of the mean is positive even for relatively small values of α, $\alpha = 1$ in this case, while, as we said before, considering only the structural balance term, the slope is negative. It converges to zero while increasing the number of agents. That means that for a given number of agents, there exists a critical value of α such that the slope of the average energy at the center of M_- becomes flat.

As represented in figure 3.25, we can think of increasing the value of α starting from zero, where the average of H_{SB} present the usual global minimum at $M_- = 0$, a local maximum M_{LM} at $0 < M_{LM} < M/2$, a negative slope at $M/2$, a local minimum M_{lm}

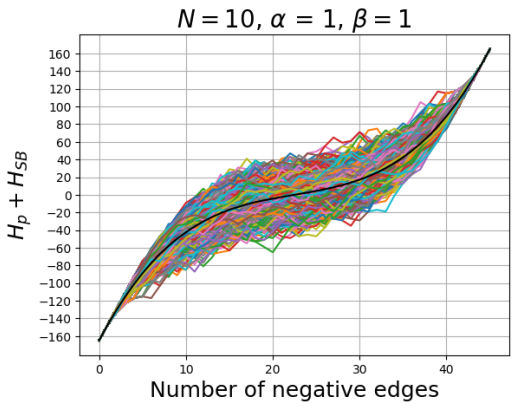


(a) $\alpha = 0$ (b) $\alpha = 0.3$ (c) $\alpha = 0.6$

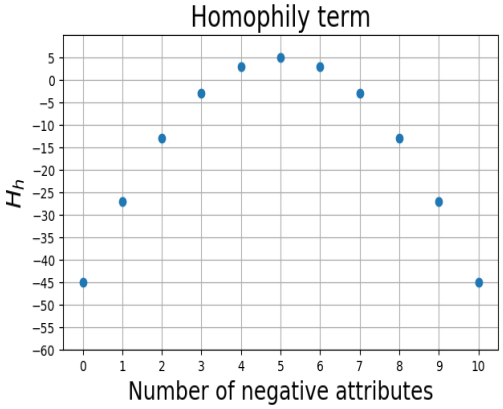
Figure 3.25: Representation of the random process for the computation of $\alpha H_p + H_{SB}$, considering a network $N = 6$ and an increasing value of α. The black line represent the mean of over all the realisations. In this case $\alpha^*(6) \simeq 0.3$.

at $M/2 < M_{lm} < M$ and, at the end, the global maximum at $M_- = M$. By increasing the value of α, we will observe M_{LM} and M_{lm} gradually move towards the center, while the slope at the center decreases. At a certain value $\alpha^*(N)$ the local maximum and minimum merge to a horizontal flex at the center, the average of $H_p + H_{SB}$ becomes monotonic and for values of α greater than $\alpha^*(N)$ the slope at the center becomes positive and steeper.
In order to have a complete analysis of the total global Hamiltonian, we need to add the third term, the homophily term H_h, to the previous study. Unlike the other two terms, H_h does not depend on the number of negative edges nor the signed topology. It depends instead on the number of negative attributes or equivalently on the difference between the number of positive and negative ones. This means that, for a fixed distribution of the attributes, the total energy of every possible signed configuration is shifted by the same amount, regardless of the number of negative edges or the topology of the signs. In order to represent the landscape of energy of the total Hamiltonian graphically, we have to add an extra dimension, namely the number of negative attributes in the system. For a matter of simplicity, the figure 3.27 shows only the average of the energy H as a function of the number of negative edges and the number of negative attributes. In fact, it is not easy to represent every realisation in such a bi-dimensional graph, but we have to keep in mind that every point in the figure corresponds to a distribution of energy levels similar to the ones in figures 3.25.

Thanks to the fact that the homophily term does not depend on the number of negative edges, and that the positivity and structural balance terms do not depend on the number of negative attributes, the two contributions sum independently. This fact can be seen by "slicing" the previous graph. If we sliced the graph horizontally, we would obtain exactly the figure 3.26a, while slicing it along the vertical axis, we would obtain figure 3.26b.



(a) Contribution coming from the positivity and the structural balance terms



(b) Contribution coming from the homophily term

Figure 3.26: Contributions coming from the positivity and the structural balance terms (3.26a) and the homophily term (3.26b)

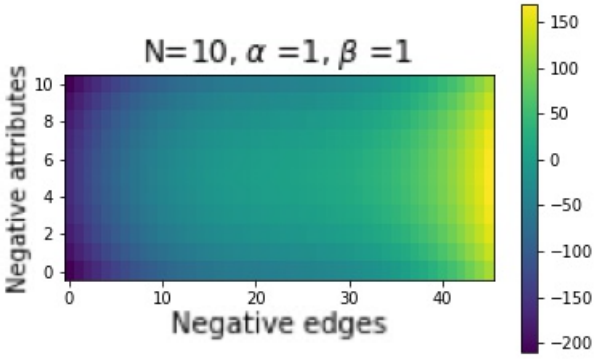


Figure 3.27: Bidimensional graph representing the mean over all the realisations of the random process for $H = H_p + H_{SB} + H_h$ as a function of the number of negative edges and the number of negative attributes.

Chapter 4

Interaction networks from relation networks

In this Chapter, we will first introduce and discuss a quantity, analogous to the energy, between two agents derived from the global Hamiltonian studied previously, in sections 4.1 and 4.2. Starting from this local Hamiltonian, we will introduce in section 4.3 a quantity that will be used as the probability of interaction between the two agents under consideration. Based on these probabilities, we will explain how to generate interactions and how to represent them through a multi-edge network. In section 4.4 we will try to compute analytically some of the topological properties of interaction networks that will be compared to the properties computed through agent-based simulations in section 4.5. In sections 4.6 we will explore the parameter space and how the properties change accordingly. Lastly, section 4.7 will be devoted to comparing the topological properties of the interaction network of particular states to their corresponding signed networks.

Apart from the definitions in section 4.5, already presented in section 2.1, everything in this Chapter is our contribution and new respect the existing literature.

4.1 Derived local Hamiltonian and limitations

It is possible to rewrite the equation 3.3 for the global Hamiltonian, defined in the previous Chapter, as follows:

$$H = \sum_{ij} \Big\{ -\frac{\alpha}{2} A_{ij} - \frac{\beta}{6} \sum_{k} A_{ij} A_{jk} A_{ki} - \frac{\gamma}{2} s_i |A_{ij}| s_j \Big\} \tag{4.1}$$

We can therefore introduce a quantity:

$$H_{ij} \quad \text{such that} \quad H = \sum_{ij} H_{ij} \tag{4.2}$$

that accounts for the "energy" between a pair of agents (i, j):

$$H_{ij} = -\frac{\alpha}{2} A_{ij} - \frac{\beta}{6} A_{ij} \sum_{k} A_{ik} A_{kj} - \frac{\gamma}{2} s_i |A_{ij}| s_j \tag{4.3}$$

This quantity takes into account the relationship between the two agents, the difference between the number of balanced and unbalanced triangles they form with common neighbours (i.e. all the other agents in our case where the signed network is fully connected) and their similarity, thanks to the homophily term and the attribute we consider. The numerical coefficients are those coming from the global Hamiltonian, and the parameters once again weigh the importance of the three different contributions.

Despite the name, the local Hamiltonian is "local" only regarding its effect, but it actually comes from global considerations. In fact, the corresponding term of the structural balance takes into account the sign of the relations between the two nodes under consideration and all their neighbours, making the quantity H_{ij} not local at all. However, since it gives us a quantity referred only to the couple (i, j), in order to make clear the difference between the global Hamiltonian 3.3, we decided to call it local, but it is clear the true meaning of this definition.

In order to simplify the discussion we can define the three independent contributions as:

$$\begin{cases} H^p_{ij} := -\dfrac{\alpha}{2} A_{ij} \\ H^{SB}_{ij} := -\dfrac{\beta}{6} A_{ij} \displaystyle\sum_k A_{ik} A_{kj} \\ H^h_{ij} := -\dfrac{\gamma}{2} s_i |A_{ij}| s_j \end{cases} \tag{4.4}$$

Such that:

$$H_{ij} = H^p_{ij} + H^{SB}_{ij} + H^h_{ij} \tag{4.5}$$

The symmetries of each individual term do not change respect to the ones of the global Hamiltonian. In fact the positivity term shows an odd parity under the transformation $A_{ij} \to -A_{ij} \ \forall \ (i, j) \in E$, as well as the structural balance term. The homophily term, as in the global case, shows an even parity under the transformation $s_i \to -s_i \ \forall \ i \in V$, as we can see from the direct calculations:

$$\begin{cases} H^p_{ij}(-\mathbf{A}) = +\dfrac{\alpha}{2} A_{ij} = -H^p_{ij}(\mathbf{A}) \\ H^{SB}_{ij}(-\mathbf{A}) = +\dfrac{\beta}{6} A_{ij} \displaystyle\sum_k (-A_{ik})(-A_{kj}) = -H^{SB}_{ij}(\mathbf{A}) \\ H^h_{ij}(-\mathbf{s}) = -\dfrac{\gamma}{2}(-s_i)|A_{ij}|(-s_j) = H^h_{ij}(\mathbf{s}) \end{cases} \tag{4.6}$$

In order to calculate the maximum and minimum values of H_{ij} we start considering the first and the third term since they can only assume two possible values:

$$H^p_{ij} = \pm \frac{\alpha}{2} \tag{4.7}$$

The minimum of H^p_{ij} corresponds to the friendship relation $A_{ij} = +1$, and therefore the maximum to $A_{ij} = -1$.

$$H^h_{ij} = \pm \frac{\gamma}{2} \tag{4.8}$$

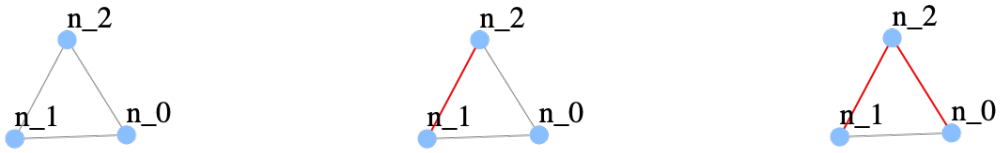
The minimum of H^h_{ij} corresponds to the configuration where the two agents i and j share the same attribute, regardless its value: $s_i = s_j$, and therefore the maximum corresponds to $s_i = -s_j$. Because of the symmetry explained above two configurations

correspond to both the maximum and the minimum of H_{ij}^h, that are therefore degenerate. Regarding the structural balance term, the minimum and maximum correspond to the case where every triangular relation, where (i, j) is included, is respectively balanced and unbalanced. Since (i, j), in a fully connected network, is included in $N-2$ triangles, the minimum value of H_{ij}^{SB} corresponds to $-\frac{\beta}{6}(N-2)$, while the maximum to $+\frac{\beta}{6}(N-2)$. The global minimum and maximum of the local Hamiltonian H_{ij} therefore corresponds to:

$$\begin{cases} \min(H_{ij}) = -\dfrac{\alpha}{2} - \dfrac{\beta}{6}(N-2) - \dfrac{\gamma}{2} \\ \max(\ H_{ij}) = \ \dfrac{\alpha}{2} + \dfrac{\beta}{6}(N-2) + \dfrac{\gamma}{2} \end{cases} \tag{4.9}$$

By taking a statistical physics perspective, we can think of configurations that minimise the quantity H_{ij} as more stable than those with a higher energy. We can therefore think to say that a link (i, j) with a lower H_{ij} is "stronger" than a link with a higher value. This consideration becomes even more meaningful if we think that this quantity will be used in section 4.3 to define a probability of interaction, and it is clear that stronger links will be related to a higher probability of interaction.

Unfortunately, this formulation contains a subtlety that can lead to wrong conclusions if seen from a sociological perspective. As said before, introducing the local Hamiltonian aims to provide a quantity related to the probability of interaction between two nodes. As we will see in the following example, the fact that a link (i, j) belongs to a balanced triangular relation does not automatically imply a larger probability of interaction. Let us provide some examples focusing on the contribution of $H_{ij}^{(SB)}$ to the strength of the bond (i, j):



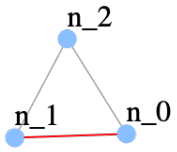
(a) n_2 common friend: balanced → stronger bond

(b) n_2 not common friend: unbalanced → weaker bond

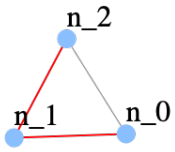
(c) n_2 common enemy: balanced → stronger bond

Figure 4.1: Sign configurations of different triangular relations always considering $(0, 1)$ positive. For every configuration the contribution of H_{01}^{SB} is described.

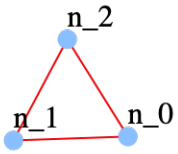
As we can see from the figures 4.1 above, when considering a positive relation (i, j) everything works as expected, i.e. common friends and common enemies increase the strength of a bond, while it decreases when that is not the case, as in 4.1b. This consideration works both from a social perspective and from our Hamiltonian perspective, but when considering a negative relation (i, j) the behaviour is different, as we can see in figures 4.2.
In this case, we can see that the contribution of H_{ij}^{SB} to the strength of the bond (i, j) behaves in a way that is against common sense and different from an usual so-



(a) n_2 common friend: unbalanced → **weaker bond**



(b) n_2 not common friend: balanced → **stronger bond**



(c) n_2 common enemy: unbalanced → **weaker bond**

Figure 4.2: Sign configurations of different triangular relations always considering $(0, 1)$ negative. For every configuration the contribution of H_{01}^{SB} is described.

cial behaviour. This is a hint that the stability of social networks and the formation of interactions are ruled by different yet related processes and assumptions. Therefore, we will continue our discussion by differentiating these two processes, highlighting their differences and similarities and, when possible, drawing some connections between them.

4.2 New local Hamiltonian and analytical properties

From the last part of the previous section 4.1, we realised that different yet related processes and assumptions rule the stability of social networks and the formation of interactions. We deduced it by observing that the straightforward derivation of the local Hamiltonian from the theory of structural balance leads, from a sociological point of view, to inconsistent results when considering it related to the formation of social interactions.

A possible change to the local Hamiltonian that provides more realistic outputs for a better social interpretation consists in not considering the nature of the relation between i and j in the second term of the local Hamiltonian, but only its existence. This translates mathematically in the introduction of the absolute value of A_{ij} in that term:

$$H_{ij} = -\alpha A_{ij} - \beta |A_{ij}| \sum_k A_{ik} A_{kj} - \gamma s_i |A_{ij}| s_j \tag{4.10}$$

Where we additionally included the numerical factors in the parameters, carrying out a formal transformation: $\alpha', \beta', \gamma' = \frac{\alpha}{2}, \frac{\beta}{6}, \frac{\gamma}{2}$ and relabelling $\alpha', \beta', \gamma' \to \alpha, \beta, \gamma$.

We can therefore consider the new modified H_{ij}^{SB} as the difference between the number of common friends/enemies and the number of agents that are friends for one and enemies for the other agent of the couple. We will still continue to call this term H_{ij}^{SB}, but it should be clear that it corresponds to the contributions of the balanced triadic relations only in the case of $A_{ij} = +1$; otherwise, it differs from the usual structural balance. With this change, the interpretation is much more realistic since the energy

decreases if the number of common friends or enemies is high, leading to a higher probability of interaction.

The computation of the maximum and minimum of the new local Hamiltonian does not show any change from the previous formulation:

$$\begin{cases} \min(H_{ij}) = -\dfrac{\alpha}{2} - \dfrac{\beta}{6}(N-2) - \dfrac{\gamma}{2} \\ \max(\,H_{ij}) = \;\dfrac{\alpha}{2} + \dfrac{\beta}{6}(N-2) + \dfrac{\gamma}{2} \end{cases} \tag{4.11}$$

The minimum and maximum value of H_{ij} do not change, but it does change the sign configuration that can correspond to them when considering (i, j) such that $A_{ij} = -1$.

Because of this modification, the new local Hamiltonian H_{ij} does not fulfil Equation 4.2 anymore. In order to study the differences introduced by this modification, we can carry out the same sum over i and j and subtract to the previous quantity:

$$\begin{aligned} \Delta H = H - \sum_{ij} H_{ij} = -\beta \sum_{ij} \Big\{ A_{ij} \sum_{k} A_{jk}A_{ki} - |A_{ij}| \sum_{k} A_{jk}A_{ki} \Big\} = \\ = -\beta \sum_{ij} \sum_{k} A_{jk}A_{ki} \big\{ A_{ij} - |A_{ij}| \big\} = -\beta \sum_{ijk} A_{jk}A_{ki} \cdot \Delta A_{ij} \end{aligned} \tag{4.12}$$

where

$$\Delta A_{ij} = \begin{cases} 0 & \text{if} \quad A_{ij} = 1, 0 \\ -2 & \text{if} \quad A_{ij} = -1 \end{cases} \tag{4.13}$$

When the number of negative edges in the system is low, the majority of the entries of ΔA_{ij} is zero, and we, therefore, expect the difference between the new and the old local Hamiltonian to lead almost to the same global properties. We will verify this statement in the followings sections when considering the interaction networks and their topological properties.

4.3 Probability of interaction and interaction networks

Starting from the local Hamiltonian H_{ij} presented and studied in the previous sections, we can introduce the probability of interaction between two agents i and j, defined as:

$$p_{ij} = \frac{\exp(-H_{ij}/T)}{\sum_{ij} \exp(-H_{ij}/T)} = \frac{1}{Z} \exp\Big(-\frac{\alpha}{T} A_{ij} - \frac{\beta}{T} |A_{ij}| \sum_{k} A_{ik}A_{kj} - \frac{\gamma}{T} s_i |A_{ij}| s_j \Big) \tag{4.14}$$

Where we define Z to be analogous to the partition function used in statistical physics: $Z = \sum_{ij} \exp(-H_{ij}/T)$, so that $\sum_{ij} p_{ij} = 1$. We also introduced T as a parameter that has the same "physical" dimension of H_{ij}. It plays the role of the temperature in statistical mechanics and can be seen as a simultaneous rescaling of all the parameters. As we will study p_{ij} and its effects for many different sets of parameters α, β, γ, we can think to rescale all the parameters by a factor $1/T$, or equivalently to set $T = 1$,

as it will be implicitly considered in the study of the properties as a function of the parameters.

Defined in this way we can interpret p_{ij} as a probability, fulfilling the following properties:

$$\begin{cases} p_{ij} \in \mathbb{R} \\ 0 \leq p_{ij} \leq 1 \\ \sum_{ij} p_{ij} = 1 \end{cases} \tag{4.15}$$

Now that we defined the probability of interaction for every couple (i, j), we can start producing interactions and build interaction networks. Two equivalent algorithms that fulfil this task will be presented below. Both of them take the same parameters as inputs and give a possible random realisation of the interaction network as an output. The inputs taken by the algorithms are:

- A signed network of size N with a scalar binary value defined on every node
- The parameters α, β and γ for the computation of H_{ij}
- The number M_c of total interactions to generate

We already discussed the meaning and the importance of the first two points above, as they basically define the probability of interaction through the local Hamiltonian H_{ij}. The last point refers only to the actual generation of the interaction network as it indicates how many interactions are to be generated. This number can be larger than the total number of possible edges M of the signed network. We aggregate the interactions as a multi-edge network, and multiple edges can be counted between the same two nodes i and j. This property allows us to represent the actual pattern of interactions in a social system, as two people can interact multiple times. Another equivalent way to represent this number is through the fraction of the total number of interactions generated and the number of edges in the fully-connected signed network:

$$M_{pc} = \frac{M_c}{M} \tag{4.16}$$

The quantities listed above define a macrostate of a set of possible interaction networks. The output given by the algorithms represents a microstate, i.e. one particular realisation of the macrostate.

The first algorithm that we used to generate interaction networks is the following:

Algorithm 3: Interaction Network Generator

- Define a Random Signed Network of size N and M_- negative links
- Define α, β and γ
- Define the number M_c of total interactions to generate
- Compute the probabilities p_{ij} $\forall i, j$
- Set $Counter = 0$

while $Counter < M_c$ **do**
- Select randomly i and $j \neq i$ uniformly from the set of agents
- Generate a random number from a uniform distribution $r \in [0, 1]$
- **if** $r < p_{ij}$ **then**
  - Create or add an edge between nodes i and j
  - $Counter = Counter + 1$
- **else**
  - Go back to agents' selection
- **end**

end

With this procedure, the probability of obtaining a specific realisation G of the interaction network is $P(G)$, and can be calculated as follows:

$$P(G) = \frac{M_c!}{\prod_{i>j} m^*_{ij}!} \prod_{i>j} p_{ij}^{m^*_{ij}} \tag{4.17}$$

Where $m^*_{ij} = G_{ij}$ is the number of interactions generated by the algorithm between nodes i and j. Remembering the form of the probability p_{ij}:

$$p_{ij} = \frac{\exp(-H_{ij})}{\sum_{ij} \exp(-H_{ij})} = \frac{1}{Z} \exp(-H_{ij}) \tag{4.18}$$

and the local Hamiltonian:

$$H_{ij} = -\alpha A_{ij} - \beta |A_{ij}| \sum_k A_{ik} A_{kj} - \gamma s_i |A_{ij}| s_j \tag{4.19}$$

we can write:

$$\begin{aligned}
\prod_{i>j} p_{ij}^{m^*_{ij}} &= \frac{1}{\left(\sum_{ij} \exp(-H_{ij})\right)^{M_c}} \prod_{i>j} \exp\left(m^*_{ij} \left[-\alpha A_{ij} - \beta |A_{ij}| \sum_k A_{ik} A_{kj} - \gamma s_i |A_{ij}| s_j \right] \right) \\
&= \frac{1}{\left(\sum_{ij} \exp(-H_{ij})\right)^{M_c}} \exp\left(\left[-\alpha \sum_{ij} m^*_{ij} A_{ij} - \beta \sum_{ij} m^*_{ij} |A_{ij}| \sum_k A_{ik} A_{kj} - \gamma \sum_{ij} m^*_{ij} s_i |A_{ij}| s_j \right] \right) \\
&= \frac{1}{Z^{M_c}} \exp\left(\alpha f_1(\boldsymbol{G}, \boldsymbol{A}) + \beta f_2(\boldsymbol{G}, \boldsymbol{A}) + \gamma f_3(\boldsymbol{G}, \boldsymbol{A}, \boldsymbol{s}) \right)
\end{aligned} \tag{4.20}$$

Where we defined:

$$\begin{cases}
f_1(\boldsymbol{G}, \boldsymbol{A}) = \sum_{ij} m^*_{ij} H^p_{ij} = -\sum_{ij} m^*_{ij} A_{ij} \\
f_2(\boldsymbol{G}, \boldsymbol{A}) = \sum_{ij} m^*_{ij} H^{SB}_{ij} = -\sum_{ij} m^*_{ij} |A_{ij}| \sum_k A_{ik} A_{kj} \\
f_3(\boldsymbol{G}, \boldsymbol{A}, \boldsymbol{s}) = \sum_{ij} m^*_{ij} H^h_{ij} = -\sum_{ij} m^*_{ij} s_i |A_{ij}| s_j
\end{cases} \tag{4.21}$$

The final formula for the probability $P(G)$ is therefore:

$$P(G) = \frac{M_c!}{\prod_{i>j} m^*_{ij}!} \frac{1}{Z^{M_c}} \exp\left(\alpha f_1(\boldsymbol{G}, \boldsymbol{A}) + \beta f_2(\boldsymbol{G}, \boldsymbol{A}) + \gamma f_3(\boldsymbol{G}, \boldsymbol{A}, \boldsymbol{s})\right) \tag{4.22}$$

After this first approach an alternative algorithm was proposed. This new algorithm takes into consideration the sum p_i of the probabilities of interaction of an agent i:

$$p_i = \sum_j p_{ij} \tag{4.23}$$

According to this probability, one agent is picked and then one edge is created or added to each of its neighbours j according to the usual probability p_{ij}.

Algorithm 4: New Interaction Network Generator

```
- Define a Random Signed Network of size N and M_ negative links
- Define α, β and γ
- Define the number M_c of total interactions to generate
- Compute the probabilities p_ij ∀i, j
- Compute p_i = Σ_j p_ij
- Set Counter = 0
while Counter < M_c do
    - Select randomly i uniformly from the set of agents
    - Generate a random number from a uniform distribution r_1 ∈ [0, 1]
    if r < p_i then
        for j in i's neighbours do
            - Generate a random number from a uniform distribution r_2 ∈ [0, 1]
            if r < p_ij then
                - Create or add an edge between nodes i and j
                - Counter = Counter + 1
            end
        end
    end
end
```

Given the greater complexity of this algorithm, the probability $P(G)$ for this case has not been calculated. However, thanks to the following considerations, we could say that the expectations should be similar to the probability found with equation 5.15. In fact, according to the new algorithm, agents that have a higher sum of interaction probabilities are selected more frequently. However, because even in the previous case, they already had larger probabilities, the expectation is that the output of the two different algorithms should not differ too much. In fact, it can be shown that interaction networks generated by this new algorithm share the same topological properties as the former one.

Therefore, in the following, we will make use of the new algorithm for the computation of the observables, as it is about four times faster in the generation of interactions than the previous algorithm, and, since the properties of the generated networks are the same, we could thus obtain the same reliable results in a more convenient way.

4.4 Analytical expectations for properties

By increasing the number of interactions created M_c, or alternatively the fraction $M_{pc} = \frac{M_c}{M}$ between the number of interactions created and the number M of the possible links between every agent, the topology of the system converges to a fully connected network since even the links with smaller probabilities p_{ij} get visited enough times that the probability of developing at least one interaction becomes non-negligible so that in the limit of $M_c \to \infty$ there is at least one interaction for every possible pair of agents and the interaction network is therefore fully connected.

If we assume, as it is for the first algorithm, a uniform random exploration of all the possible edges, the average number of visitations per edge is exactly $\frac{M_c}{M} = M_{pc}$. The probability of generating an interaction at least once in M_{pc} visitations is equal to 1 minus the probability of not generating a link in M_{pc} times, i.e. $P(G_{ij} > 0) = 1 - (1 - p_{ij})^{M_{pc}}$, where G_{ij} is the number of interactions between nodes i and j. In a network with a fixed number of nodes also M is constant and therefore $M_c \to \infty \implies M_{pc} = \frac{M_c}{M} \to \infty$ and since p_{ij} comes from exponential function, it is always larger than zero and it means that:

$$\lim_{M_{pc}\to\infty} (1 - p_{ij})^{M_{pc}} = 0 \tag{4.24}$$

and

$$\lim_{M_{pc}\to\infty} P(G_{ij} > 0) = \lim_{M_{pc}\to\infty} \left[1 - (1 - p_{ij})^{M_{pc}}\right] = 1 \tag{4.25}$$

Therefore in this limit, any interaction network built with our procedure converges to a fully connected network. Of course, a very high value of H_{ij} may lead to a p_{ij} close to zero. In this case, the number M_c needed to observe a fully connected network could be too large to compute numerically, but nevertheless, the network would result highly connected.

In the following section, we will verify this expectation using this limit for the computation of different observables.

4.5 Simulations and computation of observables

In order to study the properties of the interaction networks, defined starting from a signed network of the relations, a set of attributes of the agents and a set of parameters, we will compute some of the aggregate statistics of the network. We will study the degree sequence and the degree distribution, and some of the observables that describe the topology of the network, such as the average shortest path length (APL), the size of the largest connected component (LCC), the global clustering coefficient (GCC) and the modularity. We will study these observables on average, over all the realisations of a certain macrostate, and we will state that a particular property belongs to that specific macrostate. Obviously, it will not always be accurate because the realisations can have very different properties within the same macrostate, particularly when considering RSNG(N, M_-). We will then study the distributions of these quantities, and we will say that a property belongs to that specific macrostate only if the values do not change too much between the different realisations and, therefore, the distribution of these observables is relatively narrow and unimodal.

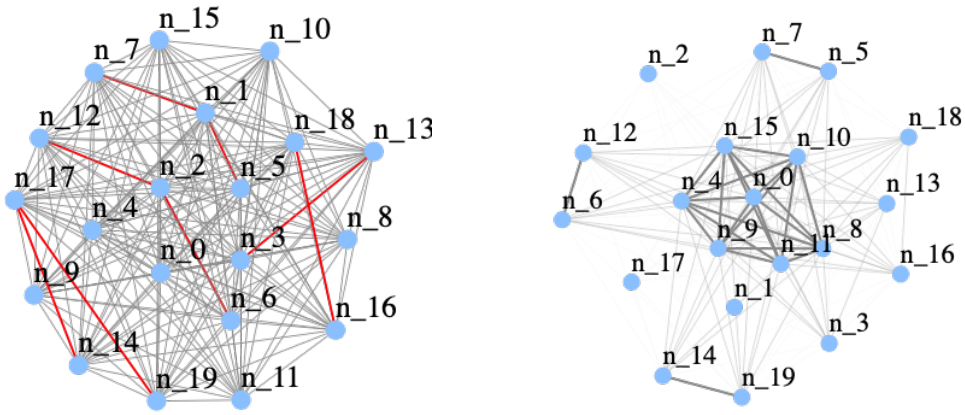
Because of the fact that we are dealing with multiple types of macrostates, i.e. the macrostate RSNG(N, M_-), defined by the number of agents and negative edges in the signed network, and the macrostate of interaction networks defined by the probability network (uniquely determined by the signed network and by the parameters α, β, γ) and M_{pc}, we have to decide which macrostate to consider for our measurements, and of course, it will depend on the observable considered.

In the first part of this discussion, we will focus only on the interactions generated by the first two terms of the local Hamiltonian 4.10, considering, therefore, $\gamma = 0$. If not expressly indicated, we will also consider $\alpha = 1$ and $\beta = 1$ in this first part. We will explore the effects of the inclusion of the homophily term in the following section.

Let us now provide an example and consider a system of $N = 20$ agents and a number $M_- = 8$ of negative relations between randomly chosen agents. Therefore, the system is one of the possible realisations of RSNG$(N = 10, M_- = 8)$. Just to picture the system, we can think of the agents as students in a high school class. They obviously know every other student in the class. We are therefore considering a fully connected network. We assume that every relationship can only be positive or negative, and all positive relations have the same importance as negative ones. This implies that no weights, except for $+1$ and -1, are used to describe the type of relation. In this example, we consider that most of the agents are friends with each other, but it is quite usual that some of the students have some kind of rivalry or enmity with some of the other students. These rivalries are represented as -1 relations in our discussion and are depicted as red links in figure 4.3a. These negative relations obviously decrease the probability of interaction between the two enemies (effect accounted in the first term of equation 4.10). However, they also decrease the probability of interaction between them and a common friend; effect accounted in the second term. In fact if we consider the relation between the nodes *n_17* and *n_19*, in figure 4.3a, not only their relation affects negatively their probability of interaction, but it also decreases the probability between nodes *n_17* and, for example, node *n_9*. That is because *n_9* is a friend also with *n_19* and since he knows that they are enemies, he reduces the number of interactions with both of them in order not to "frustrate" the other one. That is the underlying reasoning behind the use of the second term in 4.10.

The following figures 4.3 show the signed network discussed above and the related probability network deduced by computing $p_{ij} = \frac{\exp(-H_{ij})}{\sum_{ij}\exp(-H_{ij})}$ for all the couples (i, j) in the signed network. This signed network and the probability network that it induces, using equation 4.14 with parameters $\alpha = \beta = 1$ and $\gamma = 0$, will be used throughout all this section as the basis from which we will build the interaction networks. If not explicitly indicated, we will therefore consider this signed network and this set of parameters as the defining properties of the macrostate of the interaction networks.

As we can see from figure 4.3b, in this example, the probability network shows a strongly connected core of nodes, while other nodes, at the periphery, are more weakly connected. As we will see later, this is a common feature of probability networks, particularly if the number of negative links is relatively small. In fact, the core is formed by nodes that only have positive relations and are therefore not included in any frustrated pattern. Among the peripheral nodes, we can see that some of them have a strong connection between other nodes of the same region. This is due to the fact that,



(a) Signed network with $N = 20,\ M_- = 8$ (b) Probability of interaction network

Figure 4.3: On the left, the representation of the signed network discussed above. The negative links are depicted in red, while the positive ones are in grey. On the right, the probability of interaction network deduced from the signed network on the left using equation 4.14 with $\alpha = \beta = 1$ and $\gamma = 0$. Even if some nodes might seem disconnected, every node has a link with every other node, but sometimes the probabilities p_{ij} can be so low that the strength of their links is too weak to be represented graphically.

despite having some negative links connected to them, they only share common friends or enemies, so they have some uncomfortable relations with nodes in the core but none with the other node in the periphery.

In the next part of this Chapter, we will study various realisations of interaction networks generated by the signed network and the probability network introduced above. We will therefore consider fixed the signed network and study the macrostate of the interactions. As explained in section 4.3, this macrostate is defined by the signed network of the relations, the parameters of the local Hamiltonian α, β, γ, but also by the number of interactions created M_c, or equivalently by the parameter $M_{pc} = \frac{M_c}{M}$. In this section, we are going to introduce the most relevant statistics and observables, and we will study them for a set of fixed parameters $\alpha = \beta = 1, \gamma = 0$ and for different values of M_{pc}. The discussion for different values of α, β and the role of γ will take place in the following section.

4.5.1 Degree sequence and distribution

The first aggregate statistics of the interaction networks that we are going to study is the degree sequence S that, as explained in section 2.1, is defined as a vector of length N, where every entry represents the number of links that are connected to the node i, i.e. its degree d_i:

$$S := \Big(d_i\Big)_{i \in V} \tag{4.26}$$

The degree can be calculated starting from the adjacency matrix A_{ij} and, since we are considering undirected networks, we can define as:

$$d_i = \sum_j A_{ij} = \sum_j A_{ji} \tag{4.27}$$

Since we are considering multi-edges networks, we have to remember that $A_{ij} \in \mathbb{N}$, because multiple (unweighted) edges can exist between node i and node j.

The other important property that has been introduced is the degree distribution. Remembering that a network is defined as $G = (V, E)$ where V is the set of the N nodes and E is the set of edges, the degree distribution $P(k)$ is defined as:

$$\begin{gathered} P(k) := \frac{N_k}{\sum_k N_k} = \frac{N_k}{N} \\ \text{where} \quad N_k := |\{i \in V : d_i = k\}| \end{gathered} \tag{4.28}$$

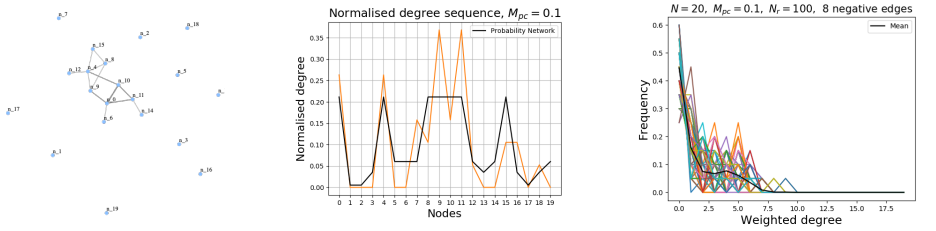
For each k, we obtain the fraction of all nodes that have degree k. Interpreted in a probabilistic way, $P(k)$ is the probability that a node chosen uniformly at random has a given degree k.
Furthermore we have introduced the mean node degree $\langle k \rangle$ in the network as the expected degree of a node chosen at random:

$$\langle k \rangle = \sum_k k \cdot P(k) = \sum_{i \in V} \frac{d_i}{N} = \frac{2 \cdot |E|}{|V|} \tag{4.29}$$

The mean degree $\langle k \rangle$ is a good estimation of the degree distribution only if the distribution is unimodal, but this could not always be the case as we are going to see.

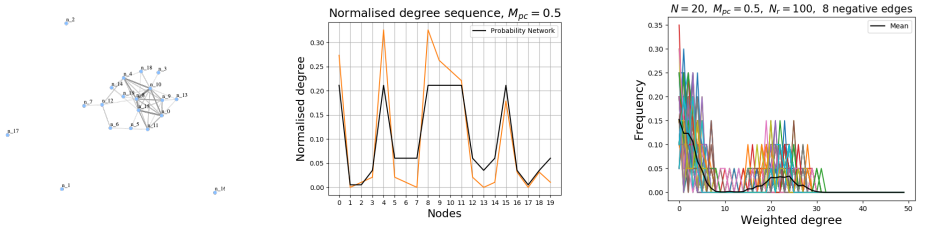
The degree distribution is a valuable aggregate statistic that can encode many properties of the network. It is actually a good compromise between the complexity of the adjacency matrix and the excessive simplicity of the mean degree. It is, therefore, an important quantity to study, but it comes with the complication that it is not a single number but rather a density function, and we will need to deal with it.

In the following figures we will show, for different values of M_{pc} an example of interaction network, its relative degree distribution, normalised with the number of interaction created $M_c = M_{pc} \cdot M_E$, and degree distribution of various random realisations, starting from the signed network in figure 4.3a with $N = 20$ and $M_- = 8$ and the same set of parameters $\alpha = \beta = 1,\ \gamma = 0$.



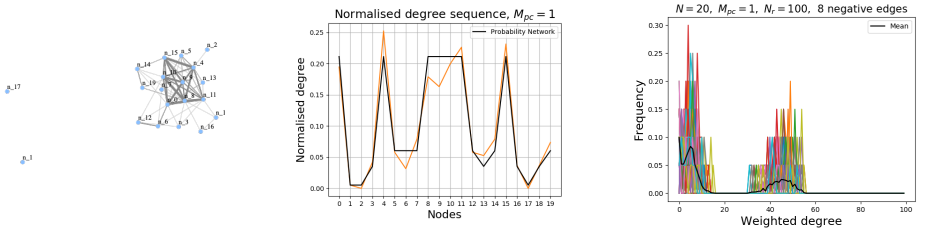
(a) Interaction network (b) Degree sequence (c) Degree distribution

Figure 4.4: $M_{pc} = 0.1$



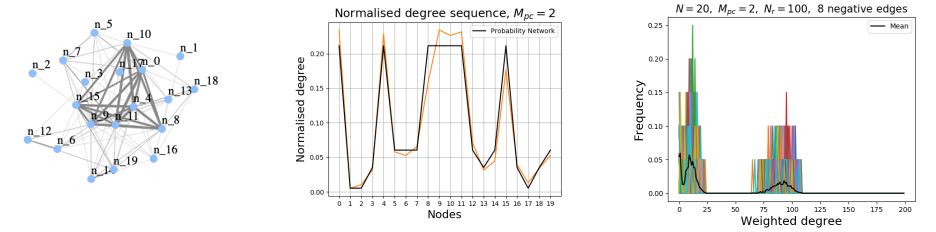
(a) Interaction network (b) Degree sequence (c) Degree distribution

Figure 4.5: $M_{pc} = 0.5$



(a) Interaction network (b) Degree sequence (c) Degree distribution

Figure 4.6: $M_{pc} = 1$



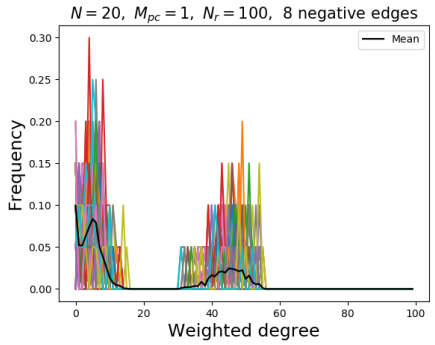
(a) Interaction network (b) Degree sequence (c) Degree distribution

Figure 4.7: $M_{pc} = 2$. Left: representation of the interaction networks G derived from the signed network 4.3a with $N = 20$, $M_- = 8$ and the same set of parameters $\alpha = \beta = 1,\ \gamma = 0$, with increasing values of $M_{pc} = 0.1,\ 0.5,\ 1,\ 2$. Thicker edges correspond to a higher number of interactions. Center: the comparison between the degree sequence (orange), normalised with the number of interactions created $M_c = M_{pc} \cdot M$ and the weighted degree sequence of the probability network (black). Right: degree distributions of 100 realisations of interaction networks (with the same parameters). The mean is represented in black. The degree is considered "weighted" because it considers the number of interactions between two nodes. The distributions show a bimodal behaviour.

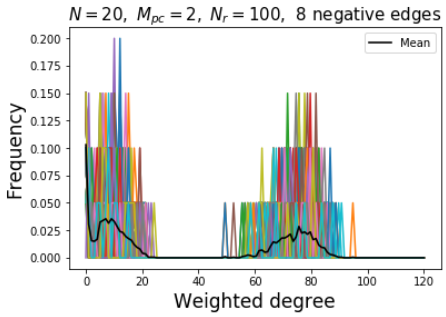
By increasing the number of interactions created, and therefore the value of M_{pc}, we can see how the interaction network become increasingly more connected, and a better-defined core is generated. Eventually, we can say that it becomes increasingly more topologically similar to the probability network. This consideration, already explained in section 4.4, is also confirmed by the degree sequence, represented in orange at the center of the figures, that, properly normalised with the number of interactions created M_c, increasingly converges to the weighted degree sequence of the probability network. On the right, instead, the degree distributions of 100 realisations of interaction networks (with the same parameters) are represented. They show, even for small values of M_{pc}, a clear bimodal distribution that can be explained by considering the fact that the interaction networks show, as the probability network, a strongly connected core. Therefore, the nodes included in the core have a higher number of connections that are far greater than the connections in the periphery. This leads to the formation of the second peak that become increasingly more defined and distant from the rest of the distribution as we increase the value of M_{pc}. Lastly, we can observe, for higher values of M_{pc}, the formation of an additional peak in the region of the distribution close to zero. This peak can be explained by looking at the peripheral nodes that, as we explained before, have negative relations with some of the nodes in the core but share only common friends or enemies with another node in the peripheral region. There are, therefore, two different kinds of peripheral nodes. The first kind of nodes are basically isolated as they do not have any frequent interactions with any of the other agents: they have at least one negative relation and do not share only common enemies or friends with any other node. They form the very first peak centred basically in zero. Then there are the peripheral nodes that have an agent, always in the peripheral region, with which they have many interactions: both of them have at least one negative relation, but they share only common enemies or friends with at least one other node. They form the second, low degree, peak. Lastly, there are the nodes in the core which have many interactions with all the other nodes in the core; they do not have any negative relation and form the third, high degree, peak.

The characterisation explained above is referred to this specific macrostate, with a specific signed network and set of parameters. It is interesting to study the behaviour of the degree distribution in the case of an interaction network generated by a different signed network with the same number of nodes $N = 20$ and the same number of negative edges $M_- = 8$ (and of course, the same set of parameters as before). We expect the degree distribution to share the same properties as the previous analysis. Suppose it will be the case, and the position of the three peaks will coincide with the ones before. In that case, we could say that all the interaction networks coming from any of the realisations of $\mathrm{RSNG}(N, M_-)$ share basically the same degree distribution regardless of the topology of the relation network's signs. In order to verify this statement, let us compute the degree distribution of an interaction network based on a random signed network generated by $\mathrm{RSNG}(N, M_-)$ with $N = 20$ and $M_- = 8$ and compare it with the case studied above, with the same set of parameters and a value of $M_{pc} = 1$:

We can observe a good similarity between the two shapes of the distributions, but unfortunately, the positions of the peaks are different. Therefore we cannot say that this specific degree distribution corresponds to the macrostate generated by $\mathrm{RSNG}(N, M_-)$ with $N = 20$ and $M_- = 8$ and the usual set of parameters, but it actually depends on



(a) Normalised degree distribution of an interaction network based on the signed network defined in figure 4.3a



(b) Normalised degree distribution of an interaction network based on the random signed network defined by RSNG(N, M_-) with the same parameters as the one defined in figure 4.3a: $N = 20$ and $M_- = 8$

Figure 4.8: Comparison between two different realisations of the interaction network based on different signed networks with the same number of nodes and negative edges. The shape is the same but the position of the peak is different. The mean of both the realisations is reported in black.

the signed topology of the relation network.

In fact, if we perform the average on a large number of realisations based on different signed networks every time, even if the parameters are the same, we can observe that the second and third peaks are much less pronounced, as we can see from figure 4.9. In this case, the position of the peaks seems not to change too much, and the average

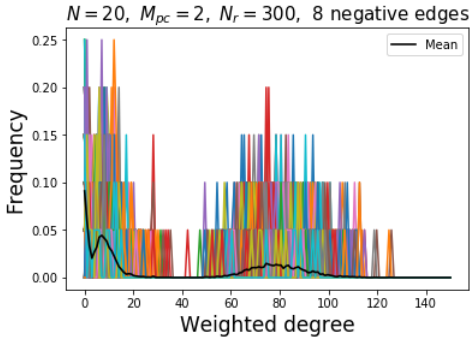


Figure 4.9: Normalised degree distribution of an interaction network based on the macrostate defined by RSNG(N, M_-) with $N = 20$ and $M_- = 8$. In black the mean over 300 realisations.

over different realisations still shows them. However, if we consider a configuration with a higher number of negative edges, these properties disappear, as shown in the figure 4.10. In fact, the average over all the realisations no longer shows any defined peak when clearly some of the realisations do.

In conclusion, when considering the degree distribution, we cannot assign a property, such as the position of the peaks, to the macrostate defined by all the signed networks with the same number of nodes and negative edges, but we can only assign it to the macrostate of the interaction networks defined by a specific fixed signed network.

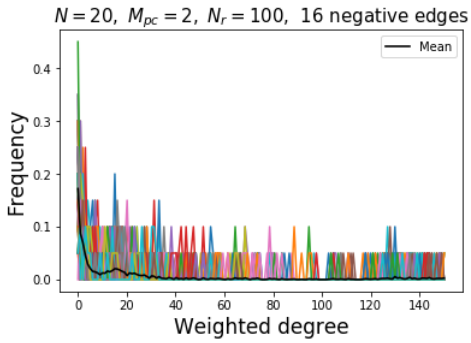


Figure 4.10: Normalised degree distribution of an interaction network based on the macrostate defined by RSNG(N, M_-) with $N = 20$ and $M_- = 16$

As we will see from the following sections, fortunately, other observables can have this useful property.

A possible step forward of this study would be the search for possible correlations between the position of the peaks of the degree distribution of a macrostate and the global structural balance of the system related to that macrostate. However, preliminary explorations in this sense do not seem to show any interesting results.

4.5.2 Average path length and largest connected component

As explained before, one of the most important concepts in a network $G = (V, E)$ is that of a path, defined as a sequence of nodes $(p_0, ..., p_l)$ where any pair of consecutive nodes is connected by a link and:

$$\begin{cases} p_i \in V \\ p_0 = i\ ,\ p_l = j \\ (p_i, p_{i+1}) \in E \quad \forall i \in [0, l-1] \end{cases} \tag{4.30}$$

The path length is defined as the number of links traversed by a path, in this case $\text{len}(p_0, ..., p_l) = l$, while the distance $\text{dist}(i, j)$ between nodes i and j is defined as the minimum length of any path between i and j. If there is no path between i and j we define the distance to be infinite: $\text{dist}(i, j) := \infty$. The definition of distance between two nodes induces us to the define a path $(p_0, ..., p_l)$ to be a shortest path if $\text{len}(p_0, ..., p_l) = \text{dist}(p_0, p_l)$. Again, it is important to stress that the shortest path is not necessarily unique, i.e. different shortest paths of the same length can exist for a given pair of nodes. The distribution of shortest path lengths between all pairs of nodes is an important topological feature of complex networks. We can see an example of the distribution of the distances of an interaction network based on the signed network in figure 4.3a, with $\alpha = \beta = 1,\ \gamma = 0$ and $M_{pc} = 1$ as in the example of figure 4.6a (but a different realisation):

In section 2.1 we have introduced two estimators for the distances distribution. The first one, the network diameter, is defined as the longest shortest path:

$$\text{diam}(G) := \max_{(i,j) \in V \times V} \Big(\text{dist}(i, j) \Big) \tag{4.31}$$

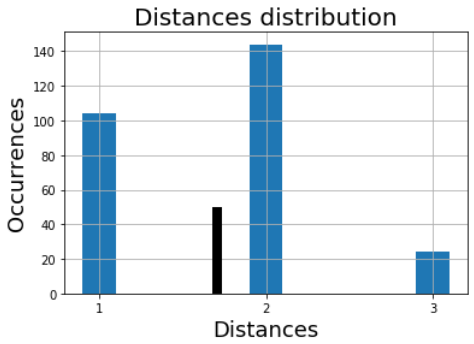


Figure 4.11: Example of the distribution of distances of an interaction network based on the signed network in figure 4.3a with $N = 20$ and $M_- = 8$. We used $\alpha = \beta = 1$, $\gamma = 0$ and $M_{pc} = 1$. The mean of the distribution is represented in black.

It is essentially the worst-case shortest path length, but it does not say anything about typical shortest path lengths. In the example of figure 4.11 $\text{diam}(G) = 3$.

The average shortest path length $\langle l \rangle$ (sometimes called average path length or simply APL) is therefore computed as the average distance between any pair of nodes:

$$\langle l \rangle := \frac{1}{N^2} \sum_{(i,j) \in V \times V} \text{dist}(i,j) \tag{4.32}$$

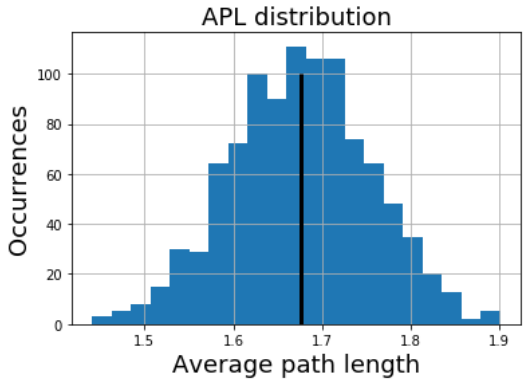
As we already said, the diameter can be large for some networks, even if many of the shortest path lengths are actually very small. The most important example of networks with this property is the small-world networks. For these cases, the average shortest path length is a better characterisation for the shortest path structures in a network. The average path length of the interaction network in the example above is represented in black in figure 4.11.

The other important characteristic, somehow related to the path lengths, is whether all network nodes are connected to each other via a path. In this case we say that the network is connected and $\text{dist}(i,j) < \infty \ \forall i,j \in V$. If a network is not connected we can always find a set of maximally connected subgraphs $G^{(i)} = (V^{(i)}, E^{(i)})$ with $V^{(i)} \subset V$ and $E^{(i)} \subset E$ called connected components, such that the union of all the connected components is equal to the starting network $G = \bigcup_i G^{(i)}$. For networks that are not connected, we are often interested in the largest connected component, i.e. the largest subset of nodes where all pairs of nodes are connected via paths. In these cases, we sometimes disregard those nodes and study the largest connected component as a network representation of the system in question.

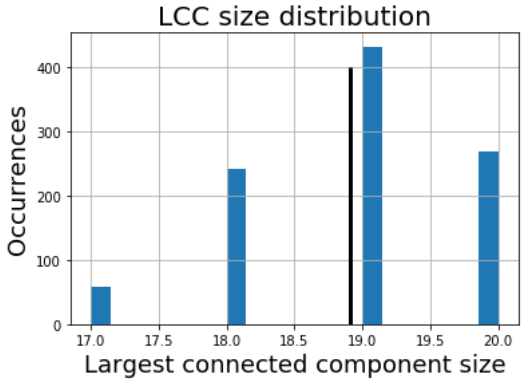
In the following computations, we will study the average path length of the interaction networks and, in the next section, how it behaves when changing the signed topology and the values of the parameters. As we saw before, the generative algorithm for the interactions can produce networks with disconnected nodes, leading to infinite values for some of the distances and, therefore, for the average path length. To avoid this possibility, we are going to calculate the APL value for the largest connected component of the system which, as we will see, sometimes can be rather small and for this reason we are going to study at the same time the distribution of the sizes of the LCCs

from different realisations of the interaction networks, in order to compare it with the distribution of the average path lengths.

In the following figures, the distributions of the average path length and the size of the largest connected component are shown for 1000 different realisations of interaction networks starting from the same signed network as before, with $N = 20$ and $M_- = 8$ and the same set of parameters $\alpha = \beta = 1,\ \gamma = 0$ and $M_{pc} = 1$.



(a) Distribution of the average path length.
In black the mean of the distribution $\langle\langle l \rangle\rangle = 1.675$.
The standard deviation is 0.079.



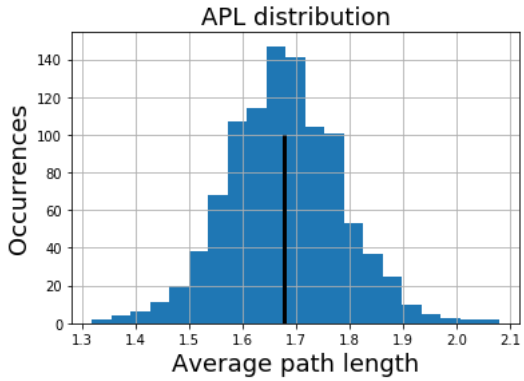
(b) Distribution of the size of the LCC.
In black the mean of the distribution $\langle N_{LCC} \rangle = 18.912$.
The standard deviation is 0.856.

Figure 4.12: Distribution of the APL and LCC size for 1000 different realisations of the same macrostate of interaction networks, defined by a fixed signed network, in figure 4.3a, with $N = 20,\ M_- = 8, \alpha = \beta = 1,\ \gamma = 0$ and $M_{pc} = 1$.

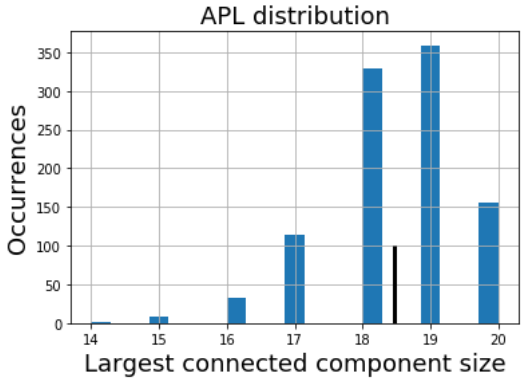
From the figure 4.12 above, we can see that the distribution for both the quantities is unimodal and rather narrow, in particular for the average path length, the distribution seems to assume a shape similar to a normal distribution with a ratio between the standard deviation and the mean of the order of less than 0.1%. Therefore, we can assign the value of the mean of such distributions to the macrostate that we are considering, i.e. the fixed signed network and the set of parameters that have been used.

By taking another step forward, we can compute the same kind of distribution but relax the constraint of the fixed signed network and study the realisations of the macrostate defined by the ensemble of signed networks with the same number of nodes N and negative links M_-. In this case, we computed the value of the average path length and the size of the largest connected component for an interaction network obtained from a random signed network generated by the algorithm $\mathrm{RSNG}(N, M_-)$, as explained in the algorithm 2, together with the usual set of parameters $N = 20,\ M_- = 8,\ \alpha = \beta = 1,\ \gamma = 0$ and $M_{pc} = 1$. In this way, the properties that we are going to compute will describe the macrostate defined only by this set of parameters and not by a specific signed topology. This will allow us to study the behaviour of some observables, for example, as a function of the number of negative edges, reducing the number of possible macrostates to study significantly. Assigning a value of an observable to a certain macrostate will be valuable only if the value efficiently estimate the distribution of the observables of the microstates that, since we saw that

RSNG(N, M_-) could generate networks with very different properties and energies, is not straightforward. To verify this we are going to compute directly this kind of distribution for the usual set of parameters $N = 20$, $M_- = 8$, $\alpha = \beta = 1$, $\gamma = 0$ and $M_{pc} = 1$, for a matter of consistency:



(a) Distribution of the average path length. In black the mean of the distribution $\langle\langle l\rangle\rangle = 1.678$. The standard deviation is 0.107.



(b) Distribution of the size of the LCC. In black the mean of the distribution $\langle N_{LCC}\rangle = 18.463$. The standard deviation is 1.05.

Figure 4.13: Distribution of the APL and LCC size for 1000 different realisations of the same macrostate of interaction networks, defined by defined by the ensemble RSNG(N, M_-) of signed network with $N = 20$, $M_- = 8, \alpha = \beta = 1$, $\gamma = 0$ and $M_{pc} = 1$.

As expected, the distribution of the average path length is wider because of the differences of the signed network between every realisation, but it still remains of the same order, and even more, the mean is basically unchanged. The distribution is still relatively narrow, with a ratio between the standard deviation and the mean lower than 0.1%. In addition, the distribution of the size of the largest connected component does not show major differences. In conclusion, we can say that every interaction network coming from any signed network with N nodes and M_- negative edges have, at random, the same properties.

Also in this case, the characterisation explained above is referred to the specific macrostate based on any signed network with a fixed number of nodes $N = 20$, negative edges $M_- = 8$ and the usual set of parameters. It can be shown that similar characteristics are present in almost all the possible realisations, referring to signed networks with a different number of nodes N, of edges M_-, or different sets of parameters. For now, we will assume this statement as true; that is, we are assuming the distribution of the observables unimodal and narrow, even for different definitions of the macrostate. In the following section 4.6 we will then verify this statement by exploring these quantities for different sets of parameters, included N and M_-.
As we already said, this result is important as it significantly reduces the number of possible macrostates to study and allows us to study the behaviour of some observables as a function of the number of negative edges.

4.5.3 Global clustering coefficient

Another important feature of the interaction network that we want to study is the property of clustering. The local clustering coefficient C_i is defined for a focal node $i \in V$, and it measures how connected are the neighbours of i. In fact, it counts how many links are between the neighbours of i, divided by the number of edges between the neighbours:

$$C_i = \frac{k(i)}{\binom{d_i}{2}} = \frac{2 \cdot k(i)}{d_i(d_i - 1)} \tag{4.33}$$

Where d_i is the degree of node i and $k(i)$ is the number of couples, neighbours of i, that have share a link between them and it is defined as:

$$k(i) := |\{(j,k) \in E : (i,j) \in E \wedge (i,k) \in E\}| \tag{4.34}$$

The local clustering coefficient basically measures the number of closed triads in which node i is included, normalised with the maximum number of triangles in which it can be included. It is clear that in a fully connected network, all the nodes in the system have a local clustering coefficient of exactly 1: $C_i = 1 \; \forall i \in V$. In the interaction networks, where the topology is less trivial, the distribution of the values of the local clustering coefficient is different. An estimator of this distribution is given by the global clustering coefficient, which is defined as the mean local clustering coefficient over all the nodes:

$$C = \frac{1}{N} \sum_{i \in V} C_i \tag{4.35}$$

The global clustering coefficient C can be seen as the expected local clustering coefficient of a node chosen at random and, from a statistical point of view, as a conditional probability which captures the probability of a link (j,k), given the condition that the two links (i,j) and (i,k) exist:

$$C = P[(j,k) \in E \mid (i,j) \in E, \; (i,k) \in E] \tag{4.36}$$

In the following figure 4.14 we can see the distribution of the local clustering coefficients for a specific realisation of the macrostate of the interaction networks defined by the usual fixed signed network in figure 4.3a and the usual set of parameters.

As we can see, the actual distribution is not too regular, but nevertheless, we can say that the mean value, i.e. the global clustering coefficient, is a good estimator for the denser part of the distribution, but there is still a portion of the distribution that corresponds to the nodes with zero clustering coefficient. Unlike the computation of the average path length, in this case, we do not consider the largest connected component but the interaction network in its entirety. Therefore, the pick at the zero of the distribution corresponds mainly to the disconnected nodes and the nodes in the periphery that are connected with only one or o few other nodes. As an example we can consider the node n_2 in in figure 4.6a, which is built with the same set of parameters, in particular $M_{pc} = 1$. It has to be said that among the different realisations of the same macrostate, the shape of the distributions can show some differences, as we can see by comparing the previous figure the following one 4.15, which share exactly the same set of parameters.

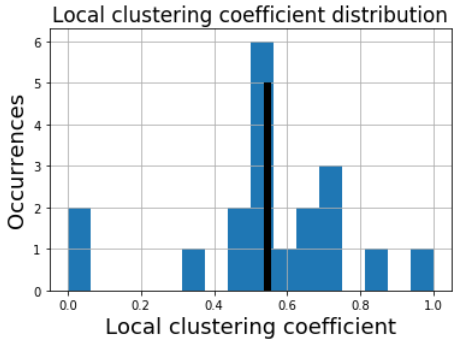


Figure 4.14: Distribution of the local clustering coefficient for a specific realisation of the macrostate of interaction networks defined by a fixed signed network, in figure 4.3a, with $N = 20,\ M_{-} = 8, \alpha = \beta = 1,\ \gamma = 0$ and $M_{pc} = 1$.

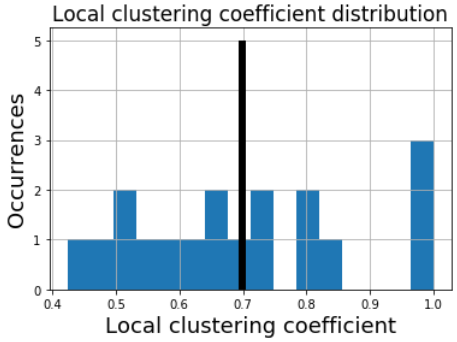


Figure 4.15: Distribution of the local clustering coefficient for a specific realisation of the macrostate of interaction networks defined by a fixed signed network, in figure 4.3a, with $N = 20,\ M_{-} = 8, \alpha = \beta = 1,\ \gamma = 0$ and $M_{pc} = 1$.

Nevertheless, the global clustering coefficient does not show huge differences. Therefore, we can proceed to compute the global clustering coefficient distribution for a large number of different realisations of the same macrostate. The results that confirm the previous statement are shown in the following figure 4.16.

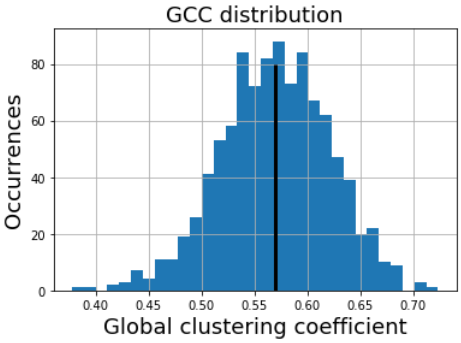


Figure 4.16: Distribution of the GCC for 1000 different realisations of the same macrostate of interaction networks, defined by a fixed signed network, in figure 4.3a, with $N = 20,\ M_{-} = 8, \alpha = \beta = 1,\ \gamma = 0$ and $M_{pc} = 1$. The mean of the distribution is shown in black and correspond to a value of 0.57, while the standard deviation is 0.052.

The distribution is, in fact, relatively narrow, as the ratio between the standard deviation and the value of the mean results of the order of 1%.

By following the same reasoning as the previous section 4.5.2 we can proceed to the computation of the distribution of the global clustering coefficient of various realisations of the macrostate of interaction networks defined, not by a fixed signed network, but by the ensemble of signed network with a fixed number of nodes and negative edges, i.e. by the RSNG(N, M_-). In order to be consistent with the previous calculations, we decided to fix $N = 20$ and $M_- = 8$ and perform 1000 different realisations with the usual set of parameters. The results can be seen in the following figure 4.17.

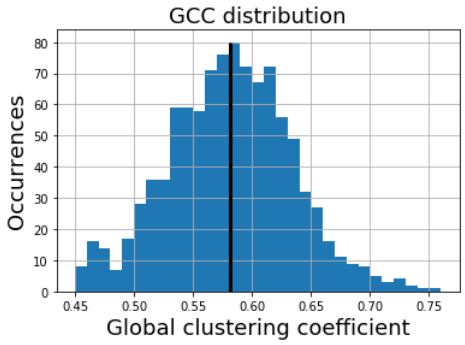


Figure 4.17: Distribution of the GCC for 1000 different realisations of the same macrostate of interaction networks, defined by the ensemble RSNG(N, M_-) of signed network with $N = 20$, $M_- = 8, \alpha = \beta = 1$, $\gamma = 0$ and $M_{pc} = 1$. The mean of the distribution is shown in black and correspond to a value of 0.58, while the standard deviation is 0.053.

As we can see, even if the shape of the distribution changed a bit, the mean value of the GCC and the standard deviation have remained basically unchanged, showing a relatively narrow shape and allowing us to study the bigger macrostate defined by RSNG(N, M_-) with the same procedure developed for the ensemble defined by a fixed signed network.

With these considerations, we can proceed to the study of these variables as a function of the number of the negative edges, as well as the parameters α, β, γ and M_{pc}.

4.5.4 Modularity

The last important concept that we introduced in order to study networks is the *modularity*, that can be defined as the quantity that measures the strength of division of a network into *communities* $C_i \subseteq V$ such that $\bigcup_i C_i = V$. In order to measure how well an undirected network G with N nodes, M links and adjacency matrix $\boldsymbol{A}$ can be divided into non overlapping communities, we have introduced the *partition quality* $Q(G, C)$ [4]:

$$Q(G, C) = \frac{1}{2M} \sum_{i,j} \left(A_{ij} - \frac{d_i d_j}{2M} \right) \delta_{c_i, c_j} \tag{4.37}$$

Where $C = \{C_1, ..., C_p\}$ is a non overlapping community partition of V and for a node i we define $c_i := k \iff i \in C_k$. For example c_i and c_j are equal if node i and j belong

to the same community, in this case δ_{c_i,c_j} in equation 4.37 would be 1. If we consider the contribution of a single edge (i,j) to Q: if $c_i = c_j$, the term $A_{ij} - \frac{d_i d_j}{2M}$ is positive if link (i,j) exists ($A_{ij} = 1$) and it is negative if the link does not exist ($A_{ij} = 0$). In other words: existing links between pairs of nodes in the same community increase Q, while missing links between nodes in the same C_i decrease Q. The resulting measure assumes values between a minimum of $Q = -0.5$ (partitions are contrary to "natural" partitions in the network) and a maximum of $Q = 1$ (partitions perfectly capture "natural" partitions in the network). For the special case of a fully connected network with N nodes without self-loops, all nodes are in a single community C_1 and the partition quality is 0.

We can define the optimal partition quality as:

$$Q_{opt}(G) := \max_{C} Q(G, C) \tag{4.38}$$

While Q is the property of both the network and the partition, Q_{opt} is the property of the network only and is called *modularity* of a network and can be computed by different kinds of optimisation algorithms.

We have seen that the modularity is sometimes normalised with the theoretical maximum of partition quality Q_{max} of network G, which is calculated pretending that all links are between nodes that are in the same community, i.e. $A_{ij} \implies \delta(c_i, c_j) = 1$:

$$Q_{max} = \frac{1}{2M}\Big(2M - \sum_{i,j} \delta(c_i, c_j)\frac{d_i d_j}{2M}\Big) \tag{4.39}$$

This leads to the definition of *community assortativity coefficient* that, given a community partition $\tilde{C}$ with optimal partition quality Q_{opt}, is defined as:

$$\frac{Q_{opt}(G)}{Q_{max}(G, \tilde{C})} \tag{4.40}$$

The community assortativity coefficient captures the nodes' preference to connect to nodes in the same community. In the following, for simplicity, we will use the term "modularity" to refer to the community assortativity coefficient as well.

Following the same procedure as the quantities above, we can compute the community assortativity coefficient for a large number of realisations of the macrostate defined by the usual fixed signed network and the usual set of parameters. As we can see in figure 4.18 the distribution is rather wide, but still well represented by the mean $\left\langle \frac{Q_{opt}(G)}{Q_{max}(G,\tilde{C})} \right\rangle$.
Same similar behaviour can be observed in figure 4.19, for the macrostate defined by the usual set of parameters and the set of random signed networks with $N = 10$ and $M_- = 8$.

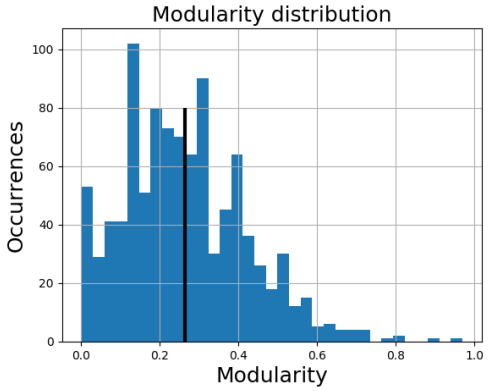


Figure 4.18: Distribution of the community assortativity coefficient (here called modularity for simplicity) for 1000 different realisations of the same macrostate of interaction networks, defined by a fixed signed network, in figure 4.3a, with $N = 20$, $M_- = 8, \alpha = \beta = 1$, $\gamma = 0$ and $M_{pc} = 1$. The mean of the distribution is shown in black and correspond to a value of 0.264, while the standard deviation is 0.157.

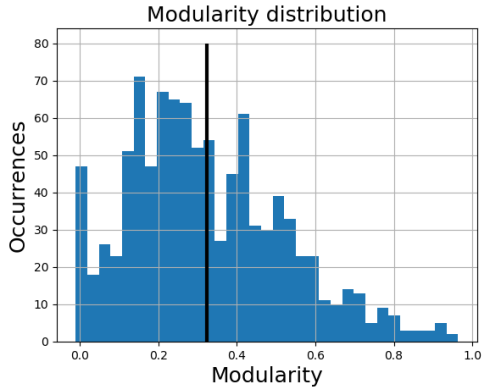


Figure 4.19: Distribution of the community assortativity coefficient (here called modularity for simplicity) for 1000 different realisations of the same macrostate of interaction networks, defined by the ensemble RSNG(N, M_-) of signed network with $N = 20$, $M_- = 8, \alpha = \beta = 1$, $\gamma = 0$ and $M_{pc} = 1$. The mean of the distribution is shown in black and correspond to a value of 0.324, while the standard deviation is 0.199.

4.6 Parameters exploration

In this section, we are going to study how the distributions of the observables that we introduced can change as a function of the parameters of the model, namely N, M_-, α, β, γ and M_{pc}. In the previous section, we found that by taking the average of the distribution of the observables of the interaction networks coming from different signed networks that belong to the same RSNG(N, M_-), it actually gives an accurate estimation of the distribution itself as, at least for some set of the parameters, the distributions are always unimodal, and the standard deviation is not too large. By taking this perspective, we decided to represent these distributions by means of these two numbers: the mean and the standard deviation for all the observables. We are therefore going to represent these two numbers as an error bar.

4.6.1 Analysis

The main parameter through which we are going to study the observables is the number of negative edges M_-. Most of the observables will be shown as a function of M_-, as parallelism with the plots of the energy H as a function of the number of negative edges of Chapter 3. We will study the behaviour of the observables for different values of parameters α, β, γ, N and M_{pc} individually by considering the other parameters fixed. Throughout the first part of this analysis, we will neglect the homophily term in H_{ij}, therefore considering $\gamma = 0$. In a second moment, we will study how the addition of the homophily term changes the observables computed before, by varying the parameter γ and the number of negative attributes.

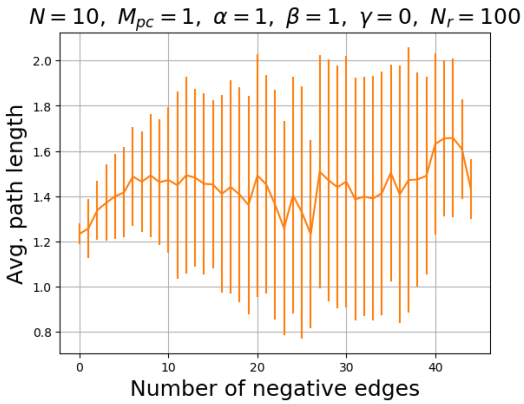
The observables studied in this part will be: the average path length (APL), the size of the largest connected component (LCC), the global clustering coefficient (GCC) and the community assortativity coefficient (sometimes simply called "modularity"). These qualities have been computed through the `python` functions included in the `pathpy` module, an Open Source Python package providing higher-order network analytics for time series data, as well as basic network analysis [1]. The community assortativity coefficient is computed by the `pathpy` function with an additional minus sign; for this reason, throughout all the analysis, the quantity reported as "modularity" correspond to $-\frac{Q_{opt}(G)}{Q_{max}(G)}$. Therefore, the global *minima* that are possibly shown in the figures actually correspond to the set of parameters that *maximise* the community assortativity coefficient.

The following figures 4.20 will give an example of this analysis, where the four observables are represented as a function of the number of negative edges, for the usual set of parameters $\alpha = \beta = 1$, $\gamma = 0$, $M_{pc} = 1$. For every value of M_-, the distribution is built by computing a number of different realisations N_r that in this case is 100. Due to the high computational cost of these simulations, for this first part of the discussion, we decided to study the properties of systems of smaller size $N = 10$.
In order to give an interpretation to these results, we have to remind the meaning of the parameters in H_{ij}. α, β and γ are the weights of the "social effects" H^p_{ij}, H^{SB}_{ij} and H^h_{ij}. The social effects are respectively: the preference for having positive relations, the preference for having common friends or enemies, and the preference for having interactions with people that are similar. The parameters measure the importance of these effects.

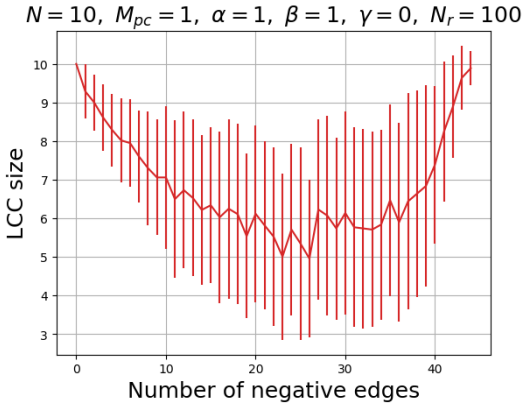
Let us take as an example the trend of the global clustering coefficient in figure 4.20c. Since in this case, we considered $\alpha = \beta = 1$ we can say that in a social system where the preference of having positive relations is as important as the preference of having common friends or enemies, a small number of negative relations in the system leads to a high value of clustering, an intermediate number of negative relations have a high probability of leading to a small value of clustering, while a very high number of negative relations can lead again to a high value of clustering. We have also to consider the fact that the value of $\alpha = \beta = 1$ is not too low (~ 0.1) nor too high (~ 10).

With the same reasoning, we can give an interpretation of the other quantities; for example, for this set of parameters, the mean of the average path length in figure 4.20a does not show any significant dependence on the number of negative edges, but

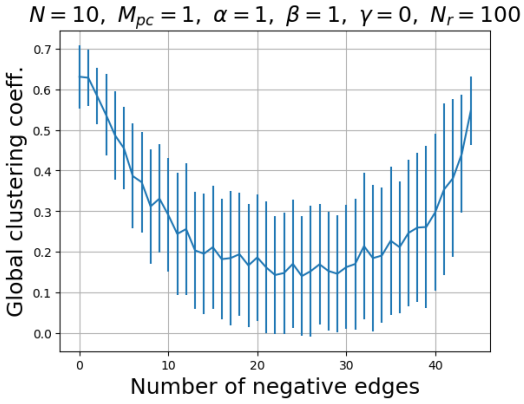
[1] https://www.pathpy.net



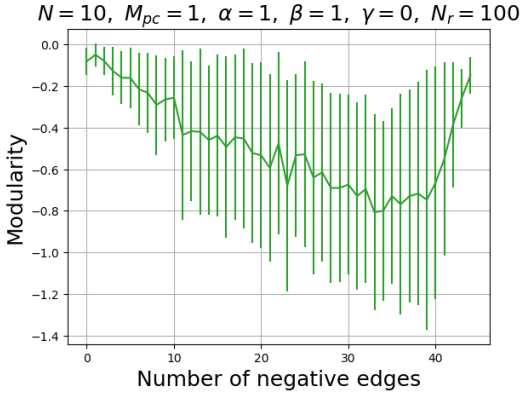
(a) Average path length of the LCC as a function of M_-.



(b) Size of the largest connected component as a function of M_-.



(c) Global clustering coefficient as a function of M_-.



(d) Community assortativity coefficient as a function of M_-.

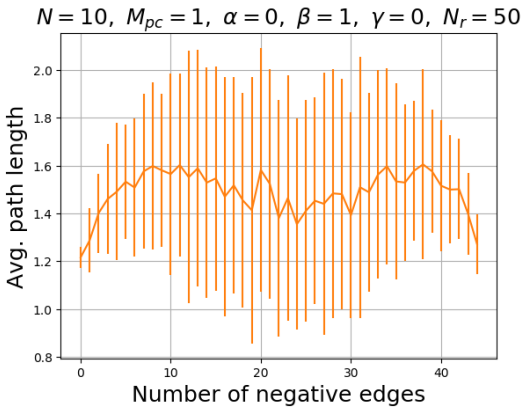
Figure 4.20: Representation of the four observables as a function of the number of negative edges M_-. Every point represents the mean of the distribution corresponding to that number of negative edges and the bar corresponds to its standard deviation. For every point the distribution is made of $N_r = 100$ different measurements of an interaction network based on the set of parameters $\alpha = \beta = 1,\ \gamma = 0,\ M_{pc} = 1$ and a different signed network generated from RSNG(N, M_-) with $N = 10$.

for intermediate values of M_- the standard deviation of the distribution increases, meaning that the APL can strongly fluctuate for those values. The size of the largest connected component follows a trend similar to the global clustering coefficient, which means that there is a correlation between the connectedness of the network and the property of clustering. We can also correlate the probability of a smaller size of LCC for intermediate numbers of M_- and the higher fluctuations of the APL in the same region. In fact, a weakly connected network can have a high value of APL, but a smaller size of the network can reduce it. These two opposite contributions probably generate the observed fluctuations in the APL. For the modularity, we can observe that for intermediate/high values of M_-, the modularity is relevant, this is probably due to the fact that the interaction networks are more weakly connected and the core is fragmented, which can lead to some sort of communities formation in that region. While for low values of M_-, the formation of a core, and therefore one only community, is highly probable. We can observe that both the GCC and the size of the LCC show some kind of symmetry respect $M_- \simeq M/2 \iff \Delta M \simeq 0$.

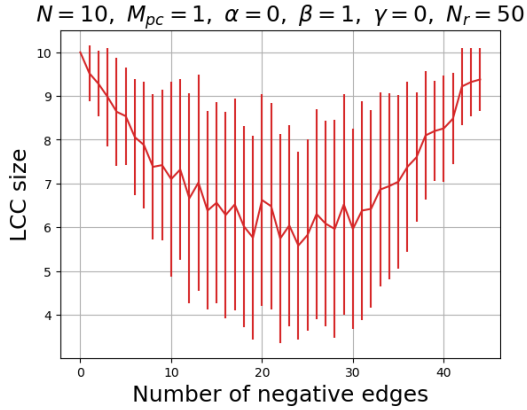
Analysis on α and β

In order to study this almost symmetrical behaviour we will show the same simulations as above for the two limit cases of $\alpha = 0, \beta = 1$ and $\alpha = 1, \beta = 0$. The limit $\alpha = 0, \beta = 1$ is shown in figure 4.21. When only the second term is considered, all the observables show the same symmetric behaviour. This is due to the fact that in the new version of H_{ij}, with the absolute value, the term becomes proportional to the square of the adjacency matrix, as $H_{ij}^{SB} = -\beta|A_{ij}|\sum_k A_{ik}A_{kj} = -\beta|A_{ij}|(A^2)_{ij}$ which, with similar considerations as before, can be seen as an even function of ΔM. From another perspective, we can say that when the number of M_- is high, also the probability of having common enemies is high and, since the term H_{ij}^{SB} does not distinguish between common enemies and common friends (more probable when M_- is low) the behaviours are the same in both the regions and the observables follow the same reasoning. The other limit $\alpha = 1,\ \beta = 0$ shows instead an asymmetric behaviour for all the observables, as we can see from figures 4.22.

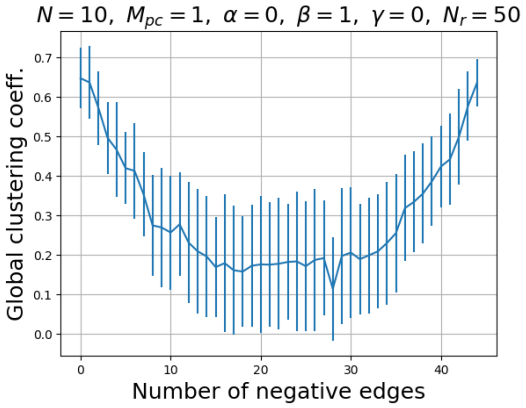
The behaviour of the observables as a function of M_-, for increasing values of α and $\beta = 1$ is shown in Appendix A in the figures A.1, A.2, A.3 and A.4. We can observe that the observables lose their property of symmetry, as expected. In particular, for high values of α and M_-, the APL shows a significant decrease related to the similar significant decrease of the size of the LCC. Another important feature that we can observe is a similar behaviour for the very end of the GCC. In fact we can see that from a value of $\alpha = 1$ and $\alpha = 2$ the GCC corresponding to very high values of M_- abruptly passes from a value of GCC $\simeq 0.5$ to 0. In order to study this behaviour, we can see in figure 4.23 the four observables as a function of the parameter α, for a few different numbers of negative edges, reported in the legend. As we can see from figure 4.23c, there exist a value of $\alpha \simeq 2$ after which for $M_- = 44$ the global clustering coefficient become exactly 0. Also for $M_- = 40$ we can observe a fast decrease. We can explain this behaviour by looking at the trends of the size of the LCC in figure 4.23b, in fact for the same values of M_- we can observe a similar behaviour with a critical α slightly higher. Since $M = 45$, considering $M_- = 44$ means that only one edge is positive, and, since the value of α is high, all the other edges generate a very low probability of



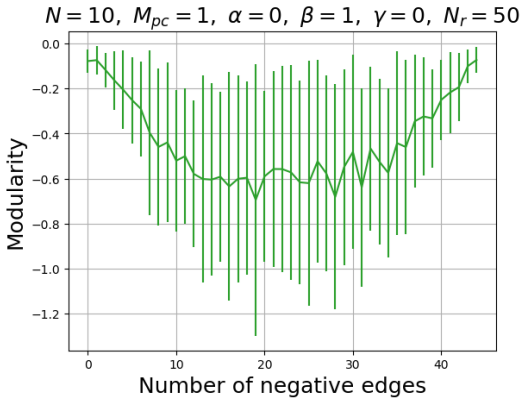
(a) Average path length of the LCC as a function of M_-.



(b) Size of the largest connected component as a function of M_-.

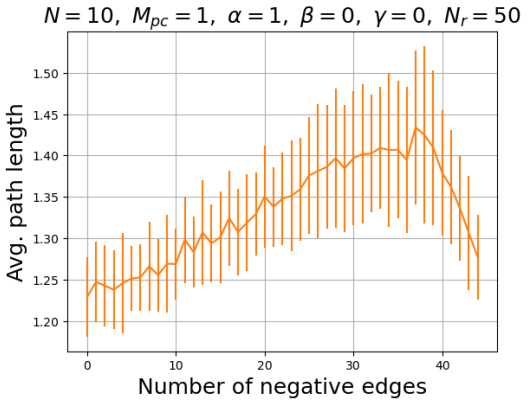


(c) Global clustering coefficient as a function of M_-.

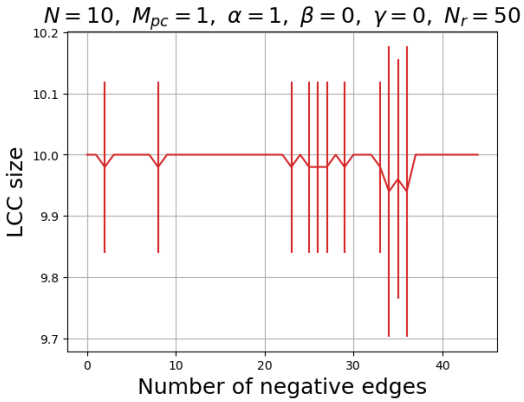


(d) Community assortativity coefficient as a function of M_-.

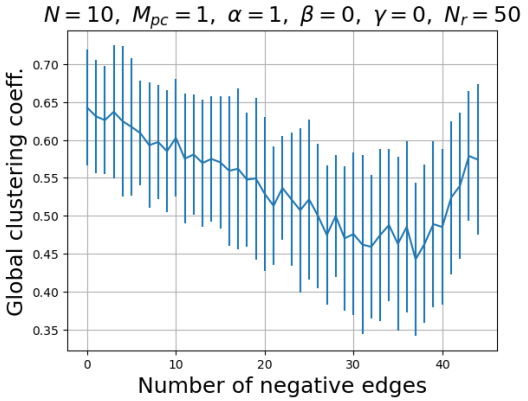
Figure 4.21: Representation of the four observables as a function of the number of negative edges M_-. Every point represents the mean of the distribution corresponding to that number of negative edges and the bar corresponds to its standard deviation. For every point the distribution is made of $N_r = 100$ different measurements of an interaction network based on the set of parameters $\alpha = 0,\ \beta = 1,\ \gamma = 0,\ M_{pc} = 1$ and a different signed network generated from RSNG(N, M_-) with $N = 10$.



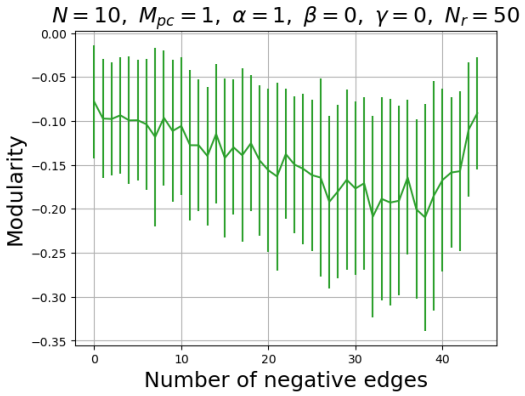
(a) Average path length of the LCC as a function of M_-.



(b) Size of the largest connected component as a function of M_-.



(c) Global clustering coefficient as a function of M_-.

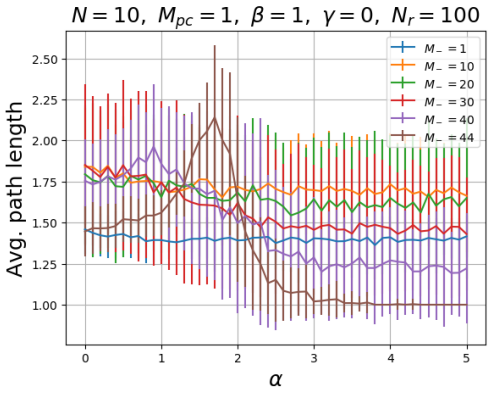


(d) Community assortativity coefficient as a function of M_-.

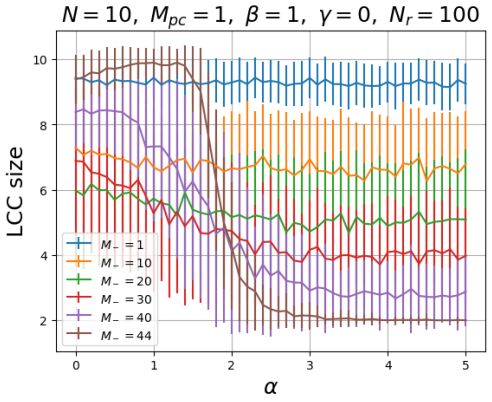
Figure 4.22: Representation of the four observables as a function of the number of negative edges M_-. Every point represents the mean of the distribution corresponding to that number of negative edges and the bar corresponds to its standard deviation. For every point the distribution is made of $N_r = 100$ different measurements of an interaction network based on the set of parameters $\alpha = 1,\ \beta = 0,\ \gamma = 0,\ M_{pc} = 1$ and a different signed network generated from RSNG(N, M_-) with $N = 10$.

interaction that reflects the formation of an interaction network of disconnected nodes, but two that are are highly connected. Therefore it follows the size 2 of the LCC and the very low values of the APL.

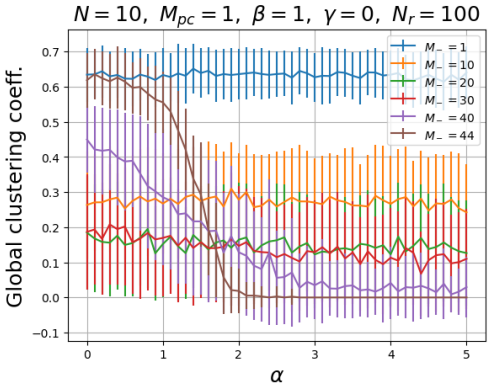
If instead we fix the value of α and study the same four variables for different β we obtain the results shown in figures A.5, A.6, A.7 and A.8, and the observables as a function of β, for few different values of M_- in figure A.9.



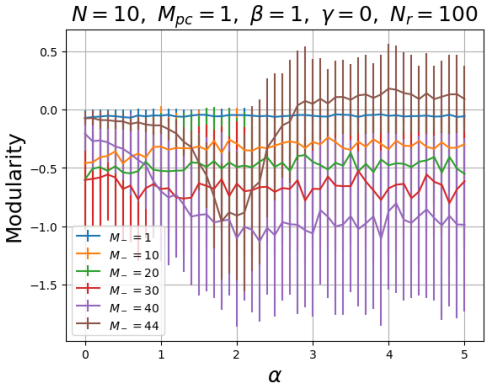
(a) Average path length of the LCC as a function of α.



(b) Size of the largest connected component as a function of α.



(c) Global clustering coefficient as a function of α.



(d) Community assortativity coefficient as a function of α.

Figure 4.23: Observables as a function of the parameter α. For every point the distribution is made of $N_r = 100$ different measurements of an interaction network based on the set of parameters $\beta = 1,\ \gamma = 0,\ M_{pc} = 1$ and a different signed network generated from RSNG(N, M_-) with $N = 10$ and different values of the number of negative edges M_- reported in the legend.

Analysis on γ

Until now we studied the interaction networks generated by using the local Hamiltonian without considering the homophily term $H^h_{ij} = -\gamma s_i |A_{ij}| s_j$. In order to add this contribution we have to define a binary scalar attribute $s_i \in \{+1, -1\}$ on each node $i \in V$. The way H^h_{ij} is defined makes the energy between two nodes increase or decrease by 2γ if $s_i = s_j$ or $s_i \neq s_j$ respectively. Let us consider a social system dominated by the homophily effect (considering $\alpha = \beta = 0$ or, equivalently, all the A_{ij} in the network to have the same exact value). We will observe the formation of a pair of clusters that have a higher probability of interaction thanks to their similarity, as we can see from the probability network in figure 4.24.

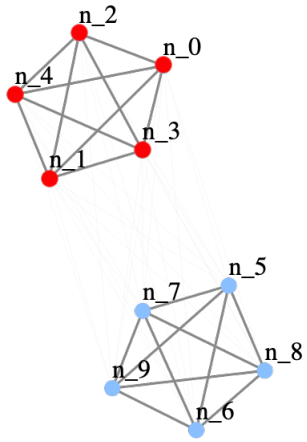


Figure 4.24: Probability network generated by considering a signed network with $N = 10$ without any negative edge. 5 nodes have positive attributes and 5 nodes have negative attributes, $\gamma = 1$. It is possible to observe also very weak connections between the two clusters, but negligible with respect to the connections inside.

If any of the social effects in H_{ij} is constant for every possible couple (i, j), its contribution can be factorised in p_{ij} and simplified with the normalisation Z in the denominator. As in the example above, let us consider each A_{ij} in the network to have the same exact value, i.e. they are all $+1$ or -1. In this case H^p_{ij} and H^{SB}_{ij} are the same for every possible couple (i, j), therefore:

$$p_{ij} = \frac{e^{-H^p_{ij} - H^{SB}_{ij}} \cdot e^{-H^p_{ij}}}{\sum_{i,j} e^{-H^p_{ij} - H^{SB}_{ij}} \cdot e^{-H^p_{ij}}} = \frac{e^{-H^p_{ij} - H^{SB}_{ij}} \cdot e^{-H^p_{ij}}}{e^{-H^p_{ij} - H^{SB}_{ij}} \cdot \sum_{i,j} e^{-H^p_{ij}}} = \frac{e^{-H^p_{ij}}}{\sum_{i,j} e^{-H^p_{ij}}} \tag{4.41}$$

Considering each A_{ij} in the network to have the same value or considering $\alpha = \beta = 0$ leads to the same conclusions. With the same reasoning, we can interpret the assumption $\gamma = 0$, kept until now, as the same as considering each node to have the same attribute.

We expect the modularity to show the most significant differences among the observables, but this is true only for high values of γ. As we can see in the figure 4.25, for $\gamma = 3$, the community assortativity coefficient assumes values close to -1, which means

that the networks generated are well divisible in two clusters, formed by the nodes with the same attribute.

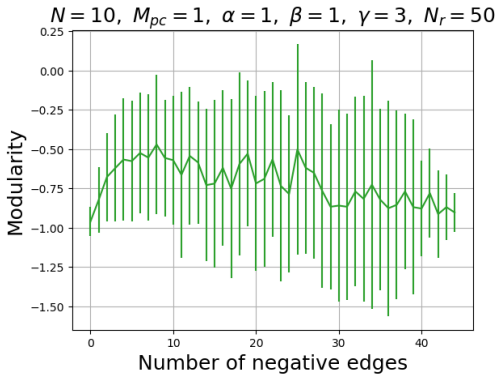
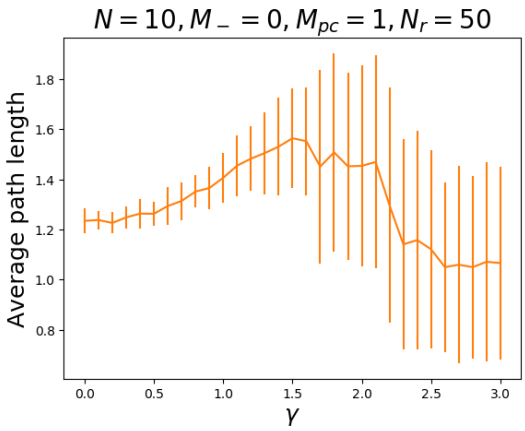


Figure 4.25: Community assortativity coefficient as a function of the number of negative edges M_-. For every point the distribution is made of $N_r = 50$ different measurements of an interaction network based on increasing values of $\alpha = \beta = 1,\ \gamma = 3,\ M_{pc} = 1$ and different signed networks generated from RSNG(N, M_-) with $N = 10$.

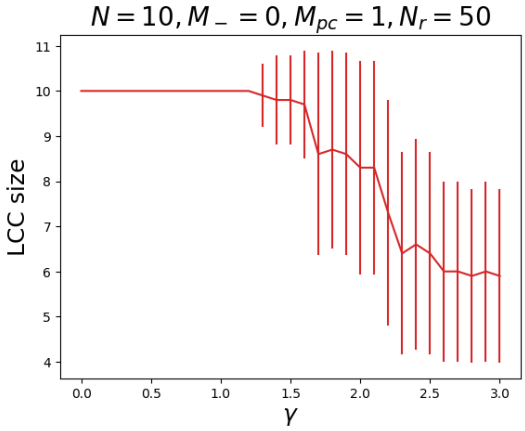
For lower values of γ, the uncertainty given by the random generation process and constraints given by the negative edges (and therefore by the contributions of the other two terms) make the modularity values not change significantly with respect to the case with $\gamma = 0$. While the figures of the observables as a function of the number of negative edges and the number of negative attributes (NNA) are reported in the in figures A.12d and A.13d, in figures 4.26 we can study the dependence on γ through the case with $M_- = 0$, which means we do not consider H_{ij}^p and H_{ij}^{SB}, and we further consider the most relevant configuration of attributes, i.e. the one that split the system into two equal subsets, in this case, NNA= 5.

For this configuration of the system, we can observe a transition from interaction networks characterised by a unique cluster of all the nodes, with a medium value of GCC, corresponding to low values of γ, to clustered interaction networks with a high clustering coefficient, for high values of γ. We can observe the transition also characterise the standard deviation of the size of the LCC, and therefore the APL. This transition happen around a critical value of $\gamma = \gamma^* \simeq 1.5$. Below this value, the topology is dominated by the randomness of the edges created; above it, the homophily term dominates. The behaviour of the observables for all the other values of M_- and γ are shown in figures A.10 and A.11.

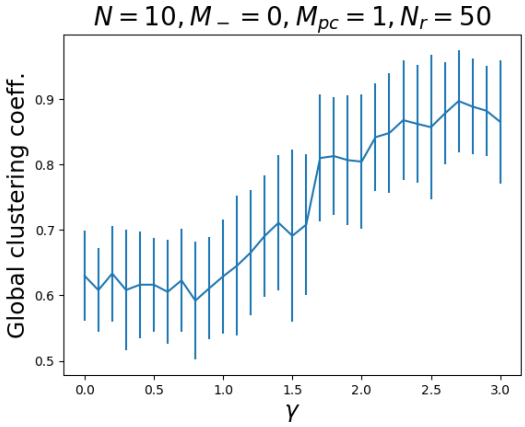
From the simulations presented here and reported in the Appendix A, we can state that the main property of the homophily term is to decrease the probability of interaction between nodes that have different attributes rather than increase the probability of interaction between two nodes with the same attribute.



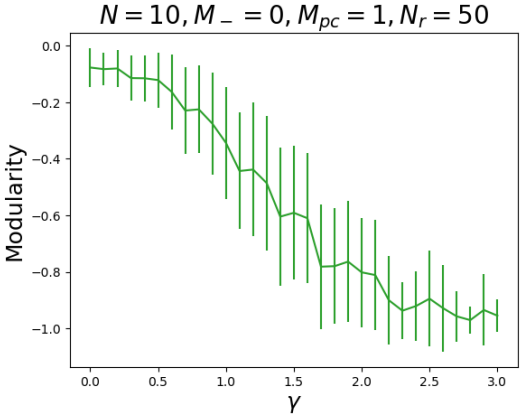
(a) Average path length of the LCC as a function of γ.



(b) Size of the largest connected component as a function of γ.



(c) Global clustering coefficient as a function of γ.

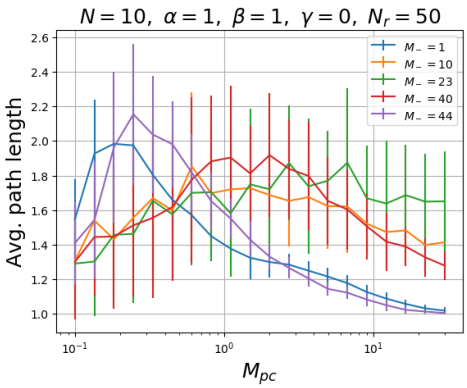


(d) Community assortativity coefficient as a function of γ.

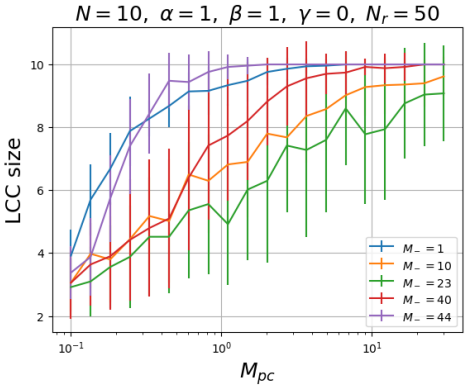
Figure 4.26: Representation of the four observables as a function of γ. Every point represents the mean of the distribution corresponding to that number of negative edges and the bar corresponds to its standard deviation. For every point the distribution is made of $N_r = 50$ different measurements of an interaction network based on the set of parameters $\alpha = \beta = 0$, $M_{pc} = 1$ and a signed network $N = 10$ with 5 negative attributes and 5 positive attributes.

Analysis on M_{pc}

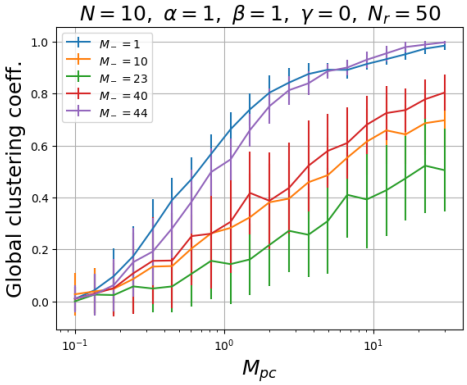
Another important variable through which we can study the observables is the number of interactions created M_c or, equivalently, $M_{pc} = \frac{M_c}{M}$. As we showed in section 4.4 and already verified in section 4.5.1, we expect the observables of the interaction networks to converge for $M_{pc} \to \infty$ to the ones corresponding to fully connected networks. As shown in figure 4.27, the expectations are fulfilled as the GCC tends to converge to 1, so as the APL, while the size of the LCC converges to $N = 10$ and the modularity to 0, since a fully connected network is considered a single community. For intermediate M_-, the convergence is slower.



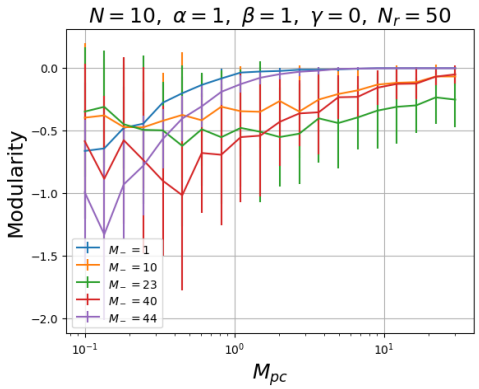
(a) Average path length of the LCC as a function of M_{pc}.



(b) Size of the largest connected component as a function of M_{pc}.



(c) Global clustering coefficient as a function of M_{pc}.

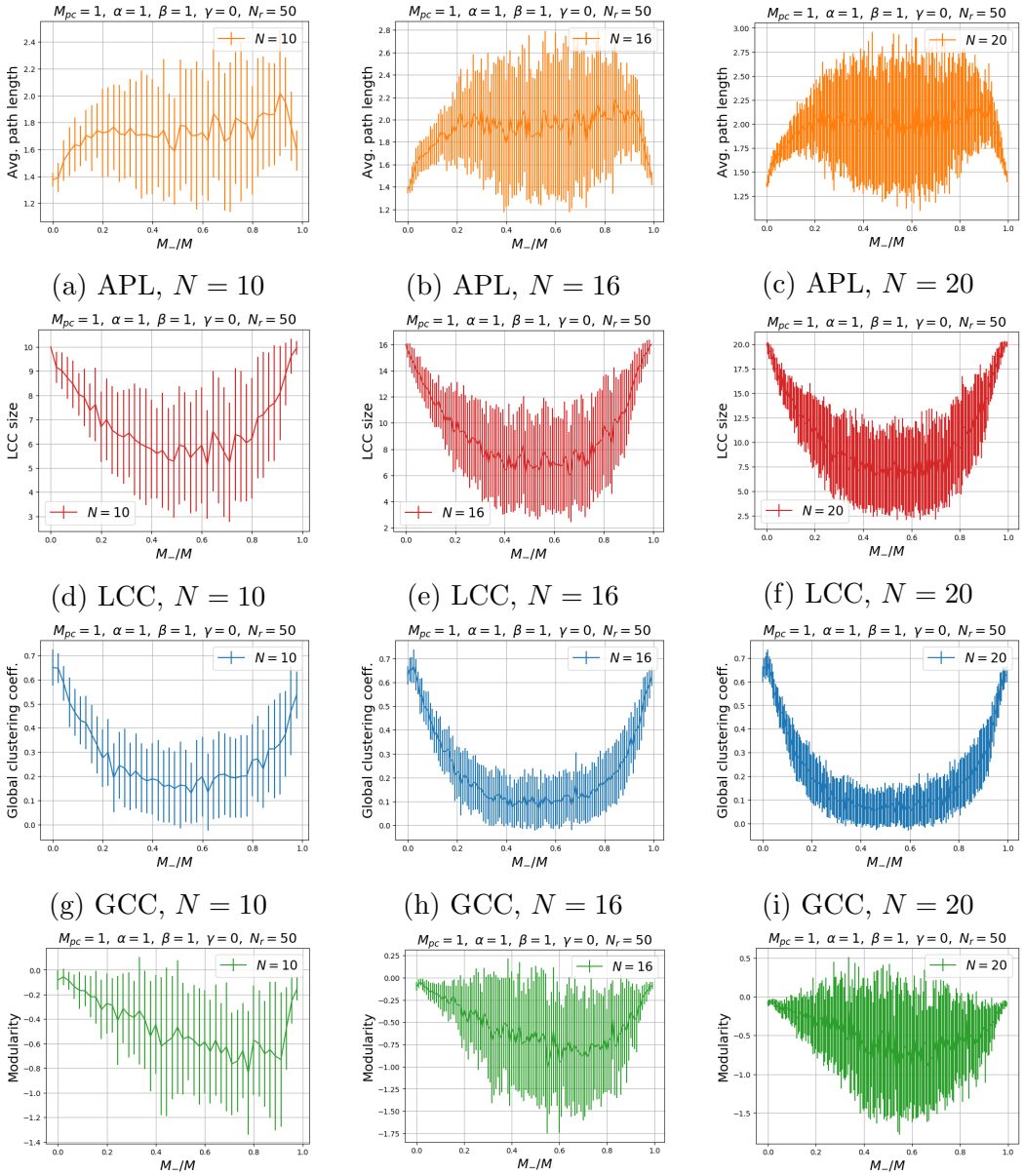


(d) Community assortativity coefficient as a function of M_{pc}.

Figure 4.27: Representation of the four observables as a function of M_{pc}. Every point represents the mean of the distribution corresponding to that number of negative edges and the bar corresponds to its standard deviation. For every point the distribution is made of $N_r = 50$ different measurements of an interaction network based on the set of parameters $\alpha = \beta = 1$, $\gamma = 0$ and a different signed network generated from RSNG(N, M_-) with $N = 10$ and different values of the number of negative edges M_- reported in the legend.

Analysis on N

The last parameter to be explored is the number N. It is the parameter that requires the largest computational time to be explored, as the number of edges increases as $\binom{N}{2} \sim N^2$. In order to study the behaviour of the system for different sizes N, we fixed the usual set of parameters and studied the observables as a function of M_- for different values of N. In order to compare the different functions, we had to normalise M_- with the number of total edges: M_-/M. We do not observe any substantial changes, but the standard deviation of the GCC becomes increasingly smaller.



(a) APL, $N = 10$ (b) APL, $N = 16$ (c) APL, $N = 20$

(d) LCC, $N = 10$ (e) LCC, $N = 16$ (f) LCC, $N = 20$

(g) GCC, $N = 10$ (h) GCC, $N = 16$ (i) GCC, $N = 20$

(j) Modularity, $N = 10$ (k) Modularity, $N = 16$ (l) Modularity, $N = 20$

Figure 4.28: Observables as a function of M_-/M. $N_r = 50$, $\alpha = \beta = 1$, $\gamma = 0$, $M_{pc} = 1$, from RSNG(N, M_-) with increasing values of $N = 10, 16, 20$.

4.6.2 Results

We saw that since H_{ij}^{SB} is symmetric with respect to $\boldsymbol{A} \to -\boldsymbol{A}$, the observables generated by this term only are symmetric with respect to $\Delta M \to -\Delta M$, while H_{ij}^{p} which have an odd parity can destroy this symmetry.

When considering the case with $\beta = 1, \gamma = 0, M_{pc} = 1$ and a high value of $M_- \simeq M$, the observables as a function of α seem to show a transition to a disconnected network, in the case of $M_- = M - 1$ and $N = 10$, the critical value of α is $\alpha^* \simeq 2$.

We showed that considering a system with a social effect constant for every edge is equal to discarding this social effect. This is due to the fact that its contribution factorises in the calculation of p_{ij} and simplifies.

We showed the results of the homophily effect in the system. More than increasing the probability of interaction between similar agents, its principal property is to decrease the interactions between agents with different attributes. Of course, this fact tends to produce clustered topologies and larger modularity. We also showed how the homophily effect has to compete with the randomness of the generation of interaction. Even discarding the other effects, H_{ij}^{h} becomes predominant only over a certain critical value of γ, which is probably dependent on the number of interactions created. In our case with $\alpha = \beta = 0, N = 10, M_- = 0, M_{pc} = 1$ and NNA $= 5$, this critical value is $\gamma^* \simeq 1.5$. Further studies on this behaviour are needed.

In general, we found that a distribution of H_{ij} uniform among all the edges leads to more connections and, therefore, high values of the size of the LCC and GCC, low values of APL and modularity. If the distribution is far from being uniform, the nodes result less connected on average. The presence of negative edges causes this effect, and it increases as the number of negative edges increases (apart from the case $M_- \simeq M$ when H_{ij}^{SB} is dominant). The effect is larger when the parameters α, β, γ become larger, decreasing as the number of interactions created M_c increases.

Lastly, an important characteristic of real-world networks, especially social networks, is the so-called small-world property, which means that the system shows a high degree of clustering (large GCC) compared to the value of the normalised APL, which is lower. In our case, the APL is almost never too large. This is also due to the fact that the system has a limited number of nodes. The GCC instead presents a large variability as the parameters change, most notably as M_- increase. As we can see from the figure below, the APL and GCC (normalised to one to be compared) shows a weak small-world regime only for small values of M_-: GCC$\simeq$ 1 and APL$\simeq$ 0.7. For other sets of parameters, the shapes change but not substantially.

Small values of M_- correspond to the cases where the signed network have low energies. We know that having a small number of negative edges is not the only condition that produces these low energy states, as we know the existence of the "war states". Since it is known that real social systems have a high structural balance and therefore low energy, we could say that configurations that are more probable to appear in the real world show a small-world regime: another known property of real social systems. In order to say that our model predicts real social networks to show a (mild) small-world regime, we have first to study the observables as a direct function of the energy. This will be done in the next section.

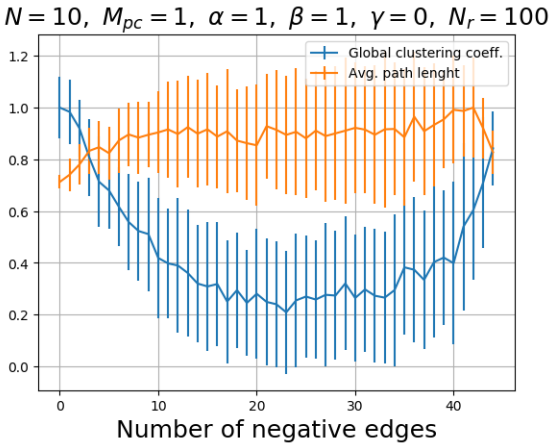


Figure 4.29: GCC and APL both normalised to the maximum of their values, as a function of M_-

4.7 Analysis and comparison with relation networks

In section 4.1 we noticed that the stability of social networks and the formation of social interactions are ruled by different, yet related, processes and assumptions. Despite this finding, it is still possible to make some connections between the observables of the interaction networks and the signed networks that generate them. One possible way it can be carried out is by computing for every realisation of the interaction network the value of the global Hamiltonian H and using it as the variable through which we represent the observables. In this way, each point no longer represents the macrostate generated by RSNG(N, M_-) and the parameters, but instead represents the macrostate of the interaction networks generated by signed networks that have the same value of H, i.e. the same "global energy", regardless the number of negative edges M_-.

The results can be seen in figures A.14 and show a different form of the observables than the case in which they are represented as a function of M_-, but the overall behaviour remains the same. This is due to the fact that the mean of the energies H is a cubic function of the number of negative edges and, in the limit of $N \to \infty$, it is a monotonic function of M_-. Therefore, we can observe a difference in the shape but not in the trend of the observables since the new variable is an almost monotonic function of the previous variable. For the example in figures A.14 we used the usual set of parameters $\alpha = \beta = 1,\ \gamma = 0,\ M_{pc} = 1$ and $N = 10$. Although $N = 10$ is not a large size, as we saw in section 3.4.2, it generates a mean of the energies that can be considered sufficiently monotonic as a function of M_-. Since the overall behaviour of the observables does not change between the two representations, we will present only the figures regarding this set of parameters.

In the previous section, we studied the small-world regime as a function of the negative edges. With the approach presented in this section, we can also study whether this property is present when considering the energy of the signed networks that generate the interactions, instead of M_- as a variable.

As we can see from the figure 4.30, the small-world regime is still present for low energies, and it also seems to be even more pronounced: the minimum energy the value

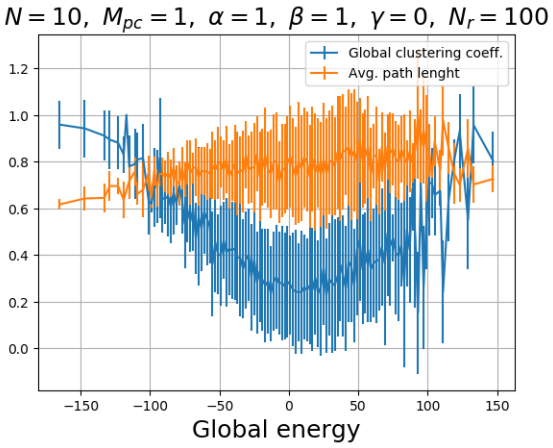


Figure 4.30: GCC and APL both normalised to the maximum of their values, as a function of the energy of the signed networks

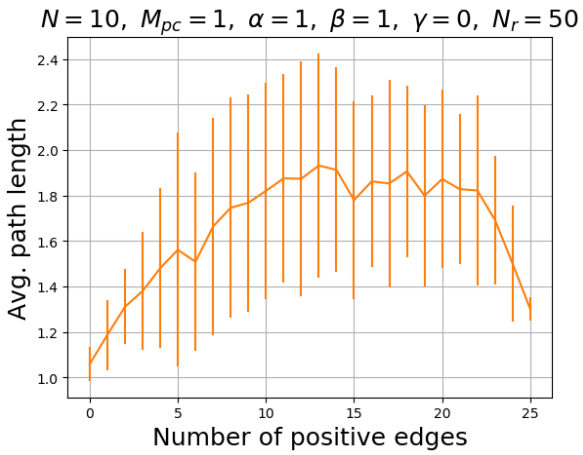
of GCC is more than 0.9, and the APL is approximately 0.6, while in the previous case GCC$\simeq$ 1, but APL$\simeq$ 0.7. This behaviour could be explained with the argument above, i.e. that lower energies correspond to lower M_{-}, but it could also confirm our hypothesis that the models predict low energy systems to generate interaction networks with small-world properties.

In order to verify this hypothesis, we will consider the cases where small energy values are not associated with a small number of negative edges.

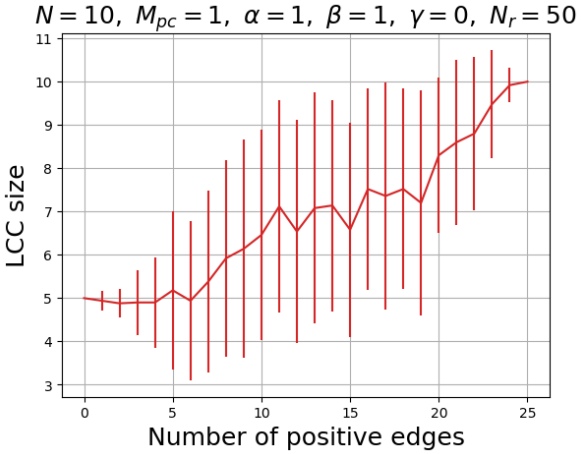
Thanks to the studies of Cartwright and Harary, we know that the global minima for the structural balance term of the global Hamiltonian H correspond to the states where there are two cliques of friends that are enemies between each other, the so-called "war states". Therefore, we can think to start with such a state and progressively switch the negative edges in between from negative to positive, in a way that from a "war state", we end up with a "paradise state" of all positive relations. In order to recreate such dynamics, we made use of a modified version of the stochastic block model, which is a model that generates clustered networks.

In the following figures 4.31 the four usual observables are represented as a function of the number of **positive** edges that are inserted in between two factions of 5 agents each, that at the beginning have only negative links between them (and only positive ones among the agents in factions).

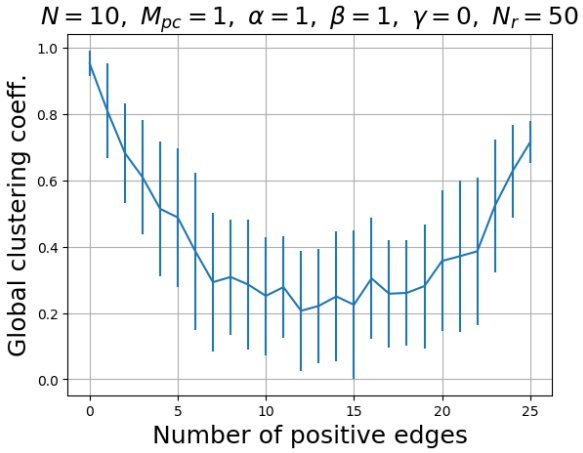
Even without comparing the two observables in the same figure, the results in figures 4.31 are explanatory enough. We can clearly see how the APL is at its minimum, and the GCC is at its maximum for values near the origin. This is clear evidence of the small-world regime for this kind of system. Also, this finding could verify our hypothesis that the model presented in this thesis can produce systems that fulfil the condition of low energy, expected by real-world social systems, and produce networks of interaction that presents the same small-world properties expected by real networks.



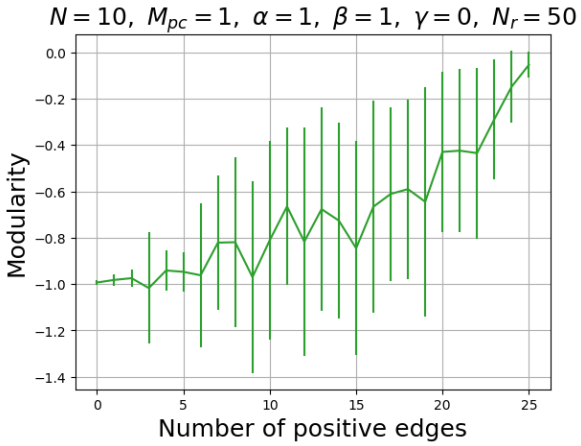
(a) Average path length of the LCC.



(b) Size of the largest connected component.



(c) Global clustering coefficient.



(d) Community assortativity coefficient.

Figure 4.31: Representation of the four observables as a function of the number of **positive** edges between two factions of 5 agents each.

Chapter 5

Possible applications and extensions

In this Chapter, we will discuss a possible description for the dynamics of signed networks, starting from the definition 3.3 of the global Hamiltonian, then we will introduce the method of the Exponential Random Graphs and show a possible application to the interaction networks, with the purpose of obtaining the values of the parameters for a given social system and, possibly, its network of relations.

5.1 Dynamics of signed networks

In real social networks, the presence of uncomfortable situations can lead to the change of the sign of the relations or their end. Since throughout all the present work, we considered only systems characterised by all-to-all connections, therefore represented by fully connected networks, in order to keep the coherence, we will consider only the former of the two possible consequences. From our point of view, the presence of uncomfortable situations can be measured by the global Hamiltonian H, and the dynamics of the relations that it generates can be seen as a relaxation towards lower energy configurations. We assume the dynamics of the relations to have different, larger, characteristic time than the time scale of the interactions, which we can therefore consider independent.

The aim of this section is, therefore, to start from the global Hamiltonian defined in equation 3.3 and provide a procedure that allows us to study the dynamics of a signed network in an analytical way. We will define a Lagrangian function $\mathcal{L}$ and some suitable generalised coordinates q, such that we can obtain the equations of motion of the signed network dynamics through the Euler–Lagrange equations:

$$\frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{q}} = \frac{\partial \mathcal{L}}{\partial q} \tag{5.1}$$

Where t is the time. Our approach is to consider $q = A_{ij}$ as generalised coordinates, but in order to do so we first have to give a definition of $\dot{A}_{ij} = \dfrac{dA_{ij}}{dt}$ and the derivative of a function respect to A_{ij}: $\dfrac{d}{dA_{ij}} f(A_{ij})$. For this purpose we need to relax the constraint of a binary value for $A_{ij} \in \{+1, -1\}$ and let it assume all the possible values in $\mathbb{R}$, considering therefore a weighted adjacency matrix, that can assume also negative values.

With the assumption $A_{ij} \in \mathbb{R}$ we can proceed to consider:

$$\dot{A}_{ij} = \frac{dA_{ij}}{dt} = \lim_{\Delta t \to 0} \frac{A_{ij}(t + \Delta t) - A_{ij}(t)}{\Delta t} \tag{5.2}$$

And also assume the standard differentiation rules to apply to $\frac{d}{dA_{ij}} f(A_{ij})$ and $\frac{d}{d\dot{A}_{ij}} f(\dot{A}_{ij})$.
We will, for example, consider:

$$\frac{dA_{kl}}{dA_{ij}} = \delta_{ik}\delta_{jl} = \begin{cases} 1 & \text{if } i = k \text{ and } j = l \\ 0 & \text{otherwise} \end{cases} \tag{5.3}$$

In this part, we will focus only on the part of H that depends only on the signed network, and therefore on A_{ij}, neglecting the homophily term H_h and considering only $H = H_p + H_{SB}$:

$$H = -\frac{\alpha}{2} \sum_{i,j} A_{ij} - \frac{\beta}{6} \sum_{i,j,k} A_{ij} A_{jk} A_{ki} \tag{5.4}$$

As we studied in Chapter 3, H shows multiple local minima that correspond to configurations that we assume to be more probable to be observed in real social systems. Therefore, we can give H an interpretation of potential energy E_p where the local minima correspond to the desired dynamics' possible final states.

In order to build a Lagrangian function capable of describing the dynamics, we need to include in $\mathcal{L}$ a term dependent on the generalised velocities $\dot{A}_{ij}$ that we interpret naively as corresponding to a "kinetic energy". Relying on this parallelism, we introduce a term analogous to the non-relativistic kinetic energy of a moving particle:

$$E_k = \frac{1}{2} \sum_{i,j} M_{ij} \dot{A}_{ij} \dot{A}_{ij} \tag{5.5}$$

Where we also consider an analogous of the mass that, in principle, depends on the link (i, j) and represent the inertia of the edge with respect to the change in its value. We can therefore introduce the Lagrangian $\mathcal{L} = E_k - E_p$ as follows:

$$\mathcal{L} = \frac{1}{2} \sum_{i,j} M_{ij} \dot{A}_{ij} \dot{A}_{ij} + \frac{\alpha}{2} \sum_{i,j} A_{ij} + \frac{\beta}{6} \sum_{i,j,k} A_{ij} A_{jk} A_{ik} \tag{5.6}$$

Starting from this we could use the Euler-Lagrange equation to derive the analogous of the equation of motion for the adjacency matrix, by using the standard differentiation rules, in particular 5.2:

$$\frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{A}_{ij}} = \frac{\partial \mathcal{L}}{\partial A_{ij}} \tag{5.7}$$

$$M_{ij} \ddot{A}_{ij} = \alpha + \beta \sum_{k} A_{ik} A_{kj} \tag{5.8}$$

This is a set of $N \cdot N$ second order ordinary and coupled differential equations.
The stationary solutions are

$$0 = \alpha + \beta \sum_{k} A_{ik} A_{kj} \tag{5.9}$$

Which can be rewritten as:

$$\sum_k A_{ik}A_{kj} = (A^2)_{ij} = -\frac{\alpha}{\beta} \tag{5.10}$$

They do not give us any interesting insight, as multiple configurations of A_{ij} can give the same $(A^2)_{ij}$. With the previous perspective of binary relations, this value could have been interpreted as the difference between the number of agents that are common friends or enemies and the ones who are not, but we have to remember that in this case, $A_{ij} \in \mathbb{R}$ does not allow such interpretation.

The equation 5.8 can be solved numerically by dividing them in two sets of first order ODEs and define two initial conditions $\boldsymbol{A} = \boldsymbol{A^*}$ and $\frac{d\boldsymbol{A}}{dt} = \boldsymbol{B^*}$:

$$\begin{cases} \dot{B}_{ij} = \alpha + \beta \sum_k A_{ik}A_{kj} \\ \dot{A}_{ij} = B_{ij} \cdot M_{ij}^{-1} \end{cases} \tag{5.11}$$

Unfortunately, preliminary results of this solutions show some problems with this approach, in fact:

- The solutions quickly diverge if not properly normalised every step
- Once normalized they approach stationary solutions which are real values **between** -1 and 1
- The solutions also show non-zero diagonal elements

These problems arise from the summation that is not limited to one and can be different from zero when $i = j$. Although this formulation does not seem to lead to any meaningful results, it would be interesting to study the possibility of a similar formulation to give the dynamics an analytical structure.

Another approach to the study of dynamics is given by numerical Montecarlo simulation. Such simulations have already been explored for similar systems, but mainly considering only the structural balance term 3.17. It would be interesting to study, in further studies, the final states of such simulations following the approach presented in our model.

5.2 ERGMs

In this section, we will use the definitions of the previous chapters in order to introduce a procedure that would allow us to study real social networks by inferring the importance of the social effects within them and, possibly, the network of their relations starting from the network of the observed interactions. As we saw in section 4.3, we can write the probability of an interaction network as:

$$P(G) = \frac{M_c!}{\prod_{i>j} m_{ij}^*!} \prod_{i>j} p_{ij}^{m_{ij}^*} \tag{5.12}$$

Where $m^*_{ij} = G_{ij}$ is the number of interactions generated by the algorithm between nodes i and j, and p_{ij} the probability of interaction:

$$p_{ij} = \frac{\exp\left(-H_{ij}\right)}{\sum_{ij}\exp\left(-H_{ij}\right)} = \frac{1}{Z}\exp\left(-H_{ij}\right) \tag{5.13}$$

With H_{ij} the local Hamiltonian:

$$H_{ij} = -\alpha A_{ij} - \beta|A_{ij}|\sum_k A_{ik}A_{kj} - \gamma s_i|A_{ij}|s_j \tag{5.14}$$

We can write the the final formula of the probability as:

$$P(G) = \frac{M_c!}{\prod_{i>j} m^*_{ij}!}\frac{1}{Z^{M_c}}\exp\left(\alpha f_1(\boldsymbol{G},\boldsymbol{A}) + \beta f_2(\boldsymbol{G},\boldsymbol{A}) + \gamma f_3(\boldsymbol{G},\boldsymbol{A},\boldsymbol{s})\right) \tag{5.15}$$

Where we defined:

$$\begin{cases} f_1(\boldsymbol{G},\boldsymbol{A}) = \sum_{ij} m^*_{ij}H^p_{ij} = -\sum_{ij} m^*_{ij}A_{ij} \\ f_2(\boldsymbol{G},\boldsymbol{A}) = \sum_{ij} m^*_{ij}H^{SB}_{ij} = -\sum_{ij} m^*_{ij}|A_{ij}|\sum_k A_{ik}A_{kj} \\ f_3(\boldsymbol{G},\boldsymbol{A},\boldsymbol{s}) = \sum_{ij} m^*_{ij}H^h_{ij} = -\sum_{ij} m^*_{ij}s_i|A_{ij}|s_j \end{cases} \tag{5.16}$$

Where $\boldsymbol{G}$ is the adjacency matrix of the interaction network and $G_{ij} = m^*_{ij}$. $\boldsymbol{A}$ is the usual adjacency matrix of the signed network and $\boldsymbol{s}$ the vector of the agents attributes.

This description fits naturally the framework given by the method of the Exponential Random Graph Models (ERGMs), first introduced in [25] (1996). An ERGM is completely defined by the conditional probability function:

$$P(G|\vec{\theta}) = \frac{1}{Z(\vec{\theta},\vec{f})}e^{-\sum_i \theta_i f_i(G)} \tag{5.17}$$

Where G is the empirical network that we want to study, $\vec{\theta} = \{\theta_i\}$ are the parameters that measure the importance of the effects (or statistics) $\vec{f} = \{f_i\}$ of the network G. Z is the normalisation calculated as the partition function:

$$Z(\vec{\theta},\vec{f}) = \sum_{G'\in\mathcal{G}} e^{-\sum_i \theta_i f_i(G')} \tag{5.18}$$

Where $\mathcal{G}$ represents the set of all the possible networks with the same number of nodes as G.

In our case the parameters θ_i are three and correspond to the parameters of the local Hamiltonian $\vec{\theta} = (\theta_1,\theta_2,\theta_3) = (\alpha,\beta,\gamma)$. The effects f_i correspond exactly to the terms defined in 5.16.

ERGMs come from the Maximum Entropy Principle where the entropy:

$$S := -\sum_G P(G)\cdot\log(P(G)) \tag{5.19}$$

is maximised in correspondence of the probability that best represents the given state of knowledge of the system. The effects f_i can be seen as the constraints of the maximisation carried on through the parameters θ_i that are therefore considered Lagrangian multipliers.

At this point, we need to estimate the parameters $\Theta_i = \theta_i$ that best fit the probability $P(G)$ of the observed empirical network. Due to the complexity of the partition function Z calculations, the analytical solution of this problem is only available for few network models, such as the $G(n, p)$ model. For an increasing number of properties f_i, in this case three, the calculations become quickly intractable, as there are no ways to simplify Z. Nevertheless a possible numerical solution is to resort to complex *maximum likelihood estimation* (MLE) algorithms, given for example by the *Markov Chain Monte-Carlo* sampling (MCMC). In fact, by generating a *random walk process* through the possible realisations G_i of networks with the same number of nodes as G, it is possible to give an estimation of $Z(\vec{\theta}, \vec{f})$ and therefore $P(G)$. The Metropolis-Hastings algorithm can implement the process that allows us to sample from an arbitrary distribution without the need to compute Z explicitly.

The last step is to determine the best parameters Θ_i. As said before, the aim is to find the parameters Θ_i that maximise the likelihood function $\mathcal{L}(\vec{\Theta}|G)$ which is basically calculated as the probability $P(G)$ as a function of the parameters θ_i. In our case:

$$\mathcal{L}(\alpha, \beta, \gamma | \boldsymbol{G}, \boldsymbol{A}, \boldsymbol{s}) = \frac{M_c!}{\prod_{i>j} m^*_{ij}!} \frac{1}{Z^{M_c}} \exp\left(\alpha f_1(\boldsymbol{G}, \boldsymbol{A}) + \beta f_2(\boldsymbol{G}, \boldsymbol{A}) + \gamma f_3(\boldsymbol{G}, \boldsymbol{A}, \boldsymbol{s})\right) \quad (5.20)$$

The idea is to combine the MCMC sampling algorithm, to calculate $\mathcal{L}(\vec{\Theta}|G) = P(G)$ with MLE, in the so-called MCMCMLE algorithm:

Algorithm 5: MCMCMLE algorithm

1. Choose initial values $\vec{\Theta}^{(0)}$
2. Compute $\mathcal{L}(\vec{\Theta}^{(0)|G})$ by generating enough microstates to compute Z.
3. Define $\vec{\Theta}^{(1)} = \vec{\Theta}^{(0)} + \delta$ with $\delta \in N(0, \sigma)$
4. Compute $\mathcal{L}(\vec{\Theta}^{(1)|G})$ by generating enough microstates to compute Z.
5. If $\mathcal{L}(\vec{\Theta}^{(1)|G}) \geq \mathcal{L}(\vec{\Theta}^{(0)|G})$ choose $\vec{\Theta}^{(1)}$ and set $\vec{\Theta}^{(0)} = \vec{\Theta}^{(1)}$, else ignore $\vec{\Theta}^{(1)}$
6. Repeat steps 2-5 until convergence

The algorithm above is just a sketch of the idea as more complexity is involved in the estimation process. The Markov Chain Monte-Carlo Maximum Likelihood Estimation process suffers from many limitations, in particular, since the drawing process produces correlated realisations, the algorithm can be numerically unstable. The convergence can be very slow for large networks, where many realisations are needed, but once it is reached, the algorithm gives as output an estimation of the desired parameters, that in our case are α, β and γ.

One last interesting further step would be to consider the network of relation $\boldsymbol{A}$ as a set parameter to infer with the same reasoning as above. The new likelihood function $\mathcal{L}(\alpha, \beta, \gamma, \boldsymbol{A} | \boldsymbol{G}, \boldsymbol{s})$ would have $\binom{N}{2}$ more parameters to explore, but the difference between α, β and γ is that A_{ij} are not real numbers but a binary scalar value. This means

that the computational time would increase, but an extension of the MCMCMLE algorithm should be possibly efficient. It would be possible, in fact, to start with a fixed $\boldsymbol{A^{(0)}}$ and compute the maximum likelihood through the MCMCMLE algorithm, then change one sign in $\boldsymbol{A^{(0)}} \rightarrow \boldsymbol{A^{(1)}}$, compute $\mathcal{L}$ again and accept the $\boldsymbol{A^{(1)}}$ if it increases the likelihood. The algorithm could continue until the maximum likelihood is found.

This algorithm could be computationally very expensive, as in principle there are 2^M different combinations of $\boldsymbol{A}$, so it could be necessary to compute the MMCMCMLE algorithm above (already rather expensive) a number of times that increases exponentially with the size of the network, but nevertheless it would be possible for relatively small social systems.

Chapter 6

Conclusions

The focus of this thesis was to study frustrated random networks and propose a model for generating interactions starting from them. The model was intended to characterise networks of relations within social systems and to reproduce interaction patterns between individuals of such systems.

The thesis is divided into four parts. The first part is dedicated to introducing the necessary definitions and concepts from network theory, geometrical frustration in physics and balance theory in social science. As highlighted by the formalism introduced by Cartwright and Harary [5], signed networks represent a powerful tool for the study of social systems, in particular when considering the network of relations within them. Moreover, Heider's approach to structural balance [12] allows us to create a connection with the theory of frustrated systems [22], used for example in the study of spin glasses.

In the second part of the thesis, we have explained the meaning of taking a statistical physics perspective to describe social signed networks quantitatively. We have introduced a utility function that measures the" cost" of a given configuration of relations and interpreted it as an energy or a pseudo-Hamiltonian. We have presented some of the previous attempts that tried the same approach by taking into account the theory of structural balance [9] [15]. We have then introduced our version of such pseudo-Hamiltonian 3.3 that quantify the "energy" of a given social system represented by a fully connected signed network. This quantity takes into account three main social effects: the preference of having positive relations, the preference of being included in balanced triadic relations and the preference to interact with individuals with similar characteristics. Precisely, we wrote:

$$H = -\frac{\alpha}{2}\sum_{i,j} A_{ij} - \frac{\beta}{6}\sum_{i,j,k} A_{ij}A_{jk}A_{ki} - \frac{\gamma}{2}\sum_{i,j} s_i|A_{ij}|s_j \tag{6.1}$$

where $A_{ij} \in \{+1, -1\}$ are the entries of the adjacency matrix of the signed network, representing the valence of the relation between i and j, while $s_i \in \{+1, -1\}$ are the attributes assigned to each agent in the system. We have studied the properties of each term of the Hamiltonian, analytically and numerically. Particularly we focused on the symmetry of the terms and the calculation of the energy levels, highlighting the maximum and minimum of the energy together with their corresponding configurations.

In equation 6.1, the structural balance of the system is captured by the second term. This term depends non trivially on the topology of the signed network and is not a function of the number of negative edges. To study its properties, we have developed three numerical methods based on the generation of random signed networks with a given number of negative edges. We found a general formula to compute the lowest possible energy value for a signed network with an arbitrary number of nodes N and a given number of negative edges $M_- < N$:

$$H_{SB}^{(\min)}(M_-) = -\binom{N}{3} + \sum_{i=1}^{M_-} 2(N-2i) = -\binom{N}{3} - 2\Big[M_-^2 - (N+1)M_- + N\Big] \quad (6.2)$$

Thanks to equation 6.2, we have been able to obtain the values of energy coming from the first N steps of the "ordered process" where the signed relations switch sequentially from positive to negative. Therefore, we describe the formation of the first local minimum analytically. We have proceeded by describing the properties of the other local minima of the energy landscape, and lastly, we presented the possible outputs of the global Hamiltonian, combining all the terms together.

The main limitation of this approach is connected to the number of different sign configurations that a system with N elements can present. In fact, for undirected relations, as assumed in this work, the number of possible configurations is $2^{\binom{N}{2}}$, which becomes intractable even for relatively small numbers of elements $N \gtrsim 10$. For this reason, the numerical processes presented to study the structural balance term can explore only a small portion of all the configurations. In Section 3.4, we have pointed out that the number of topologically different configurations is actually much lower. Further analytical studies are necessary to leverage this property and provide procedures to explore this extremely large configuration space.

In the third part of the thesis, we have developed a procedure to simulate the formation of interactions within a social system. By carefully analysing the proposed pseudo-Hamiltonian, we found that it cannot be directly used to model this process. Therefore, we have introduced a modified version of the quantity H_{ij}, that account for the "energy" between a pair of agents (i, j):

$$H_{ij} = -\alpha A_{ij} - \beta |A_{ij}| \sum_k A_{ik} A_{kj} - \gamma s_i |A_{ij}| s_j \quad (6.3)$$

Based on H_{ij}, we have defined a probability of interaction p_{ij} and two different algorithms that use these probabilities to generate interactions. We have aggregated these interactions and represented them using a multi-edge network where multi-edges capture the number of generated interactions between agents. To study the properties of these multi-edge networks, we have introduced some standard statistics and observables that describe the topology of networks. We have computed these quantities for many different sets of parameters of the model. The most important parameters are α, β and γ that reflect the importance of the" social effects" captured in H_{ij}. We have studied how the symmetries of such terms affect the properties of the interaction networks and how their characteristics change as a function of the parameters.

For example, we have found that for some sets of parameters, characterised by a high value of M_-, the interaction networks presents a transition to disconnected networks above a critical value of α.

Another transition was found as a function of γ. Above a critical value, it starts producing clustered networks. In this case, the effect of the homophily term has to overcome the randomness produced by the generating process of interactions.

By analysing the observables and comparing the average path length to the global clustering coefficient, we have found that our systems show a mild small-world regime only when there are very few negative relations. Note that, nevertheless, such states are also the ones more probable to happen as they have lower energy according to 6.1. Starting from this consideration, we were able to formulate the hypothesis that the model presented in this thesis can produce interaction networks that present the same small-world properties expected by real networks, starting from systems that fulfil the condition of low energy, expected by real-world social systems.

To verify this hypothesis, we have tried to draw some connections between the interaction networks and the values of the signed networks' energy that generates them. We did it in two different ways. First, we have computed the usual observables as a function of the energy of the signed network that generates the interactions. We observed the same behaviour of the observables computed as a function of the negative edges, presenting only slight differences in the shape. In particular, for low energy values, the results showed the same small-world regime as the previous case. This behaviour is compatible with the fact that the mean value of the energy is an almost monotonic function of the number of negative links. In fact, as observed in Section 3.4, systems with small M_- tend to have on average lower energies, while the opposite holds for systems with a high number of negative edges.

In the second study, we tested our hypothesis for the cases where small energy values were not associated with a small number of negative edges. We have measured the properties of a set of interaction networks generated starting from a clustered signed network, which we know is one of the local minimum configurations. We have sequentially changed the negative edges between the two clusters to positive until every link was positive.

We found that the small-world regime for the states near the local minima was significant, and our hypothesis was confirmed. The overall shape of the global clustering coefficient and the average path length, as a function of the number of positive links, was very similar to the usual case as a function of the number of negative links but with more extreme values at the origin. This behaviour can be explained by the fact that, in this case, increasing the number of positive edges generates an increase in the number of non-common friends, which leads to a not homogeneous distribution of H_{ij} and a weaker connectedness. Again, it is important to underline that when the number of positive edges is low, the interaction networks present a small-network behaviour, as the APL appears to be relatively lower than the GCC in that region. As we said, this is a characteristic property of social systems, and in this case, it refers to two highly connected clusters with few interactions between them.

This study's main limitation is the large number of different parameters used in the model, which make the analysis more complicated. Moreover, large network analysis suffers from the same combinatorial explosion of the possible configurations as the previous part.

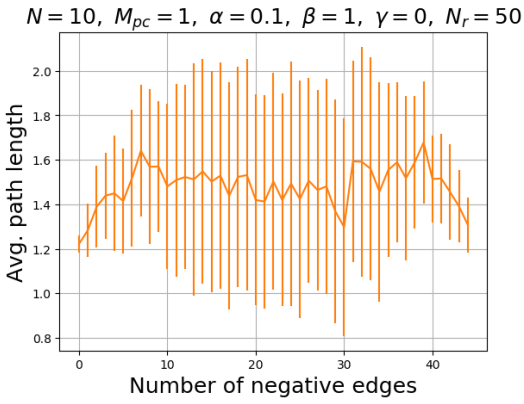
The fourth and last part of the thesis is devoted to the possible applications end extensions of the model. In particular, we have discussed how to introduce an analytical

procedure to describe the dynamics of the signed relation using the global pseudo-Hamiltonian 6.1. We have introduced a Lagrangian formulation and found that the solutions quickly diverge if not properly normalised. Also, we have shown that non-zero values appear in the diagonal of the matrix of relations $\boldsymbol{A}$. Further studies in this direction are needed. Lastly, we have created a connection between our model for generating multi-edge networks and the Exponential Random Graph Models (ERGMs). Despite their limitations, ERGMs are a standard method to study real-world networks. In particular, they allow us to infer interpretable parameters of network models by a fitting procedure. Hence, given a set of interactions among individuals together with their relations, we can use ERGMs to infer the model parameters α, β and γ of 6.3. Future research is needed to explore this possible application of our model on real-world data.

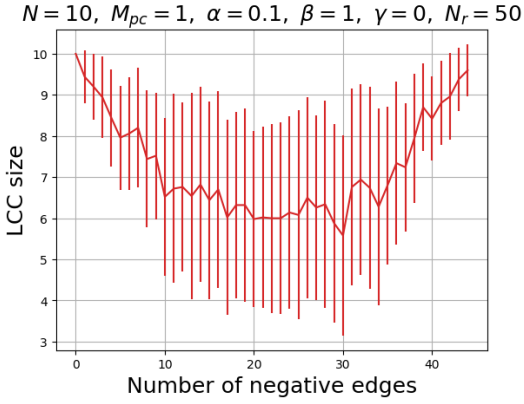
Appendix A

Figures

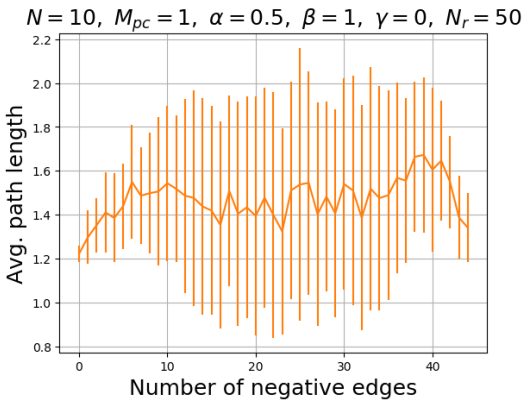
The Appendix A will present the figures relating to the discussion of the observables of the interaction networks in Section 4.6.



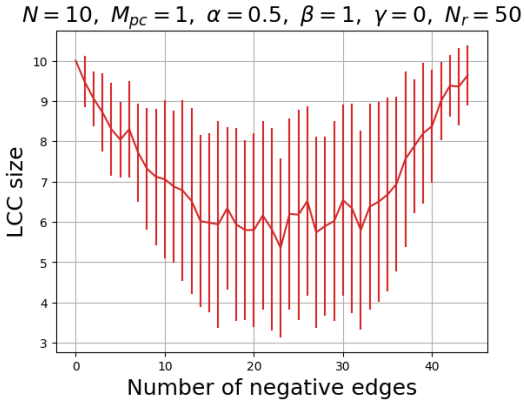
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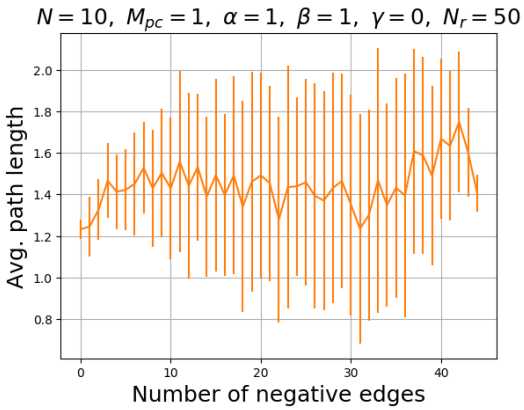
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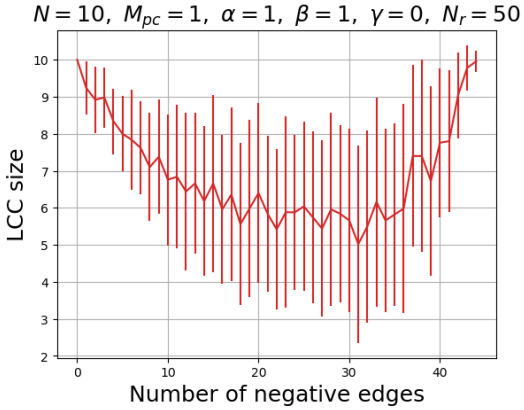
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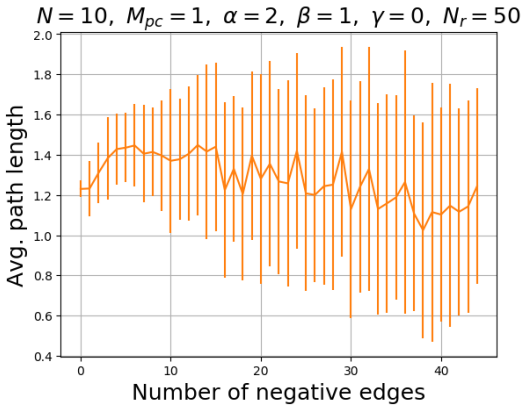


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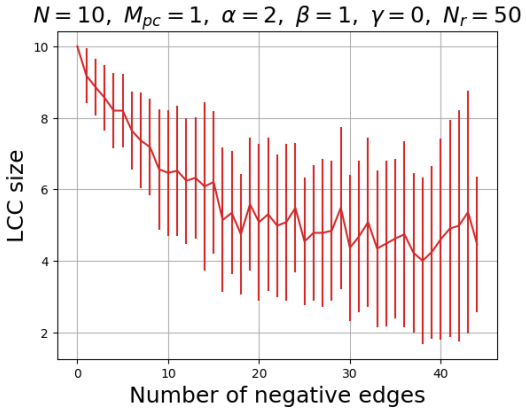


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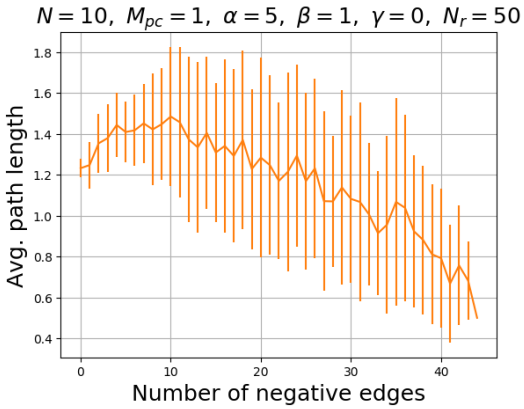
Figure A.1: Representation of the four observables as a function of the number of negative edges M_-. Every point represents the mean of the distribution corresponding to that number of negative edges and the bar corresponds to its standard deviation. For every point the distribution is made of $N_r = 50$ different measurements of an interaction network based on increasing values of α, $\beta = 1$, $\gamma = 0$, $M_{pc} = 1$ and different signed networks generated from RSNG(N, M_-) with $N = 10$.



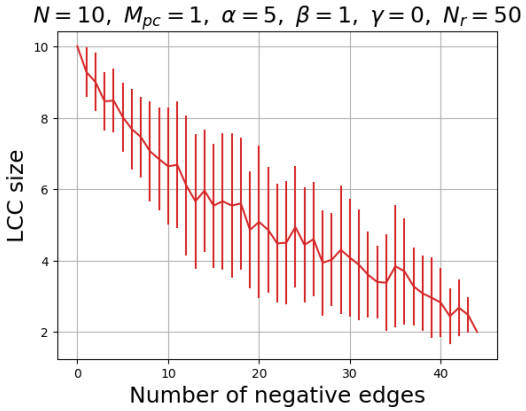
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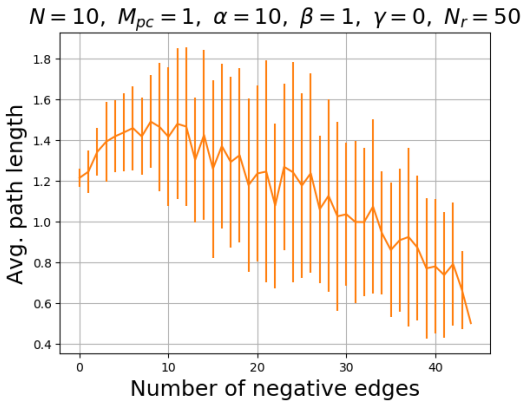
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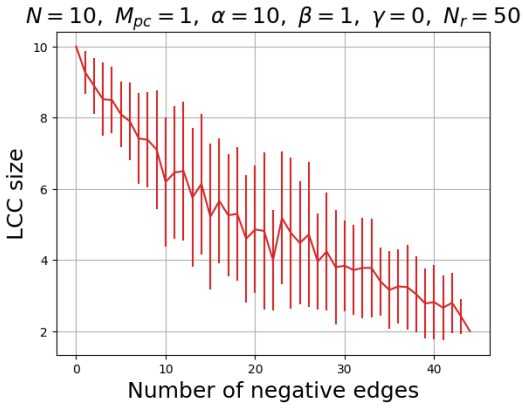
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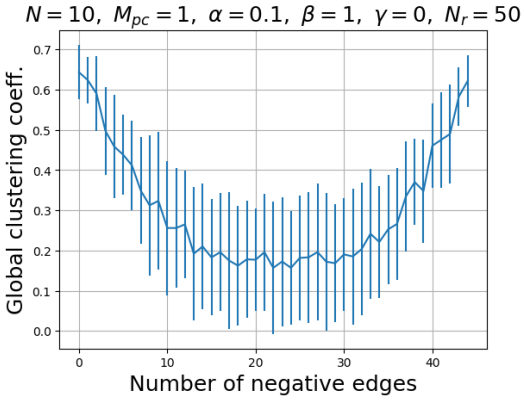


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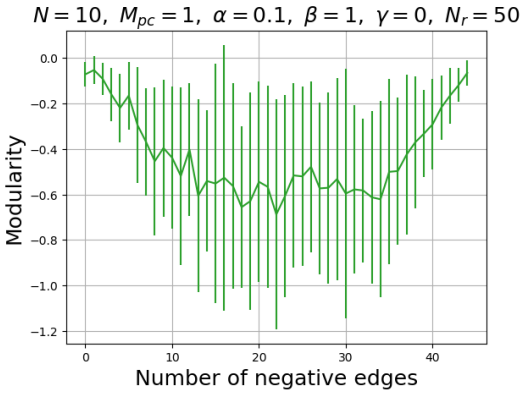


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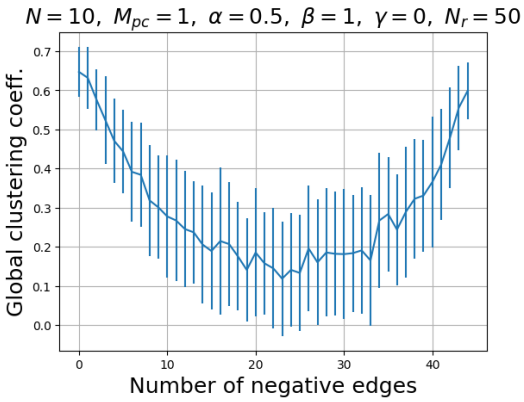
Figure A.2: Representation of the four observables as a function of the number of negative edges M_-. Every point represents the mean of the distribution corresponding to that number of negative edges and the bar corresponds to its standard deviation. For every point the distribution is made of $N_r = 50$ different measurements of an interaction network based on increasing values of α, $\beta = 1$, $\gamma = 0$, $M_{pc} = 1$ and different signed networks generated from RSNG(N, M_-) with $N = 10$.



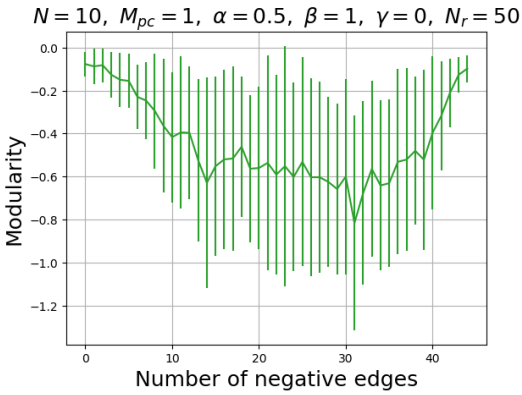
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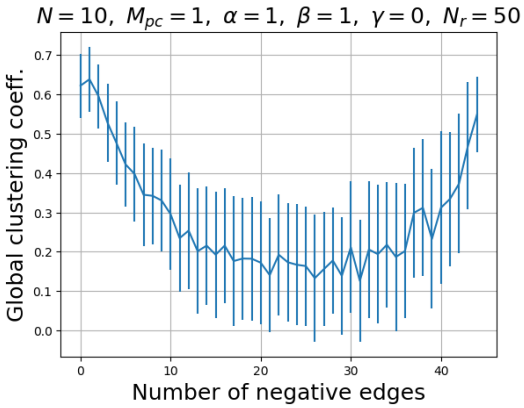
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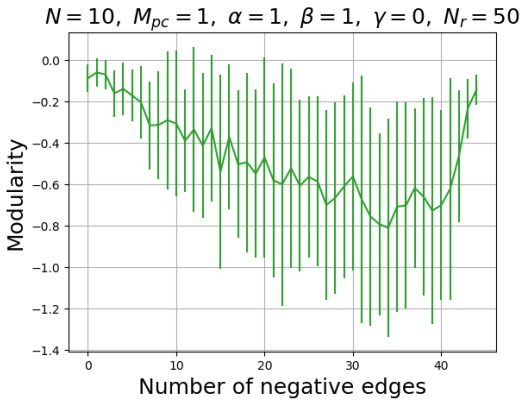
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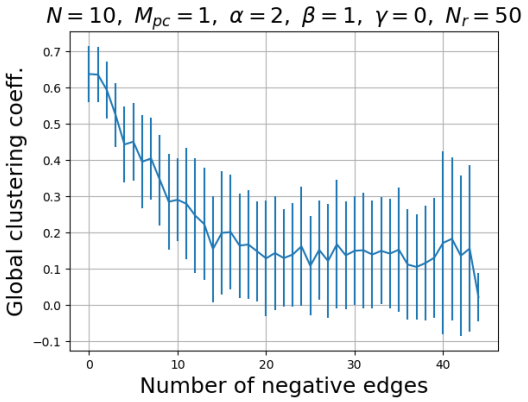


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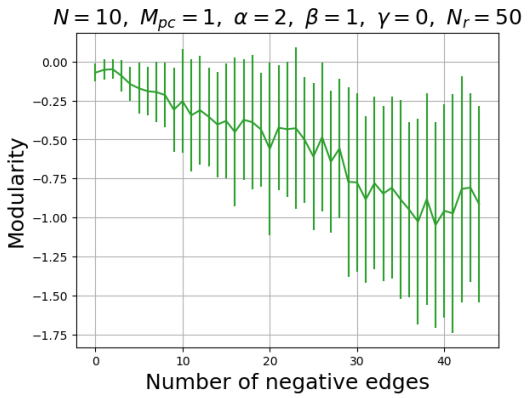


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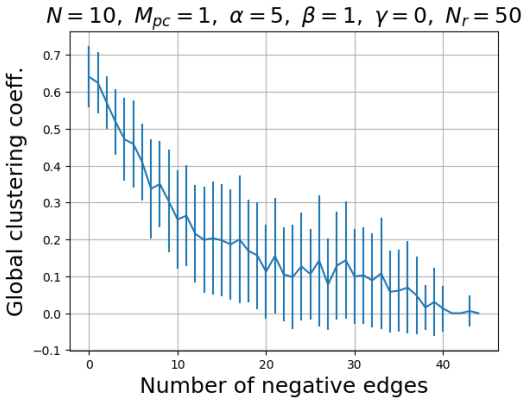
Figure A.3: Representation of the four observables as a function of the number of negative edges M_-. Every point represents the mean of the distribution corresponding to that number of negative edges and the bar corresponds to its standard deviation. For every point the distribution is made of $N_r = 50$ different measurements of an interaction network based on increasing values of α, $\beta = 1$, $\gamma = 0$, $M_{pc} = 1$ and different signed networks generated from RSNG(N, M_-) with $N = 10$.



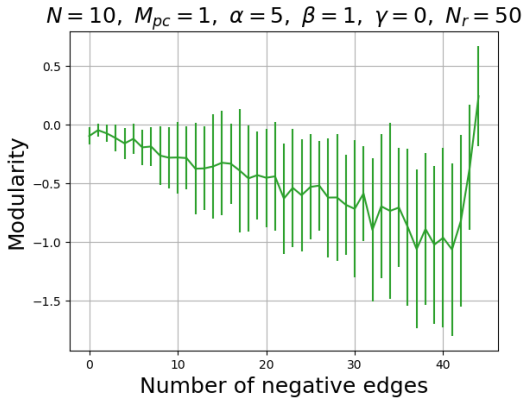
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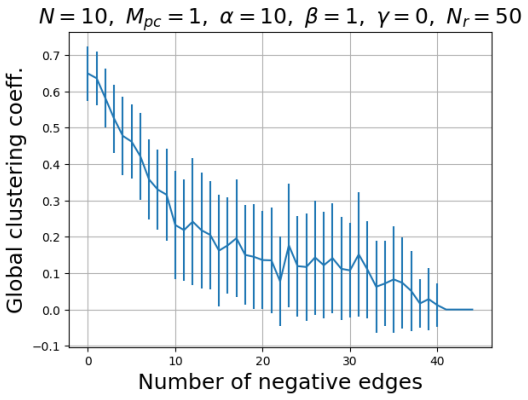
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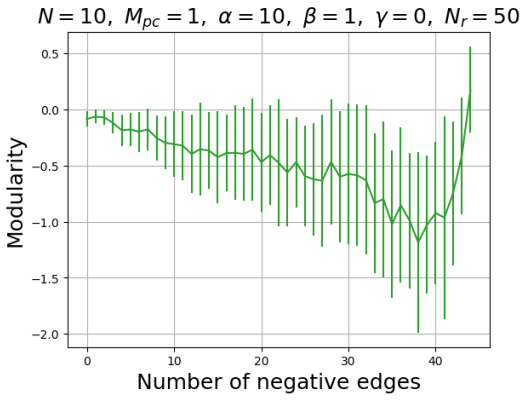
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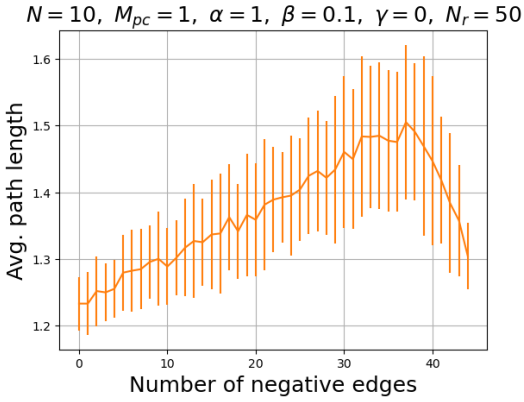


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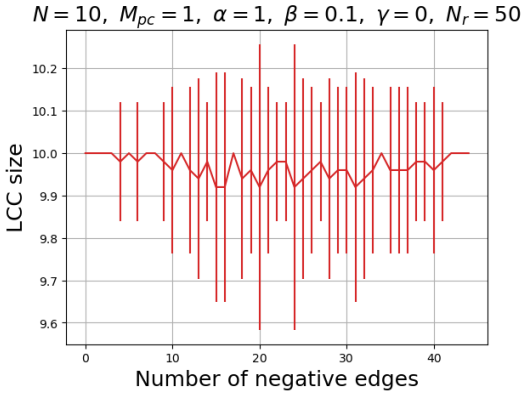


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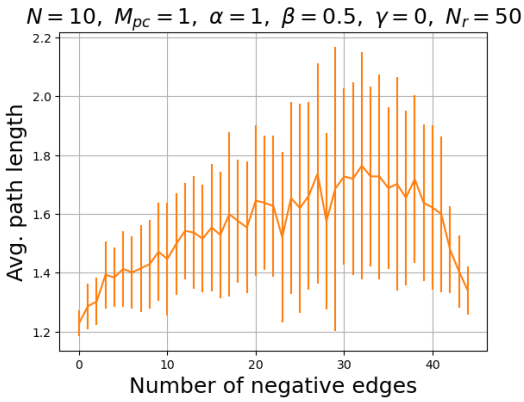
Figure A.4: Representation of the four observables as a function of the number of negative edges M_-. Every point represents the mean of the distribution corresponding to that number of negative edges and the bar corresponds to its standard deviation. For every point the distribution is made of $N_r = 50$ different measurements of an interaction network based on increasing values of α, $\beta = 1$, $\gamma = 0$, $M_{pc} = 1$ and different signed networks generated from RSNG(N, M_-) with $N = 10$.



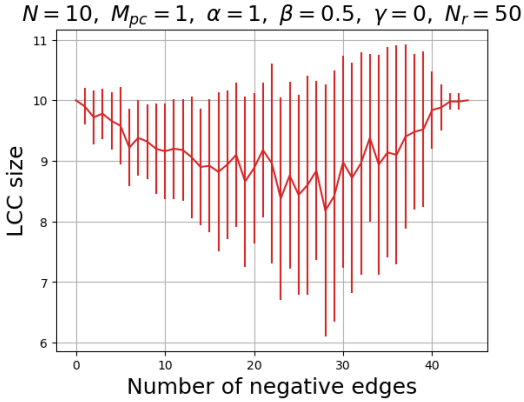
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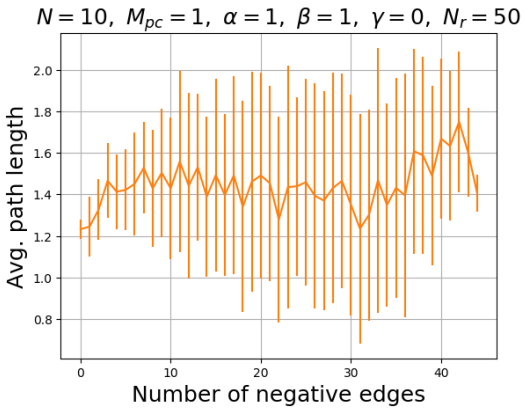
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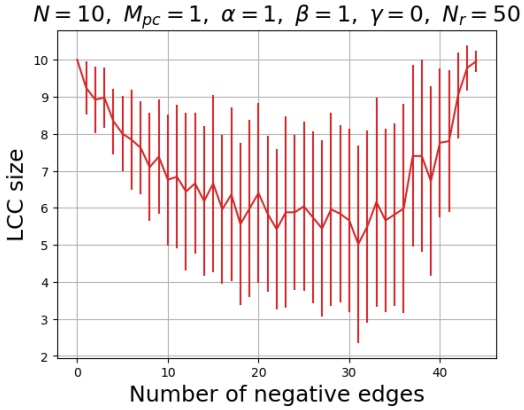
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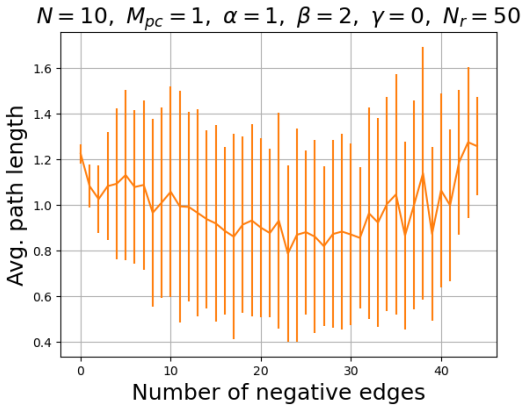


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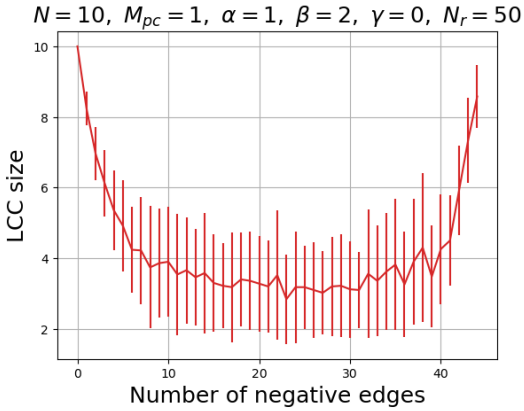


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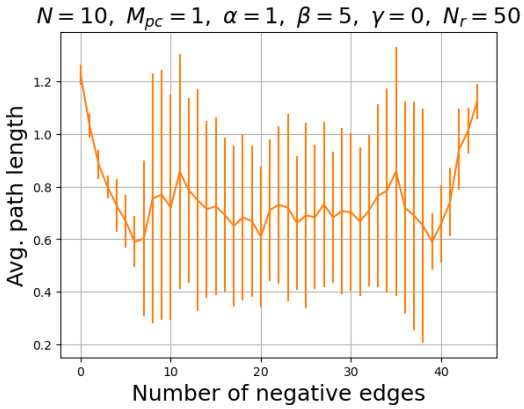
Figure A.5: Representation of the four observables as a function of the number of negative edges M_-. Every point represents the mean of the distribution corresponding to that number of negative edges and the bar corresponds to its standard deviation. For every point the distribution is made of $N_r = 50$ different measurements of an interaction network based on increasing values of β, $\alpha = 1$, $\gamma = 0$, $M_{pc} = 1$ and different signed networks generated from RSNG(N, M_-) with $N = 10$.



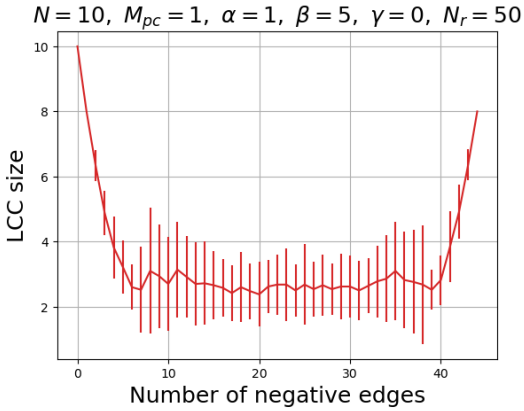
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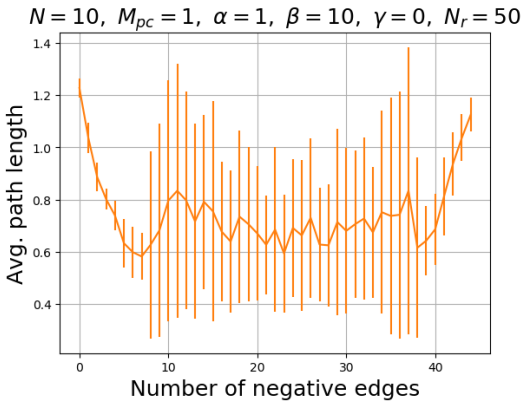
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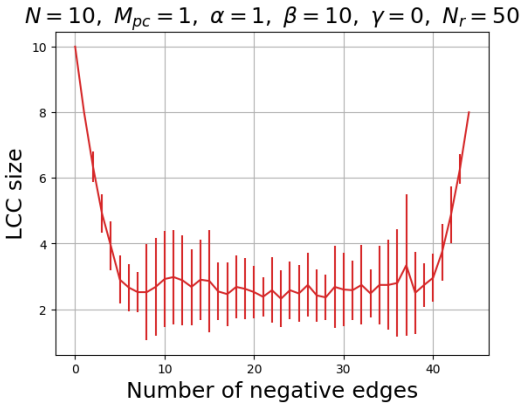
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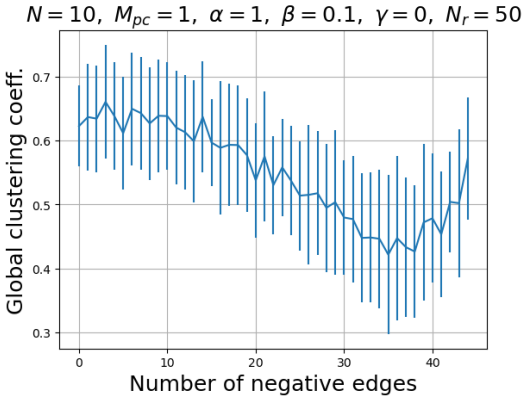


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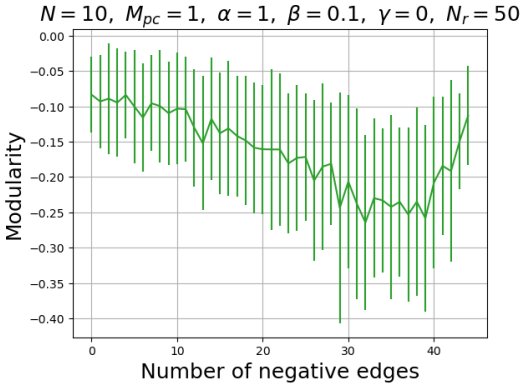


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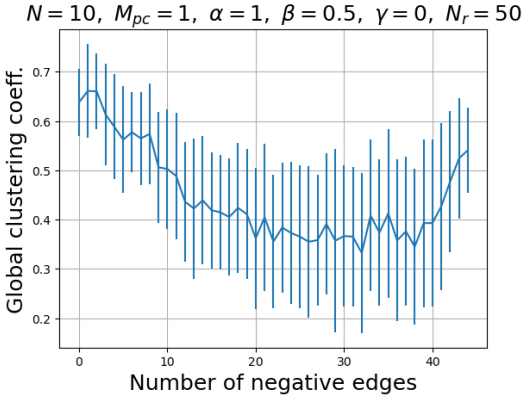
Figure A.6: Representation of the four observables as a function of the number of negative edges M_-. Every point represents the mean of the distribution corresponding to that number of negative edges and the bar corresponds to its standard deviation. For every point the distribution is made of $N_r = 50$ different measurements of an interaction network based on increasing values of β, $\alpha = 1$, $\gamma = 0$, $M_{pc} = 1$ and different signed networks generated from RSNG(N, M_-) with $N = 10$.



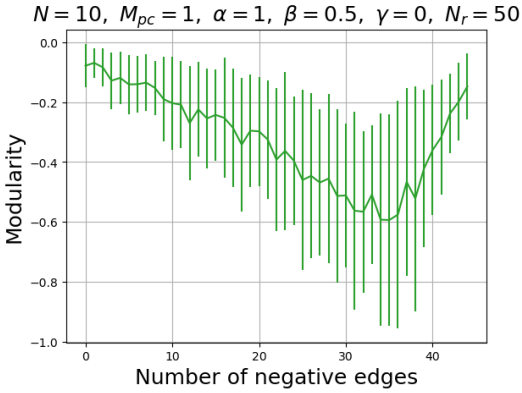
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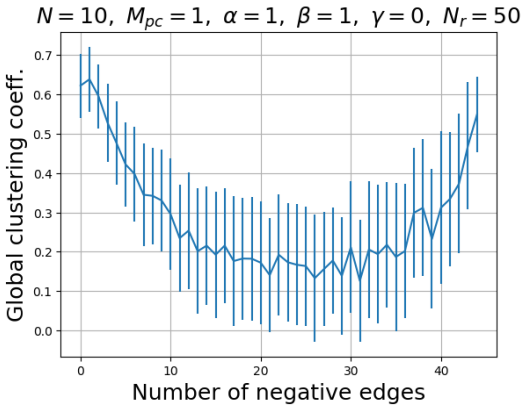
(b) $\beta = 0.1$.



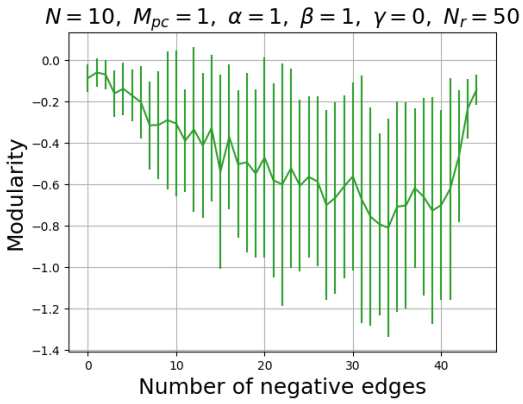
(c) $\beta = 0.5$.



(d) $\beta = 0.5$.

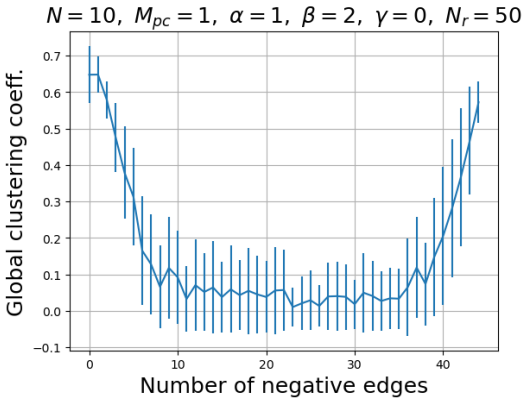


(e) $\beta = 1$.

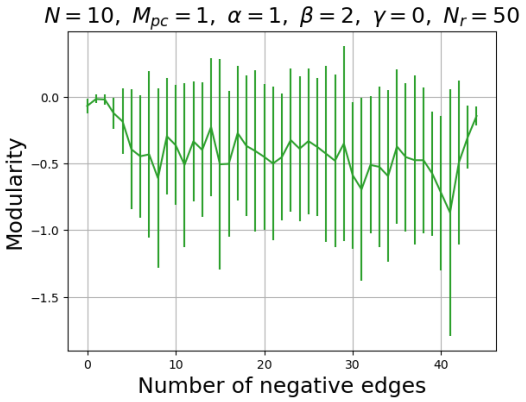


(f) $\beta = 1$.

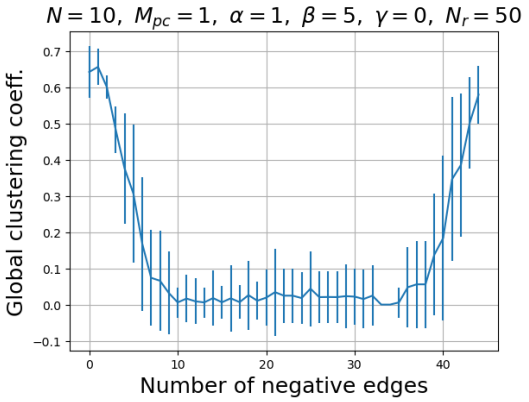
Figure A.7: Representation of the four observables as a function of the number of negative edges M_-. Every point represents the mean of the distribution corresponding to that number of negative edges and the bar corresponds to its standard deviation. For every point the distribution is made of $N_r = 50$ different measurements of an interaction network based on increasing values of β, $\alpha = 1$, $\gamma = 0$, $M_{pc} = 1$ and different signed networks generated from RSNG(N, M_-) with $N = 10$.



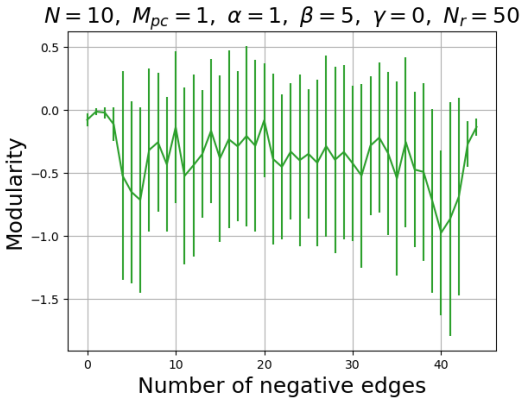
(a) $\beta = 2$.



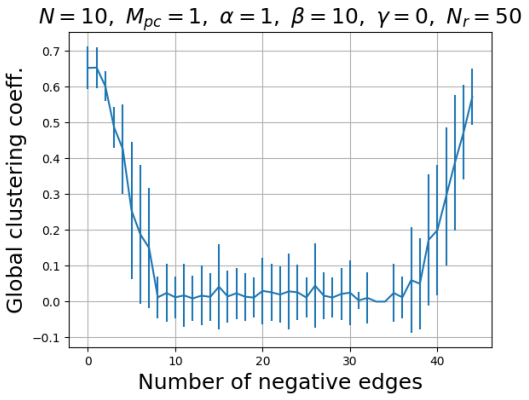
(b) $\beta = 2$.



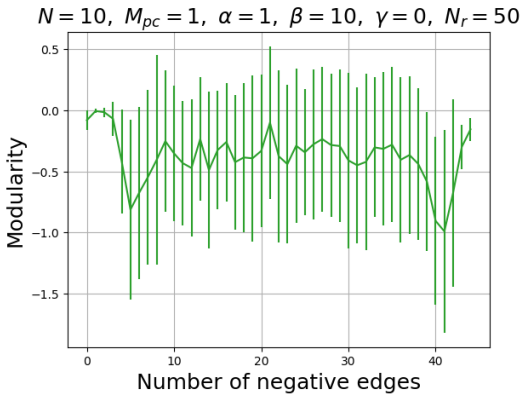
(c) $\beta = 5$.



(d) $\beta = 5$.

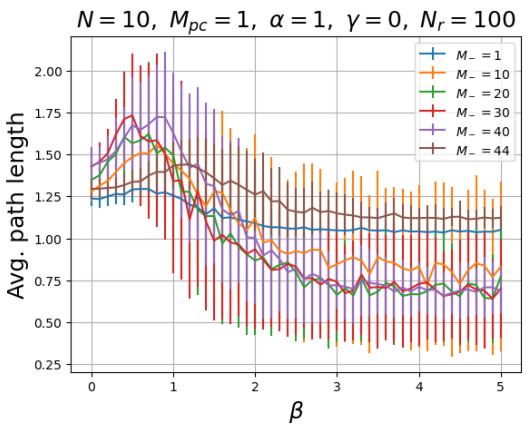


(e) $\beta = 10$.

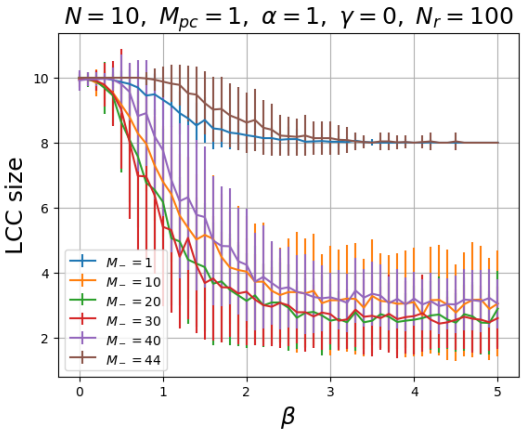


(f) $\beta = 10$.

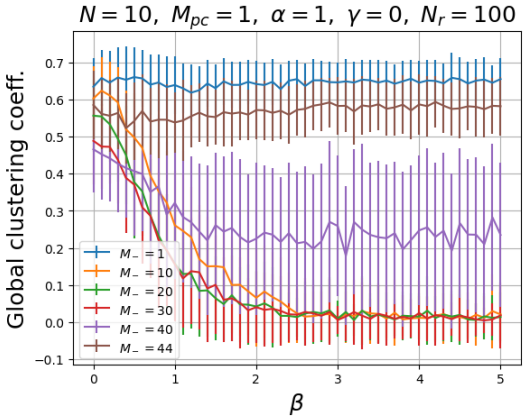
Figure A.8: Representation of the four observables as a function of the number of negative edges M_-. Every point represents the mean of the distribution corresponding to that number of negative edges and the bar corresponds to its standard deviation. For every point the distribution is made of $N_r = 50$ different measurements of an interaction network based on increasing values of β, $\alpha = 1$, $\gamma = 0$, $M_{pc} = 1$ and different signed networks generated from RSNG(N, M_-) with $N = 10$.



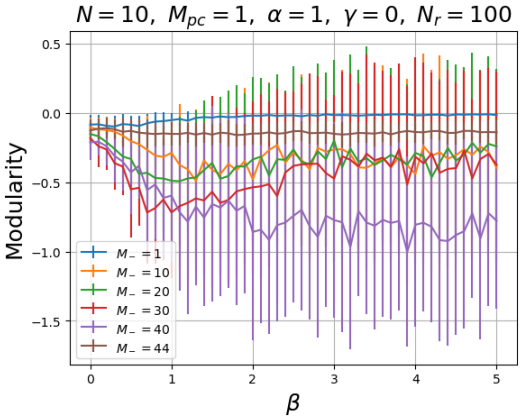
(a) Average path length of the LCC as a function of M_-.



(b) Size of the largest connected component as a function of M_-.

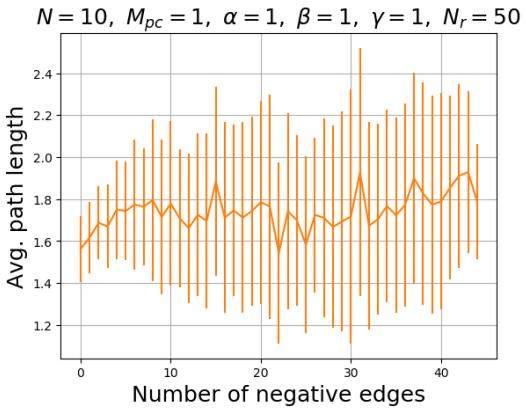


(c) Global clustering coefficient as a function of M_-.

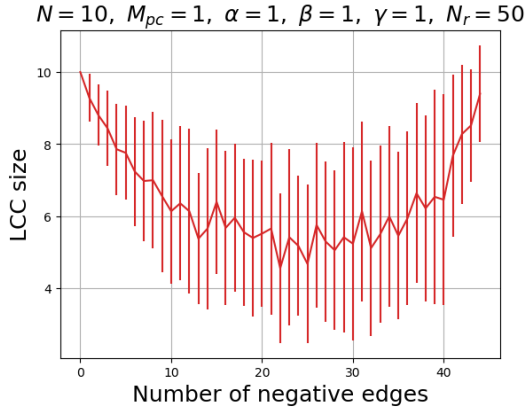


(d) Community assortativity coefficient as a function of M_-.

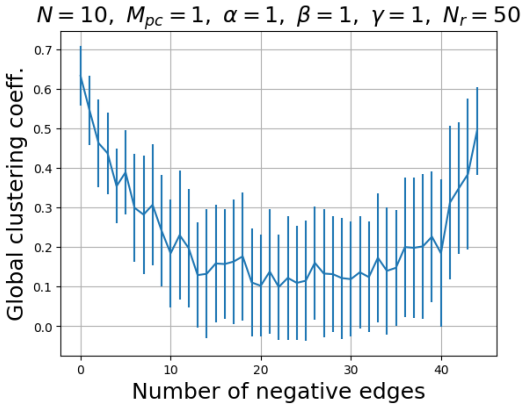
Figure A.9: Representation of the four observables as a function of the parameter α. Every point represents the mean of the distribution corresponding to that number of negative edges and the bar corresponds to its standard deviation. For every point the distribution is made of $N_r = 100$ different measurements of an interaction network based on the set of parameters $\alpha = 1,\ \gamma = 0,\ M_{pc} = 1$ and a different signed network generated from RSNG(N, M_-) with $N = 10$ and different values of the number of negative edges M_- reported in the legend.



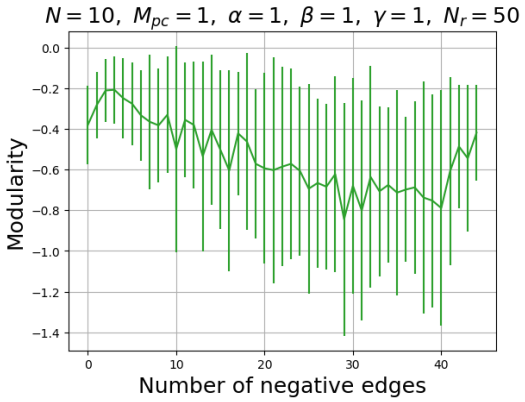
(a) Average path length of the LCC as a function of M_-.



(b) Size of the largest connected component as a function of M_-.

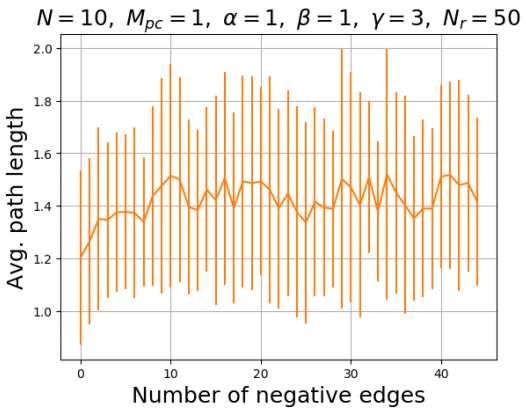


(c) Global clustering coefficient as a function of M_-.

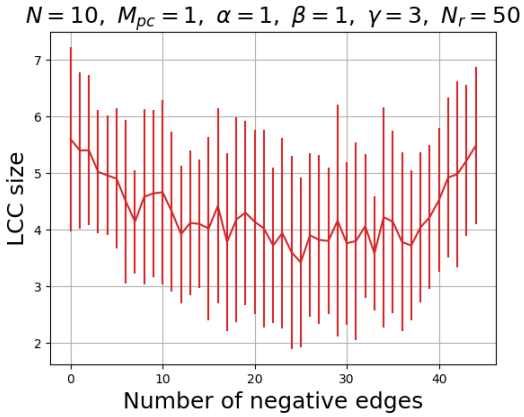


(d) Community assortativity coefficient as a function of M_-.

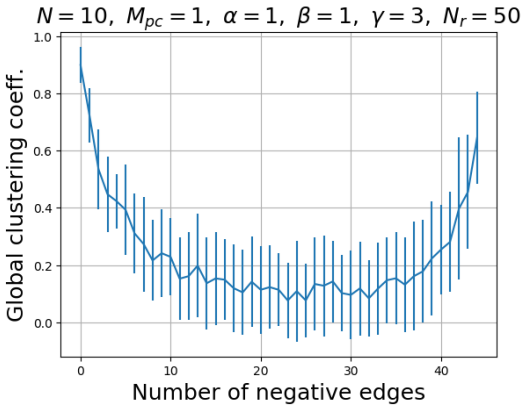
Figure A.10: Representation of the four observables as a function of the number of negative edges M_-. Every point represents the mean of the distribution corresponding to that number of negative edges and the bar corresponds to its standard deviation. For every point the distribution is made of $N_r = 50$ different measurements of an interaction network based on the set of parameters $\alpha = \beta = \gamma = 1,\ M_{pc} = 1$ and a different signed network generated from RSNG(N, M_-) with $N = 10$.



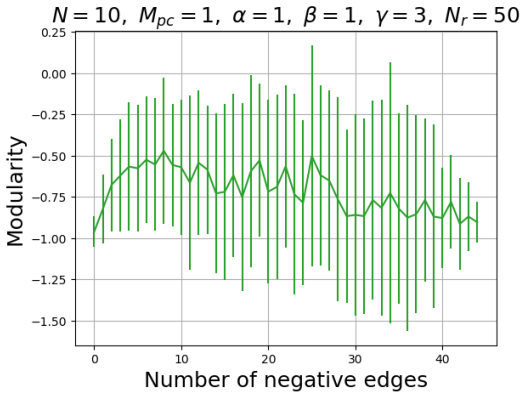
(a) Average path length of the LCC as a function of M_-.



(b) Size of the largest connected component as a function of M_-.

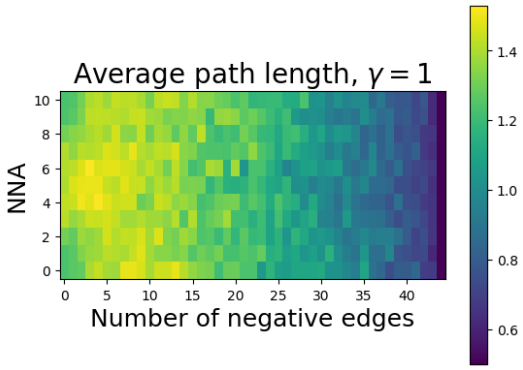


(c) Global clustering coefficient as a function of M_-.

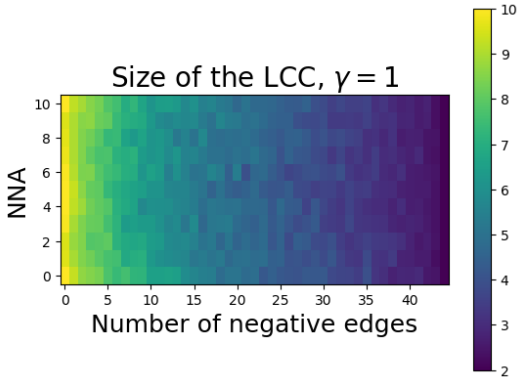


(d) Community assortativity coefficient as a function of M_-.

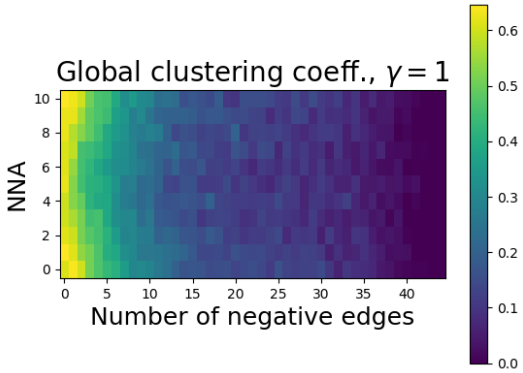
Figure A.11: Representation of the four observables as a function of the number of negative edges M_-. Every point represents the mean of the distribution corresponding to that number of negative edges and the bar corresponds to its standard deviation. For every point the distribution is made of $N_r = 50$ different measurements of an interaction network based on the set of parameters $\alpha = \beta = 1,\ \gamma = 3,\ M_{pc} = 1$ and a different signed network generated from RSNG(N, M_-) with $N = 10$.



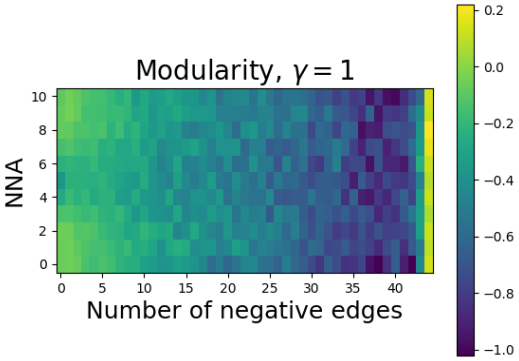
(a) Mean value of the average path length of the LCC as a function of M_- and NNA.



(b) Mean value of the aize of the largest connected component as a function of M_- and NNA.

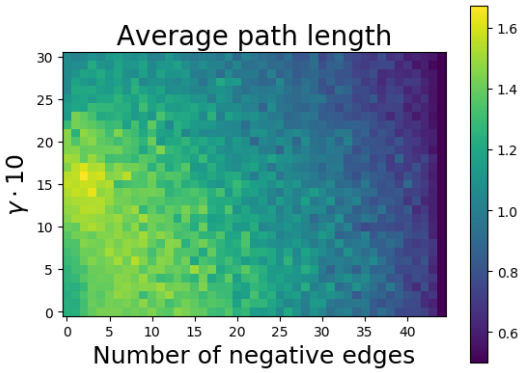


(c) Mean value of the global clustering coefficient as a function of M_- and NNA.

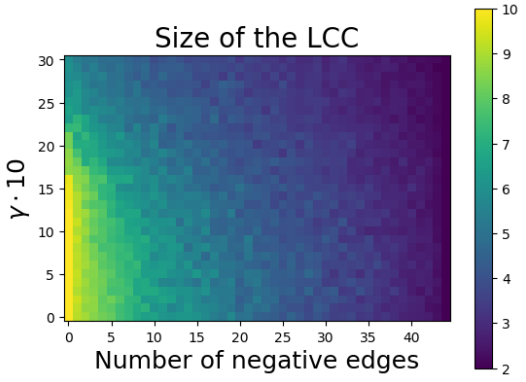


(d) Mean value of the community assortativity coefficient as a function of M_- and NNA.

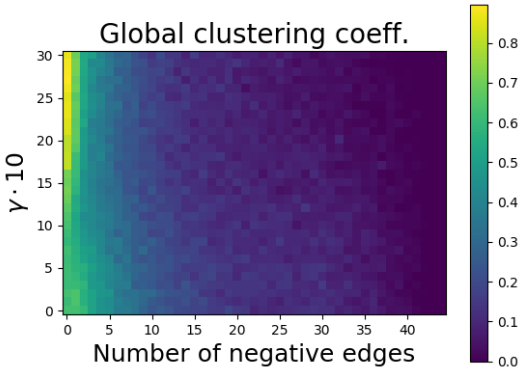
Figure A.12: Representation of the four observables as a function of the number of negative edges M_- and the number of negative attributes NNA. Every point represents the mean of the distribution corresponding to that number of negative edges and attributes. For every point the distribution is made of $N_r = 50$ different measurements of an interaction network based on the set of parameters $\alpha = \beta = 1,\ \gamma = 1,\ M_{pc} = 1$ and a different signed network generated from RSNG(N, M_-) with $N = 10$.



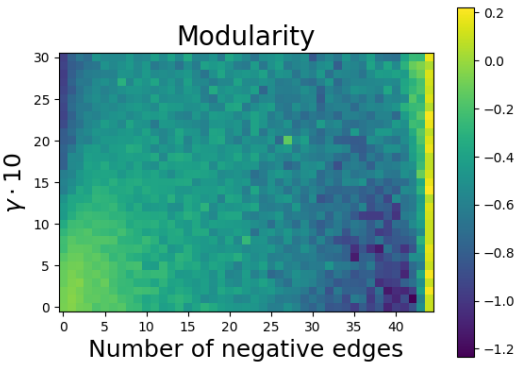
(a) Mean value of the average path length of the LCC as a function of M_- and γ.



(b) Mean value of the size of the largest connected component as a function of M_- and γ.

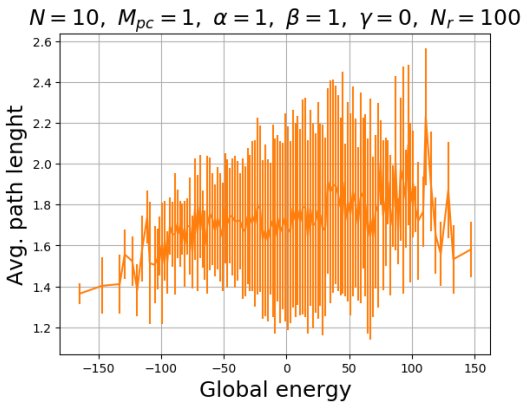


(c) Mean value of the global clustering coefficient as a function of M_- and γ.

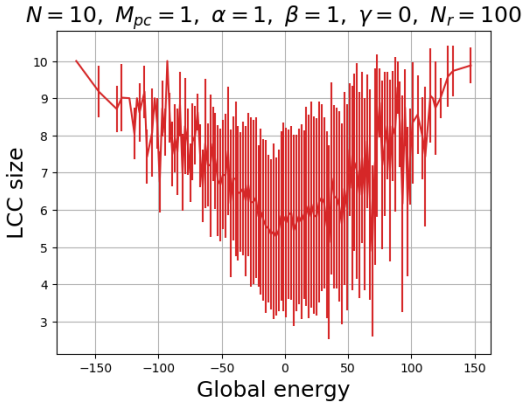


(d) Mean value of the community assortativity coefficient as a function of M_- and γ.

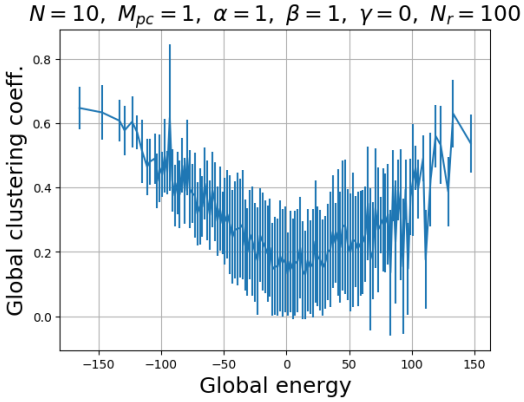
Figure A.13: Representation of the mean values of the four observables as a function of the number of negative edges M_- and $\gamma \cdot 10$, this means that, for example, the value 15 correspond to $\gamma = 1.5$. Every point represents the mean of the distribution corresponding to that number of negative edges and that value of γ. For every point the distribution is made of $N_r = 50$ different measurements of an interaction network based on the set of parameters $\alpha = \beta = 1$, $M_{pc} = 1$, a different signed network generated from RSNG(N, M_-) with $N = 10$ and a fixed number of negative attributes NNA=5.



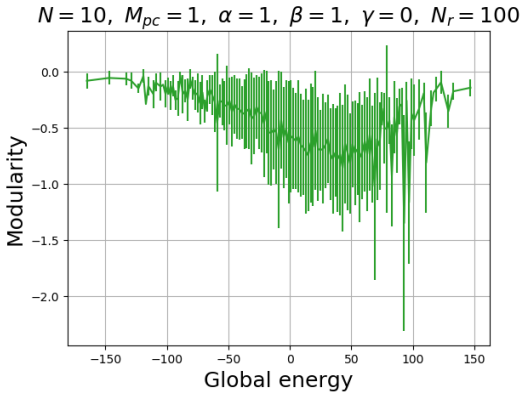
(a) Average path length of the LCC as a function of the value of H of the signed network.



(b) Size of the largest connected component as a function of the value of H of the signed network.



(c) Global clustering coefficient as a function of the value of H of the signed network.



(d) Community assortativity coefficient as a function of the value of H of the signed network.

Figure A.14: Representation of the four observables as a function of the value of the global Hamiltonian H of the signed network. Every point represents the mean of the distribution corresponding to the signed networks with same value of H and the bar corresponds to its standard deviation. For every point the distribution is made of $N_r = 100$ different measurements of an interaction network based on the set of parameters $\alpha = \beta = 1,\ \gamma = 0,\ M_{pc} = 1$ and a different signed network generated from RSNG(N, M_-) with $N = 10$.

Appendix B

Codes

```
import numpy as np
import matplotlib.pyplot as plt
import pathpy as pp
from tqdm.notebook import tqdm
from copy import deepcopy
from collections import defaultdict
from itertools import combinations
from collections import Counter
import scipy.special as ss
import itertools
import networkx as nx
from matplotlib import rcParams
```

```
def calcH1(net):
    A=net.adjacency_matrix().todense()
    return -np.sum(A)/2

def calcH2(net):
    A = net.adjacency_matrix().todense()
    return -np.sum([(A*A*A)[i,i] for i in range(A.shape[0])])/6

def calcH3(net):

    # network's nodes have to have a scalar attribute 's'

    A=net.adjacency_matrix().todense()
    sumv = 0
    for i, v in enumerate(net.nodes):
        for j, w in enumerate(net.nodes):
            if i != j:
                    sumv += net.nodes[i]['s']*net.nodes[j]['s']
```

```
    return -sumv/2


def calcH(net, gamma=1, alpha=1, beta=1):

    A = net.adjacency_matrix().todense()
    N = len(A)

    H1 = -np.sum(A)/2

    H2 = 0
    for i in range(N):
        for j in range(N):
            for k in range(N):
                H2 += A[i,j]*A[j,k]*A[k,i]
    H2 = -H2/6    # divided by 3

    # network's nodes have to have an attribute 's'
    # be sure that the vector 's' has the same dimension for all the nodes

    H3 = 0
    for i, v in enumerate(net.nodes):
        ls = len(net.nodes[i]['s'])
        for j, w in enumerate(net.nodes):

            if i != j:
                for k in range(ls):
                    H3 += net.nodes[i]['s'][k]*net.nodes[j]['s'][k]
                    # if the network is not fully connected use:
                    # H3 += net.nodes[i]['s'][k]*np.abs(A[i,j])*net.nodes[j
    H3 = -H3/2

    return gamma*H1 + alpha*H2 + beta*H3


def calcH12(net, alpha=1, beta=1):

    A = net.adjacency_matrix().todense()
    N = len(A)

    H1 = -np.sum(A)/2

    H2 = calcH2(net)

    return alpha*H1 + beta*H2
```

```
def nT(N):
    '''Number of triangles '''
    return int(N*(N-1)*(N-2)/6)

def nE(N):
    '''Number of edges '''
    return int(N*(N-1)/2)



def rsng(N, p1=0.5, self_loop=False, undirected=True):
    '''Random Sign Network Generator '''

    net=pp.Network(directed=not(undirected))
    for i in range(N):
        net.add_node(i)

    if undirected:
        for i in range(N):
            for j in range(i,N):
                if self_loop==False:
                    if i!=j:
                        r = np.random.random()
                        if r <= p1:
                            net.add_edge(i,j,weight=1)
                        else: net.add_edge(i,j,weight=-1)
                else:
                    r = np.random.random()
                    if r <= p1:
                        net.add_edge(i,j,weight=1)
                    else: net.add_edge(i,j,weight=-1)


    # for directed graphs
    else:
        for i in range(N):
            for j in range(N):
                if self_loop==False:
                    if i!=j:
                        r = np.random.random()
                        if r <= p1:
                            net.add_edge(i,j,weight=1)
                        else: net.add_edge(i,j,weight=-1)
                else:
                    r = np.random.random()
                    if r <= p1:
```

```
                    net.add_edge(i,j,weight=1)
                else: net.add_edge(i,j,weight=-1)

    return net


def rsng2(N, m1, attr=False, Arand=True, NNA=0):
    ''' Random Sign Network Generator
    (Complete, Undirected and without Self Loops)
    with a fixed number m1 of negative signs '''

    net = pp.Network(directed=False)

    nedges = np.int(N*(N-1)/2)

    edgelist = [1] * nedges

    negedges = np.random.choice(range(nedges), size=m1, replace=False)

    for ni in negedges:
        edgelist[ni]=-1

    k=0
    for i in range(N):
        for j in range(i+1,N):
            net.add_edge(i, j, weight=edgelist[k])
            k+=1

    if attr:
        if Arand:
            for v in net.nodes:
                net.nodes[v]['s'] = np.random.choice([-1,1])
        else:
            s=np.ones(N)
            for j in range(NNA):
                s[j]=-1
            for j, v in enumerate(net.nodes):
                net.nodes[v]['s'] = s[j]

    return net


def addA(net):

    for v in net.nodes:
        net.nodes[v]['s'] = np.random.choice([-1,1])
```

```
    return net


def fcng(numn):

    '''Fully Connected Network Generator, undirected '''

    net = pp.Network()

    for i in range(numn):
        for j in range(i+1,numn):
            net.add_edge(i, j, weight=1)

    return net



def weiDeg(net, undirected=True):
    "Weighted Degree Sequence"

    if undirected:
        deg=np.zeros(net.ncount())
        for i, v in enumerate(net.nodes):
            deg[i]=net.nodes[v]['inweight']

    return deg


def prop_seq(net, prop = 'inweight'):
    '''Property sequence '''
    return [net.nodes[x][prop] for x in net.nodes]

def prop_dist(net, prop = 'degree', density=True, func=None):
    '''Property distribution, only for integer properties '''

    if func==None:
        if int(net.nodes[1][prop]) != net.nodes[1][prop]: print('Only for i
        p_k = Counter([int(net.nodes[x][prop]) for x in net.nodes])
    else:
        p_k = Counter([int(func(net, x, density=density)) for x in net.node

    dist = [0]*max(p_k)

    if density:
        for x in range(max(p_k)):
            dist[x] = p_k[x]/net.ncount()
    else:
```

```
        for x in range(max(p_k)):
            dist[x] = p_k[x]

    return dist


def local_cc(network, v, density=True):
    if network.nodes[v]['degree']<2:
        return 0
    k_i = 0
    neighbours = list(network.successors[v])
    for idx, i in enumerate(neighbours):
        for j in neighbours[idx+1:]:
            if (i,j) in network.edges:
                k_i += 1
    if density:
        return 2*k_i/(network.nodes[v]['degree']*(network.nodes[v]['degree'
    else:
        return k_i

def global_cc(network):
    s = 0
    for v in network.nodes:
        s += local_cc(network, v)
    return s/network.ncount()


def local_sb(network, v):
    ''' It works only with fully connected signed (not weighted) network'''

    if network.nodes[v]['degree']<2:
        return 0
    k_i = 0
    neighbours = list(network.successors[v])
    for idx, i in enumerate(neighbours):
        for j in neighbours[idx+1:]:
            if (i,j) in network.edges:
                k_i += -1 + network.edges[(v,i)]['weight']*network.edges[(v
    return -k_i/2

def global_sb(network):
    ''' It works only with fully connected signed (not weighted) network'''

    s = 0
    for v in network.nodes:
        s += local_sb(network, v)
    return s/3
```

```
def visNeg(net, edgelist=False, colorneg='red', colorpos=None, widneg=0.6,
    '''Visualize relation networks with negative edges in red'''

    nedg = {}
    nwid = {}
    net2 = deepcopy(net)

    for ei in net2.edges:

        if net2.edges[ei]['weight'] == -1:
            nedg[ei] = colorneg
            nwid[ei] = widneg
            net2.edges[ei]['weight'] = weineg
            net2.add_edge(ei[0], ei[1])
        else:
            if colorpos != None:
                nedg[ei ] = colorpos
            if widpos != None:
                nwid[ei] = widpos

    pp.visualisation.plot(net2, edge_width = nwid, edge_color = nedg, label

    if edgelist == True:
        return net.edges


def visWei(net, mult=.5, edgelist=False, weiForce=True):
    '''Visualize interaction networks with weighted/multi edges '''

    nwid={}
    net2 = deepcopy(net)
    if weiForce:
        for ei in net2.edges:
            nwid[ei]=net2.edges[ei]['weight']*mult
    else:
        for ei in net2.edges:
            nwid[ei]=net2.edges[ei]['weight']*mult
            net2.add_edge(ei[0],ei[1],weight=1)

    pp.visualisation.plot(net2, edge_width=nwid, label_size='12px')

    if edgelist==True:
        return net2.edges
```

```
def calcHij(net, i, j, Abs=True, alpha=1, beta=1, gamma=0):

    A = net.adjacency_matrix().todense()
    N = net.ncount()
    Hk = 0
    for k in net.nodes:
        Hk += A[i,k]*A[k,j]

    if Abs: Hk = np.abs(A[i,j]) * Hk
    else: Hk = A[i,j] * Hk

    Hs=0

    if gamma != 0:
        Hs = np.abs(A[i,j]) * net.nodes[i]['s'] * net.nodes[j]['s']

    return -(alpha * A[i,j]  + beta * Hk  + gamma * Hs )



def Prob_Gen(net, Abs=True, alpha=1, beta=1, gamma=0, norm=True, T=1):
    ''' Probability of Interaction Network Generator '''

    N=net.ncount()
    N2=N*(N-1)/2
    net2 = pp.Network(directed=False)
    net2.nodes = net.nodes
    eHij=np.zeros((N,N))

    for i in net2.nodes:
        for j in net2.nodes:
            if i != j:
                eHij[i,j] = np.exp(-calcHij(net, i, j, Abs=Abs, alpha=alpha

    eHtot = np.sum(eHij)/2    # NEW !!!

    if norm:
        for i in net2.nodes:
            for j in net2.nodes:
                if i != j:
                    net2.add_edge(i, j, weight = eHij[i,j]/eHtot)
    else:
        for i in net2.nodes:
            for j in net2.nodes:
```

```
                if i != j:
                    net2.add_edge(i, j, weight = eHij[i,j])


    return net2 #, eHtot]



def Int_Gen(Sign_net, M_pc=1, Abs=True, alpha=1, beta=1, gamma=0, norm=True
    '''Generate interaction network starting from a relation network'''

    pnet3 = Prob_Gen(Sign_net, Abs=Abs, alpha=alpha, beta=beta, gamma=gamma

    N = Sign_net.ncount()
    M = np.int(np.round(M_pc * Sign_net.ecount())) # number of edges create


    inet3 = pp.Network()
    for v in Sign_net.nodes:
        inet3.add_node(v)

    m = M

    nodelist = range(N)
    edgedict = defaultdict(lambda: 0)

    while m > 0:
        v = np.random.choice(nodelist)
        w = np.random.choice(nodelist)
        if v != w:
            r = np.random.random()
            if r <= pnet3.edges[v,w]['weight']:
                edgedict[(v,w)]+=1
                edgedict[(w,v)]+=1
                m -= 1

    #inet3=pp.Network()
    for ei in edgedict:
        inet3.add_edge(ei[0], ei[1], weight = edgedict[ei])

    return inet3


def Int_GenNew(Sign_net, M_pc=1, Abs=True, alpha=1, beta=1, gamma=0, norm=T
    '''Generate interaction network starting from a relation network with a

    pnet3 = Prob_Gen(Sign_net, Abs=Abs, alpha=alpha, beta=beta, gamma=gamma
```

```
    N = Sign_net.ncount()
    M = np.int(np.round(M_pc * Sign_net.ecount())) # number of edges create

    inet3 = pp.Network()
    for v in Sign_net.nodes:
        inet3.add_node(v)

    m = M

    nodelist = range(N)
    edgedict = defaultdict(lambda: 0)

    ndist = weiDeg(pnet3)
    ndist = ndist/np.sum(ndist)

    while m > 0:

        v = np.random.choice(list(inet3.nodes))
        r1 = np.random.random()
        if r1 < ndist[v]:
            for w in pnet3.successors[v]:
                r = np.random.random()
                if r <= pnet3.edges[v,w]['weight']:
                    edgedict[(v,w)]+=1
                    edgedict[(w,v)]+=1
                    m -= 1

    #inet3=pp.Network()
    for ei in edgedict:
        inet3.add_edge(ei[0], ei[1], weight = edgedict[ei])

    return inet3


def ord_process(N):
    ''' Ordered process '''
    if type(N) is int:
        net = fcng(N)
    elif type(N) is pp.classes.network.Network:
        net = N
        N = net.ncount()
    else:
        print('Error: type(N) has to be an int or a pathpy Network')

    M = net.ecount()
    elist = list(net.edges)
```

```
    vh = np.zeros(M+1)
    vh[0] = calcH2(net)
    for ei in (range(M)):
        net.add_edge(elist[ei][0], elist[ei][1], weight=-1)
        vh[ei+1] = calcH2(net)

    return vh

def rand_process(N, nrep):
    ''' Random process '''
    if type(N) is int:
        net = fcng(N)
    elif type(N) is pp.classes.network.Network:
        net = N
        N = net.ncount()
    else:
        print('Error: type(N) has to be an int or a pathpy Network')

    M = net.ecount()
    elist = list(net.edges)

    vh = np.zeros((M+1, nrep))
    vh[0, :] = calcH2(net)

    for ri in tqdm(range(nrep)):
        net1 = deepcopy(net)
        elist1 = deepcopy(elist)
        np.random.shuffle(elist1)
        for ei in (range(M)):
            net1.add_edge(elist1[ei][0], elist1[ei][1], weight=-1)
            vh[ei+1, ri] = calcH2(net1)

    return vh


def minE(N, Mm):
    Em = -ss.binom(N, 3)
    for i in range(1, Mm+1):
        Em += 2*(N-2*i)
    return Em

def maxE(N,Mm):
    if Mm <= N/2:
        return -ss.binom(N,3)+2*Mm*(N-2)
    else: return -ss.binom(N,3)+2*(Mm-2)*(N-2)+2*2*(N-3)
```

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